Designing AI Information Artifacts for Teams

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Abstract

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Collaborative teams rely on information artifacts such as notes, recordings, recaps, and recommendations to stay aligned and coordinate work. Traditionally, what a team could preserve was limited by the effort team members could spare to write it down. Generative AI changes that constraint by making artifacts cheaper to produce, but it does not expand the attention available for teams to consume them. In this thesis, I ask how generative AI information artifacts can help groups repair and maintain common ground when attention is finite. I argue that such artifacts must select information worth carrying forward, ground new information in the group's social context, and time contributions so they support rather than interrupt collaboration.

I develop this argument through three projects. First, Meeting Bridges examines how people use meeting information after synchronous discussion. Through interviews with 13 remote workers, a survey of 198 participants, and a co-design study with 16 workers across five groups, I show that meeting artifacts should not be treated only as records for recall but should instead be designed to carry common ground into asynchronous work. Second, Social-RAG extends retrieval-augmented generation from factual grounding to social grounding. I instantiate the Social-RAG technique in PaperPing, an AI agent that retrieves social facts from group interaction

history to generate paper recommendations grounded in the group's past conversations, members, and reactions. PaperPing was deployed for three months across 18 Slack channels reaching 500+ researchers. Third, ProSKI evaluates when proactive agents should contribute external knowledge to group conversations. Built from 384 participant-authored messages across 35 real group conversations, it shows that human timing judgments follow a distribution rather than a single point, and that current models achieve human-level placement only with hindsight. Under deployable turn-by-turn replay conditions, model performance falls to roughly a third of its retrospective value, with premature contributions as the dominant error.

Together, these projects contribute to the social computing and human-AI interaction literature by envisioning a future in which AI systems not only generate information accurately but also communicate well in social contexts, augmenting how people work together rather than obliging people to adapt to the systems around them.

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Chapter 1

Introduction

In modern societies, people rarely work alone. Nearly every workplace innovation, scientific advancement, and routine daily task depends on a group of people pooling knowledge and coordinating their efforts over time. Across different forms of collaboration, whether co-located or distributed, synchronous or asynchronous, effective coordination depends on the group building and maintaining a shared understanding [22] that brings individual knowledge into group synergy. More formally, this shared understanding rests on the group's *common ground*, defined as the knowledge, beliefs, and assumptions its members know they hold in common [44, 43]. Common ground allows members of a team to interpret one another and to coordinate interdependent work, supplying a shared sense of what has happened, what has been decided, and what should happen next [174, 152].

Instead of accumulating on its own [22], common ground is built and repaired through deliberate effort, and much of that effort runs through the information artifacts a group produces. People take notes, draft summaries, and write status updates, continually deciding what is important to record and what can be omitted. Through these artifacts, groups carry forward what they have learned collectively to guide later behavior [130].

The effort of creating and consuming such information artifacts has become harder to afford as collaboration has grown in scale. Over the past decade, organizations have seen a substantial rise in collaborative activity: more meetings, more group messages, more shared documents. This rise has contributed to growing "collaboration overload" and a deficit of human attention [197]. In an information-rich environment, attention is the scarce resource: each person can read, watch, and think about only so much before something else goes unattended [21, 166, 65, 231]. As the volume and pace of communication rise, maintaining common ground therefore becomes a question of design: *how should the information artifacts (e.g., documents, messages, and recordings) through which a team sustains its common ground respect the limits of that team's attention?*

1.1 Designing Information Artifacts for Collaborative Work

Contemporary collaboration is often conducted either partially or fully through digital media such as video calls, chat, and email, and it therefore leaves behind a continuous record of *raw collaborative traces* [214, 83]: transcripts, recordings, and message histories. These raw traces accumulate quickly, often without selection of what is important to keep. As a result, the information that matters to a collaboration's common ground can be drowned in much that does not matter. An hour-long recording may include the 90 seconds when a team settled on its direction, but also the wait time for a late participant to join and tangents about other tasks. Moreover, modern communication tools have become increasingly capable of capturing more and richer raw traces, from document workflow histories [83] to the edit and execution records of shared computational notebooks [242].

Turning raw traces into something a group can use therefore takes deliberate effort and design [8]. To do it manually, someone has to read through the traces, judge what mattered and what can be set aside, and render that judgment in a form others can digest (e.g., turn raw

meeting transcripts into readable meeting notes). This labor of keeping collaboration legible to itself is what the Computer-Supported Cooperative Work (CSCW) literature defines as *articulation work*: the work of organizing collaboration, which competes directly with the time available to carry that collaboration out [227, 229, 225]. One common outcome of articulation work is an *information artifact*: *a designed representation through which information becomes available to be inspected, shared, revisited, or acted upon in collaboration.* Besides the selection of content, how the information is presented [66, 242], when it appears [186, 37], and where it is understood to have come from [138] all shape how well members of a group can fold it into their common ground.

Indeed, CSCW research has long established that artifacts are embedded in socially organized work and that maintaining them is itself an often invisible labor [210, 225, 229, 227]. Earlier collaborative systems commonly managed this labor by reducing complexity through predefined structure: designers specified the fields to extract, the templates in which to place them, and the rules that determined when a notification appeared [210, 69]. Such designs are effective but limited as collaboration scales, because decisions about how to organize information have to be made before any particular interaction occurs, and producing their content still requires human articulation work. As a result, the binding constraint on information artifacts was historically *creation.* A group could preserve only what someone had time to write down, and could take in new information only through the fixed channels a system had been built to deliver.

1.2 Information Artifacts in the Age of Generative AI

Since the early 2020s, rapid advances in generative models and their wide adoption in everyday work tools have changed what an information artifact costs to make [26]. These models can synthesize scattered and even multimodal traces into a single account [11, 235], move fluidly between levels of detail [230, 99], and render the same material in different formats for different

audiences [13]. Artifacts that no person would have had time to make in the past, such as a tailored recap for each absent member or a news digest of what might be relevant to the collaborative task, can now be produced on demand. In this dissertation, I use *generative AI information artifact* to refer to an artifact whose content is generated from a group's traces or from sources beyond them using generative AI models, rather than authored in advance by a person or assembled into a fixed schema.

With the cost of making an artifact falling, the binding constraint on information artifacts is no longer creation but consumption. Within a group, each member has a finite amount of attention available to consume information about the group's common ground. While generative AI models can produce artifacts that are more informative and useful than any person could have produced in the same amount of time, they do not expand the attention available to the group to consume this information.

Poorly designed artifacts that cost little to produce could eventually flood the common information space that the group relies on to sustain its common ground. AI-generated content that has the appearance of useful work but lacks the substance to advance a task can produce "AI slop", creating burdens for whoever receives it [170, 117, 18]. Whether generative AI strengthens or erodes a group's common ground thus depends on intentional design choices beyond the model's inherent capabilities. Realizing the promise of generative AI for collaboration requires decisions about what content to select, how to present it, and when to place it in front of groups.

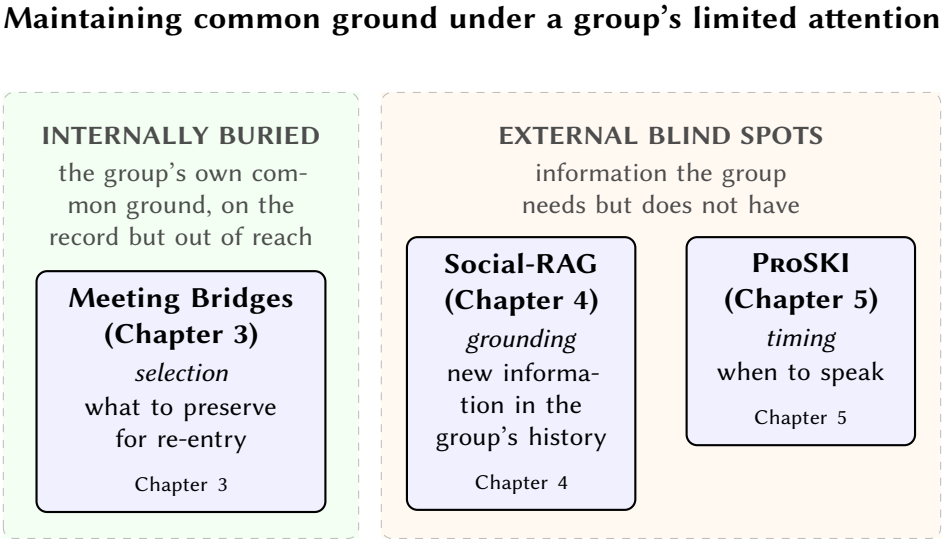


Figure 1.1: Common ground could break down in two directions. On the left, what the group has already established sits in its own traces but cannot be reached, making *selection* the key question. On the right, information lies beyond the group's own activity and arrives uninvited, making *grounding* and *timing* essential to earning the group's attention.

1.3 Designing Generative AI Information Artifacts to Repair and Maintain Common Ground

This dissertation asks how generative AI can help groups maintain their common ground when the attention available to consume information is finite. Specifically, I take up the two conditions under which common ground fails: when what a group has already established becomes *internally buried*, and when the group has *external blind spots*. Three projects take up these conditions, as Figure 1.1 summarizes.

1.3.1 Internally Buried Common Ground

As collaboration scales, a group can produce more raw traces than its members can keep up with, so the shared understanding captured in the record can become unreachable for those who

missed a meeting or lost the thread of a busy channel. While hand-curated artifacts can prioritize information that is worth the attention, a curated artifact embodies one person's judgment of what the group would need, fixed at the moment of writing, and everyone who arrives later inherits that same selection whatever purpose brought them to it [210, 69]. A reader whose need falls outside it must go back to the raw traces, and most do not, so the common ground stays buried. With generative AI, the same traces can now create personalized artifacts tailored to readers' needs. Yet flexibility with nothing steering it is precisely the recipe for workslop. The question is therefore *selecting* what is worth capturing from a collaboration in the first place.

Meeting Bridges (Chapter 3)

Meeting Bridges takes up the selection problem in the context of online meetings, where much of a team's common ground is built and then buried. In a survey of 198 remote workers and interviews with 13 more workers, we found that people lack trust in notes written by others, who "might write things that I don't care about, or they might not write things that I care about," and that most never return to recordings, so the fallback to raw traces is largely unavailable. When asked what they actually do with meeting information afterward, participants highlighted the need to use information generated during the meeting to support subsequent work. This observation motivates us to reframe the design goal of a meeting record from producing a faithful record that helps people recall common ground to *carrying that common ground forward*, which we call a *meeting bridge*. We develop five design principles for *meeting bridges* and illustrate them with concrete examples.

1.3.2 External Blind Spots

The second condition is an *external blind spot*: information the group needs but does not yet have, because no group can monitor everything relevant beyond its own activity. In collaboration, groups commonly need to search for or pursue information together [158, 61]. However, finding the needed information and fitting it to a specific group at a specific moment often exceeds what anyone can spend [91], so groups fall back on generic channels or do without the information. Generative AI makes this task tractable, because searching and synthesizing large volumes of outside material is precisely what these models do well [26]. When a system places what it finds in the group's shared space and makes that material available to the group as something its members know they hold in common, it fills a blind spot of the group's common ground.

However, the difficulty is contributing the external information so that a group takes it in rather than scrolling past it or resenting the interruption. A blind spot is by definition something nobody thought to ask about, so the contribution is uninvited, and unlike an artifact a person consults on demand, it spends the group's attention the moment it is sent. The external information contributed by the system thus needs to satisfy two criteria so that it can fill the group's blind spot rather than become noise. The first is *grounding*: if people cannot readily see how new information connects to what the group already knows and cares about, they treat it as noise, since group conversation is a low-attention setting in which people judge what merits their attention from surface cues and framing rather than by evaluating content on its merits [186, 37]. The second is *timing*: a contribution has to arrive when the group can still act on it. Without both, a generated contribution only adds to the pile the group is already struggling to consume. The two projects below take up one property each, and both study contributions posted into a group's shared channel.

Social-RAG (Chapter 4)

Social-RAG focuses on grounding in the context of an AI agent recommending relevant research papers for research group conversations. In a formative study with 39 researchers, we find a strong preference for explanations that connect a paper to the group's own context. However, contextualizing work is currently effortful "free labor" for researchers and limits the volume of external information that can be shared within a group. We propose a computational technique called Social-RAG to automatically contextualize a given body of knowledge within a group's ongoing context. Specifically, Social-RAG curates selected interaction traces into a social knowledge base and extracts *social signals* from the group's raw collaboration traces, such as whether a paper cites one the group has discussed or whether a member has shown sustained interest in its topic. The design is a deliberate analogue to retrieval-augmented generation, retrieving social facts where conventional RAG retrieves factual knowledge, and because those facts are inferred from a partial record, the workflow surfaces the traces a given explanation rests on.

We instantiated Social-RAG in a Slack application, PaperPing, which was deployed for at least three months across 18 channels reaching over 500 researchers. PaperPing's socially grounded messages helped researchers judge a paper's relevance to their group more effectively than generic summaries did. However, the deployment also exposes the limits of grounding. Sparse histories and ambiguous signals produce speculative inferences, and a fixed posting schedule that suited some channels led PaperPing to dominate others, converting a discussion space into a passive feed.

ProSKI (Chapter 5)

To tackle the challenge of timing when contributing external information to build group common ground, we first need to understand what humans expect of an agent's contributions. To do

so, we collected ProSKI, a benchmark comprising 384 candidate messages across 35 real group conversations, each written by a participant to fill a blind spot in their own discussion, and each carrying timing judgments from every member of the group that held it.

Through this dataset, we found that timing ground truth is a distribution rather than a point: after deliberation, annotators land within two turns of one another on only 54% of messages, and only 29.7% of messages have a single moment every annotator independently agreed on. We evaluate 15 models in 28 configurations under three task framings that differ in how much of the conversation is visible before the model commits. Given the whole conversation, models place contributions about as well as a group member does with hindsight in the offline task framing. However, this competence does not survive the realistic online turn-by-turn task framing: accuracy falls to a third of its retrospective value, 46.4% of predictions fire before any annotator's window has opened, and 1.0% arrive after every window has closed. Reasoning traces show that in 92.8% of firing justifications, the model argues only that the content fits at the current turn, while an explicit weighing of whether a later moment would serve better appears in 0.14%. Our findings indicate that models have learned where contributions belong retroactively; they have not learned to wait for that place to arrive in real time.

Taken together, these projects argue that **Generative AI information artifacts sustain a team's common ground under limited attention when they (1) select information for the work that comes next, (2) ground it in the group's own interaction, and (3) surface it at the right moment.**

1.4 Dissertation Organization

In the following chapters, Chapter 2 reviews related work on maintaining common ground and the selectivity of curated artifacts, attention as the constraint on their consumption, AI-generated

artifacts and their limitations, grounding new information in a group's context, and proactivity and the problem of timing. Chapter 3 presents Meeting Bridges, which addresses selection under internally buried common ground. Chapter 4 and Chapter 5 present Social-RAG and ProSKI, which address grounding and timing under external blind spots. Chapter 6 synthesizes findings across the three projects and discusses future directions.

1.5 Relevant Publications

Though I am the primary author of the research detailed in this dissertation, it is a product of collaboration with my advisor and collaborators. Chapter 3 is based on a prior publication at CSCW 2024, in collaboration with Lin Qiu, Justin Cranshaw, and Amy X. Zhang. Chapter 4 is based on a prior publication at CHI 2025, in collaboration with Xinyi Zhou, Lin Qiu, Joseph Chee Chang, Jonathan Bragg, and Amy X. Zhang. At the time of writing this dissertation, Chapter 5 is in submission, in collaboration with Brian Yu, Xinyi Zhou, Liwei Jiang, Joseph Chee Chang, Jonathan Bragg, Pao Siangliulue, and Amy X. Zhang. I use the first-person plural ("we") in these chapters to reflect our collective contribution.

Chapter 2

Related Work

In this chapter, I review literature on themes that cut across the dissertation as a whole. I begin with common ground and the labor of maintaining it, which establishes why groups produce information artifacts at all (Section 2.1). I then examine why curated artifacts are selective, and why their selections become fixed at the moment of creation (Section 2.2). From there, I turn to attention as the binding constraint once generative AI reduces the cost of producing artifacts (Section 2.3), and to the limitations of the AI-generated artifacts this shift makes possible (Section 2.4). The remaining sections take up two questions that organize the dissertation's treatment of external blind spots: how a new contribution can be *grounded* so that its relevance to a particular group is visible (Section 2.5), and how a system can decide *when* to contribute at all (Section 2.6). Chapter-specific background appears in the corresponding chapters.

2.1 Common Ground and the Cost of Maintaining It

Common ground is the knowledge, beliefs, and assumptions that members of a group mutually believe they share. Teams establish it incrementally by grounding new information in shared con-

text during interaction [44, 43]. Computer-Supported Cooperative Work (CSCW) has long studied systems and artifacts that support this process because common ground enables groups to coordinate, cooperate, and communicate [174], especially when collaboration is tightly coupled [152]. The work required to maintain common ground also depends on the communication medium: distributed, asynchronous, and remote work are harder to ground than co-located work [174].

One way groups manage that effort is by externalizing common ground in artifacts such as documents, records, messages, and shared tools [210, 124, 146]. These artifacts hold what a group knows outside any one member and beyond any one interaction, allowing shared understanding to be returned to, negotiated, and iterated rather than rebuilt each time [130]. How common ground is encoded in artifacts partly determines what a group can learn from itself [108]. Yet an artifact's meaning is not fixed when it is first made; it is constructed through the collaborative activity that accumulates around it [45, 48]. Common ground is therefore an ongoing achievement, not a state that persists on its own [19].

Much of the effort required to maintain group common ground falls under "articulation work," defined as the necessary work that unites distributed yet interdependent work elements [227, 229, 225]. Articulation work manifests as tasks such as "ensuring the flow of resources," "making arrangements about the division of labor," or "linking or meshing otherwise divided tasks" [227, 225, 177]. These tasks are often less visible than the work they coordinate, but they are indispensable to collaboration [206]. They also increase as work becomes more complex, driven by new tasks and technologies [206]. As remote and asynchronous work have become more common, articulation can consume an increasing share of workers' time and divert people from their primary tasks [197].

Researchers have supported articulation work in two broad ways. Some systems avoid articulation needs by keeping work separated [90]; others facilitate articulation by introducing standardization and formal structure [229, 210]. Standardization, whether through protocols, procedures, or schemes, reduces the complexity of articulation [27, 4]. Schmidt and Bannon,

for example, proposed managing shared information spaces through curatable and accessible information objects [210]. Field studies have observed such spaces in high-stakes settings, including medical care [196] and scientific collaboration [101]. A contemporary version is often a group chat. In an enterprise Slack deployment, Wang et al. find that channels differentiate into distinct categories with measurably different communication styles rather than forming one undifferentiated space [243]. What counts as an appropriate contribution is therefore a property of the particular channel, not only of the medium.

2.2 Curated Artifacts Are Selective, and Their Selections Are Fixed

Because articulation work is costly, the artifacts a group produces record only what somebody had time to record, in a form chosen for that person's purpose. The standardization and formal structure that CSCW recommends for managing articulation cost [210, 229, 27] work by fixing those choices in advance: fields to extract, templates to fill, indices to browse. This can be effective, but it also bounds artifacts by decisions made before any particular interaction takes place. Whoever comes to the artifact afterward inherits a selection made for someone else's purpose at an earlier moment. Even the original author may later struggle with that selection. Lin et al.'s study of micronotes, the quick informal notes people write as reminders for themselves, finds that they are often hard to interpret after the fact *by the person who wrote them*, as abbreviations, shorthand, and the surrounding situation that made the note legible fall away with time [138]. A selection is bound to the moment of its making even when maker and reader are the same person.

Dourish and Bellotti raised an early objection to supplying coordination through dedicated machinery of this kind. Examining systems that maintain awareness of who is doing what through

active mechanisms kept separate from the shared workspace, they observe that such designs penalize the people who must generate the information, assume that what is generated will be relevant to whoever receives it, and make the result awkward to reach [60].

Meeting artifacts illustrate the pattern. Systems for accessing meeting information have largely assumed what Tucker and Whittaker term *index-centric random access*: the artifact preserves the record and provides structure through which a user can locate and recall a piece of it [234]. Recall is a real need, but it treats the meeting as a single point in time rather than as one step in a continuing collaboration [154, 69]. Studies of recurring meetings make this limitation visible: participants experience meetings as a cycle of transitional activities before and after each session, and the meeting's instrumental goals are pursued across that cycle rather than within any one sitting [171]. The preservation assumption is strained from another direction as well. Virtual meeting platforms capture far more contextual data than their in-person predecessors, and records management research finds that this increased documentation challenges traditional recordkeeping models rather than satisfying them [71]. A record that captures everything defers the question of what is worth keeping instead of answering it. Abundance of capture is therefore not a substitute for selection but a reason selection matters more. Chapter 3 argues from empirical evidence that recall accounts for only one of five recurring uses of meeting information, and that the remaining four are better described as carrying context forward into subsequent work.

2.3 Attention as the Constraint on Consuming Artifacts

If creating artifacts is one side of articulation work, consuming them is the other, and it is governed by a different scarcity. Simon's observation that a wealth of information creates a poverty of attention [220] names the constraint that CSCW research has repeatedly rediscovered in collaborative settings: the growth of meetings, channels, and shared documents produces

documented "collaboration overload" [197] and, in remote work specifically, fatigue and diminished capacity to engage [21, 166, 65, 231]. Attention is also the resource that any unsolicited contribution spends. Work on interruption shows that the cost of an interruption depends heavily on when it arrives relative to the structure of the ongoing activity, with interruptions at coarse task boundaries substantially less disruptive than those mid-task [5, 17, 185, 53]. This literature has largely studied individuals interrupted at their own work; a contribution posted into a shared group space spends the attention of everyone in it at once. The corresponding question of when a system should claim a *group's* attention is much less well characterized.

Generative AI makes this constraint more consequential. Models can synthesize heterogeneous traces, move between levels of abstraction for different audiences, and generate language fitted to the immediate context of an interaction [179]. Artifacts that no person would have had time to produce can now be produced on demand. The cost of creation falls; the group's attention does not grow. Early accounts of the result describe "workslop," AI-generated output with the appearance of good work but without the substance to advance a task, which shifts interpretive burden onto whoever receives it [170]. Survey evidence also finds workplace AI adoption often accompanied by stress rather than the promised productivity gains [237]. Three design questions therefore become central: what is worth representing, how it connects to what a group already knows, and when it should appear.

2.4 AI-Generated Artifacts and Their Limitations

The rise of generative AI has spurred interest in AI-Mediated Communication (AI-MC), systems that generate messages or contributions on behalf of humans in interpersonal or group contexts [86]. Within this framework, AI systems may produce entirely new messages, interpret multimodal inputs, or operate in real time with varying degrees of autonomy. AI-MC research

shows that these systems can shape self-presentation, impression formation, and trust [73, 155, 132]. These concerns become especially important when AI acts as a conversational surrogate or collaborator.

AI-generated meeting summaries have moved quickly from research prototypes [248] to widely deployed product features. Yet most systems still focus on content summarization: generating condensed accounts of what was discussed without asking how individuals experienced the meeting or how relational dynamics shape interpretation. Recent work calls attention to affective and interpersonal nuance in summaries [270]. AffectiveSpotlight, for instance, surfaces audience members' affective responses to a presenter who would otherwise be speaking into a void, on the premise that knowing how a group is reacting matters as much as knowing what it said [161]. Aoyama et al. similarly argue that LLMs can transform communication to help create shared understanding rather than merely transmit information [7]. However, these explorations remain largely limited to dyadic text chat and do not address the multimodal, asynchronous, and socially situated nature of real-world collaboration.

AI agents also act as communication proxies and participants. Work on agents that represent users when they are absent [129, 41] shows their value as proxies, but also the difficulty of representing a user accurately and keeping that user aware of what the agent has said. Interactive agents have also been deployed in group spaces to support conversation summarization [263], idea generation [194], collaborative search [14], conflict resolution [81], and balanced participation [57]. Schiavo et al. treat the *manner* of intervention as a design variable, contrasting overt directives to named participants with subtle cues that steer the group without singling anyone out [209]. These systems regulate the group's own interactional dynamics. The systems studied in this dissertation instead carry information the group did not already have, so they enter the conversation as additional participants: interjecting to offer facilitation support [57] or to share information for decision-making [273].

That entry can be disruptive. AI interjections in group spaces are often judged by the timing, relevance, and delivery of their content [16], and they can affect the trajectory of ongoing group discussion [273]. Uneven reception is not peculiar to AI. When instant messaging and chat were first introduced into workplaces, Herbsleb et al. found that adoption and perception varied across groups and were shaped by existing practices rather than by the medium's affordances alone [88]. An AI agent entering a group channel inherits the same unevenness. Researchers therefore call for AI agents that are socially aware and able to adapt to the context in which they operate [194, 87, 6, 57, 211]. The challenge is partly the *sociotechnical gap*: CSCW systems often lack the flexibility to reflect complex social needs [3].

Existing approaches model pieces of this social context. Some systems attend to group dynamics such as member participation, including speech frequency [58] and gaze [188]; others draw on individual or collective preferences, such as expressed opinions [42, 33]. Agents that show awareness of human behavior and learn from it can be perceived as more natural and trustworthy [265, 128, 102], and thus more likely to positively affect team dynamics [273]. Avula et al. find that proactive agents should also provide clear justifications for their actions, since unprompted suggestions may not be immediately understood or valued by group members [16]. Even so, developing socially aware agents remains difficult because human social interaction is hard to capture and respond to through an AI system [16, 194].

2.5 Grounding New Information in a Group's Context

2.5.1 Groups Seek Information Together

Filling a blind spot is not a need that systems invented. Information seeking has long been understood as a collaborative rather than a solitary activity, and a review of the first decade

of collaborative-search research confirmed it as a recurring practice with its own coordination demands [158]. Kraut et al.'s early study of scientific research collaboration shows where the raw material comes from: informal and unplanned contact brings researchers into contact with work they had no way of knowing they needed, and distance disrupts that contact [120]. Later accounts specify the practice more closely. Researchers report identifiable triggers that prompt them to turn to colleagues rather than continue alone [61], and studies of collaborative information seeking document both the coordination difficulties that follow and the tools built to relieve them [91].

Two findings from this literature bear on what follows. First, a group's information needs are legible in its conversation: Shiga et al. model needs expressed during collaborative search tasks and show that categories of need can be detected from the talk itself [217]. This is the premise on which a system might notice a gap the group has not stated. Second, support for collaborative seeking works best when placed inside the group's own space rather than beside it, whether as search embedded in a conversational platform [15] or as a search agent participating in the channel [14]. What these systems do not take on is the step this dissertation studies: contributing information the group never asked for.

2.5.2 Why Making Relevance Visible Matters

The *visibility* of an artifact's relevance is a design concern in its own right, not just a matter of picking better items to surface. Dual-process accounts of persuasion distinguish a systematic route, in which people evaluate a message on its merits, from a heuristic route, in which they judge it from surface cues about its source, framing, and apparent fit [186, 37]. Which route a person takes depends on motivation and capacity at that moment. An ongoing group conversation is close to the limiting case for both: attention is divided, the message was not requested, and the cost of ignoring it is low. A useful message can therefore be discarded as noise simply for arriving

unframed. Work on framing makes the complementary point that how information is presented shapes what recipients take it to be about and what they do with it [66]. Consistent with this, Avula et al. find that proactive agents in group search settings need to provide clear justifications for their actions, because unprompted suggestions are not immediately understood or valued by group members [16].

Existing personalization and retrieval approaches only partly address this problem. Personalization frameworks and conversational agents can adapt to individual preferences through explicit elicitation or interaction histories, but group preferences are often implicit, emergent, and socially distributed. Group norms are rarely stated directly; they appear in turn-taking patterns, tone, reactions, and long-term interactional rhythms. Traditional LLMs, even when augmented with retrieval, face a related limit: they can retrieve relevant knowledge but lack mechanisms for representing or reasoning about social facts, the structured traces of how groups relate, respond, and coordinate over time. Because group conversation is a low-attention setting in which relevance is often judged heuristically [186, 37], retrieving the right item is not sufficient. The design implication that Chapter 4 pursues is that an artifact filling a group's blind spot must carry, in the artifact itself, an account of why it belongs here.

2.6 Proactivity and the Problem of Timing

Grounding settles why a contribution belongs in a setting. It does not settle when it should appear, a question that arises only once a system acts without being asked.

Timing is a problem groups already have before any system is involved. Norris et al. describe distributed teams as living with a condition of *sociotemporal disorder*, in which members working across different rhythms, availabilities, and planning horizons must continually reconcile whose sense of the present governs a piece of shared work [173]. A contribution arriving from outside

the group enters that negotiation with no standing in it, which is what makes its moment a design problem rather than a scheduling one.

2.6.1 Initiative as a Design Variable

Horvitz's principles of mixed-initiative interaction framed the core trade-off early: an agent that acts on its own behalf must weigh the expected value of acting against the cost of interrupting, and must be able to defer or hand control back when that calculus is unfavorable [92]. Subsequent work on proactive assistants developed mechanisms for deciding what to volunteer [162]. The recent turn to LLM-based agents has revived the question at a larger scope, including agents that ask clarifying questions before planning [264], agents that generate and act on their own inner deliberation [142], and general frameworks for human-centered proactive conversational agents [55, 136].

Deployed proactive agents show that the same behavior can be received very differently depending on delivery: proactive support agents are valued when their contributions land well [141, 200] and resented when they do not [137, 260]. Across this literature, initiative is treated as a property to be calibrated rather than maximized. That calibration, however, is usually expressed as a policy parameter, such as a schedule, confidence threshold, or frequency cap, rather than as a judgment about a specific moment in a specific conversation. A recent line of work does treat the moment as a judgment by separating a gate that decides whether to speak from a model that decides what to say [62, 23, 40], and by supplying corpora that label a speak-or-stay-silent decision at each turn [180, 163]. Those corpora are synthesized or built around a planted trigger, so the moment they score is one the construction procedure installed. Chapter 5 takes up what is missing from them: whether a system's chosen moment agrees with the people whose conversation it enters.

2.6.2 When a Contribution Belongs in a Conversation

Conversation analysis supplies the vocabulary for that judgment. Turn-taking is organized around transition-relevance places, points at which a change of speaker becomes legitimate, and contributions that ignore this organization read as interruptions regardless of their content [202]. Grice's account of conversational cooperation makes the parallel point about substance: a contribution is judged by whether it is appropriately informative, relevant, and timely for the current purpose of the exchange, not merely by whether it is true [82]. Empirical work in collaborative settings finds that interruption timing measurably affects group performance and perceived disruption [185, 16], and that agent interjections shape the trajectory of a group's attention rather than simply adding to it [273].

What this work does not provide is a way to evaluate whether a system can make the timing judgment itself. Existing benchmarks for conversational models assess what a system says: factual accuracy, helpfulness, safety, and, more recently, social knowledge and social intelligence [**?** 134, 250]. They score a response once a turn has been allocated to the model. They do not ask whether the model should have taken a turn at all. Chapter 5 takes up that gap.

2.7 Research Opportunities

Across these strands of research, the opportunity is not simply to generate more information for groups. It is to design AI information artifacts that respect the social and attentional conditions under which groups can use information at all. The gaps line up with the three questions that organize this dissertation.

Selection. Prior CSCW work explains why groups produce artifacts and why those artifacts are costly to maintain, but the dominant design assumption for artifacts that carry a past interaction

forward remains recall: preserve the record and provide structure for locating parts of it. Generative AI makes selection revisable rather than fixed at the moment of creation, which raises the question of what an artifact should select *for*.

Grounding. Existing personalization and retrieval approaches can identify relevant content, but group relevance is social as well as semantic. In low-attention group conversation, an artifact must make the basis of its relevance visible to the people whose attention it asks for.

Timing. Work on proactive agents shows that initiative must be calibrated, and conversation research shows that timing shapes reception. What remains underdeveloped is an evaluation of whether a model's sense of the right moment corresponds to the group's own.

Together, these gaps call for AI artifacts and agent behaviors that incorporate social context, learn from interaction histories, and produce contributions that are both factually correct and socially appropriate. They also call for evaluations that can tell whether those contributions land as useful additions to common ground rather than as more noise for the group to manage.

Chapter 3

Meeting Bridges

3.1 Introduction

As work-based collaborations increasingly occur remotely, people are spending more time in online meetings [49, 12, 195]. A 2021 survey finds that the time information workers spend on voice and video calls doubled from 7 to 14 hours after the 2020 onset of the pandemic [197], and the change persisted with the more recent shift to hybrid work [25]. While synchronous meetings play an important role in team planning, coordination, and decision making, the sharp increase in meeting load has contributed to problems such as "Zoom fatigue" [72], where prolonged videoconferencing becomes physically and mentally exhausting, and "collaboration overload" [197], where individual task accomplishment suffers because of collaborative group activities. These phenomena negatively affect worker productivity (e.g., [21, 197, 166]) and well-being (e.g., [21, 197, 65, 231]).

To achieve a more productive and healthier practice of remote work, many organizations are developing practices to shorten meetings and make them more intentional [34] or exploring strategies to eliminate meetings altogether, such as no-meeting days [34, 122, 31]. These strategies can reduce time spent in online meetings, but they often do so by shifting more work to periods

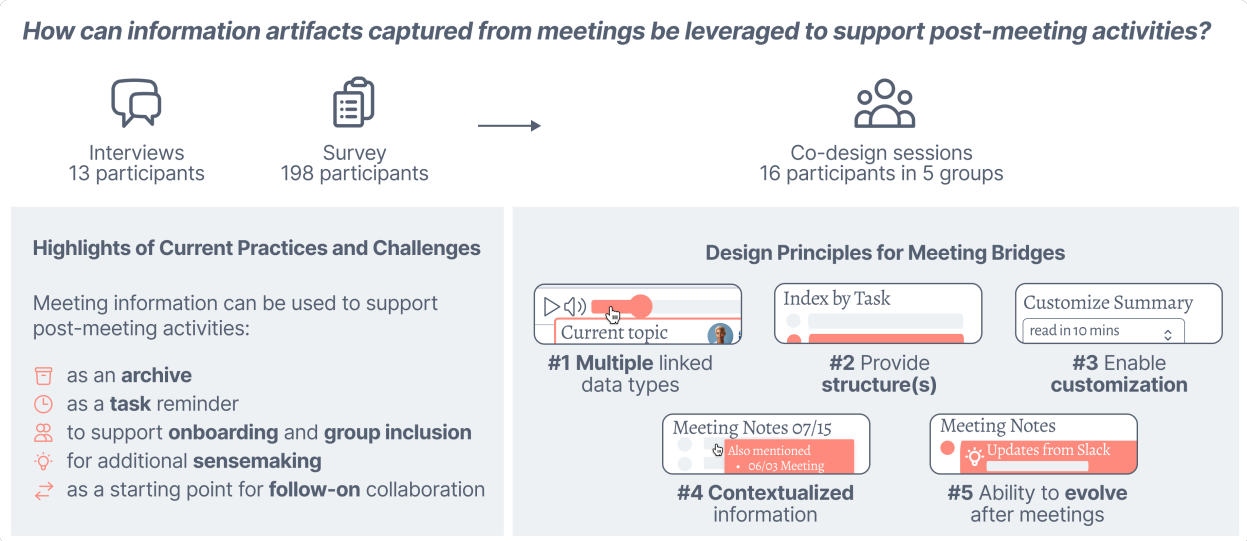


Figure 3.1: Based on interviews and a survey about how people use meeting information after meetings end, we propose the design concept of *meeting bridges*. Through co-design sessions, we distilled five design principles and created mockups that illustrate how information artifacts can help users access, transform, and engage with meeting information after the meeting.

before and after the meetings themselves. Studies show that online workplace collaboration shifted work from synchronous communication to asynchronous or decentralized work during the pandemic [251, 32, 104]. In a survey with Microsoft employees, over half of participants report spending more time preparing for meetings than they did pre-pandemic, including through pre-reads, agenda and goal setting, and asynchronous and synchronous discussions [21]. The same survey shows that follow-ups to synchronous meetings are now prevalent: 94% of workers spend more time than before on post-meeting follow-up activities through chats or additional meetings [21].

Given the prevalence of asynchronous work that occurs *around* meetings, the opportunity to alleviate meeting burden may not lie only in reducing meeting time. It also lies in helping people bridge from synchronous remote meetings to asynchronous activities. Inspired by classical and recent CSCW work on the role of digital artifacts in team coordination and communication [12, 69,

227, 229, 124, 27], we ask how *enhanced information artifacts captured from meetings can support post-meeting activities.*

We launched our inquiry by identifying U.S. information workers' current practices of using information captured *during* remote meetings *after* the meeting is over. We interviewed 13 people and surveyed 198 additional individuals who frequently participate in remote meetings. We find that attendees use meeting information in five key ways after meetings end: as an archive, as task reminders, to onboard or support inclusion, for additional sensemaking, and as a launching point for follow-on collaboration. However, we also find that existing information artifacts captured from meetings, such as notes and recordings, have drawbacks when used to fill these post-meeting roles, including a lack of support for lightweight but rich ways to capture meeting information, post-meeting difficulties in understanding and filtering relevant information, and hesitancy to share meeting content with others due to trust and privacy concerns.

Our results highlight an opportunity that informs our basic research question: **can new artifacts from synchronous meetings be designed to more effectively support asynchronous collaboration?** These *meeting bridges* could simplify team efforts to synthesize meeting content and engage with it in post-meeting activities, regardless of whether a team member participated in the synchronous meeting. To explore the design space of meeting bridges, we conducted a co-design study with 5 groups of 16 U.S. participants. We then distilled five design principles and created example mockups that enable participants to asynchronously explore and make sense of rich content from meetings, share customized meeting summaries with others, and continue to update the artifact as collaboration evolves. Figure 3.1 gives an overview of our methods and findings. We conclude by discussing how existing artifacts map to the design principles, how our characterization of meeting bridges can generate additional design ideas for tools to support remote collaboration, and the feasibility of implementing these design ideas in real systems.

Overall, our work makes the following contributions to prior literature: (1) we contribute

empirical observations that clarify users' needs and challenges in revisiting and engaging with meeting information beyond the duration of remote meetings; (2) we provide a set of design principles for meeting bridges, highlighting the opportunity to support users in customizing, updating, and sharing meeting information after meetings end. To show how one can apply our findings, we provide interface mockups that exemplify the possibilities in this design space, offering a timely answer to researchers' call to extend meetings beyond "a single point in time" [154].

3.2 Related Work

From the thesis-wide review in Chapter 2, this chapter especially builds on CSCW accounts of articulation work and shared information artifacts [227, 210, 69]: meetings are a form of coordination overhead, and digital artifacts can transfer, translate, and transform knowledge across people and time. Meeting Bridges asks how those roles apply when the artifact is produced *from* a remote meeting and then used in asynchronous follow-up. Below we review the chapter-specific background: shifts that make post-meeting work more salient, and systems designed to support revisiting meeting information.

3.2.1 Recent Shifts in Remote Work

In response to issues such as Zoom fatigue [72] and meeting overload [197], researchers are calling for more intentional use of meetings [199, 59, 123] and a reduction of unnecessary meetings [34, 122, 31] in remote work. However, they also warn that merely restricting meetings could hinder employees' opportunities to connect and socialize [157] and make workers fear being "left behind" when not participating in meetings [154].

As a result, prior work has highlighted the potential of augmenting synchronous meetings with asynchronous communication, such as chat workspaces or asynchronous brainstorming

tools, to make meetings more efficient and let workers choose when to engage [198, 228, 213, 147, 123]. Research also shows that the shift to remote work has increased asynchronous modes of communication, such as emails and chat messages [251], and that many workers perceive asynchronous communication as a useful way to follow up after meetings [21]. To support these shifting patterns, some researchers propose viewing meetings as a *digital artifact* that persists beyond a single point in time, allowing workers to engage when it best suits their needs [154].

Building on the trend of supporting asynchronous activities around meetings, we conduct a survey and interview study to explore users' current practices and challenges in engaging with meeting information beyond the meeting itself. Our observation that current tools fall short in supporting post-meeting needs also led us to explore design solutions that support a smoother transition from remote meetings to asynchronous follow-up activities.

3.2.2 Supporting the Use of Meeting Information after Meetings

Online meetings are indispensable for remote team communication, coordination, and decision-making. Much CSCW work focuses on supporting the in-meeting experience by making meetings more productive [139, 223, 153, 38], effective [161, 156, 10], and equitable [106, 204, 95, 203]. Recent research on recurring meetings urges a broader approach that considers users' need to revisit and continue engaging in asynchronous meeting conversations after meetings end [171]. In their experiential account of recurring meetings in a large organization, Niemantsverdriet and Erickson observe that a recurring meeting is oriented toward instrumental goals that no single session completes, and that much of the work of pursuing those goals happens in the activities that transition into and out of each session. On this view, the boundary of a meeting is drawn by the calendar rather than by the work.

However, rich and unstructured multimedia data captured from remote meetings are challeng-

ing for users to consume afterward. Records management research reaches a similar conclusion from a different tradition: virtual meeting platforms enable the creation, capture, and sharing of considerably more contextual data than in-person meetings ever generated, and this increased documentation challenges traditional recordkeeping models rather than being absorbed by them [71]. To address this, many researchers have investigated ways to support users in accessing information captured from meetings. Tucker and Whittaker categorize systems into four groups: *audio* (e.g., an audio browsing tool with speaker segmentation [116], audio interfaces with visual indices [89], and SpeechSkimmer, which lets a listener skim recorded speech at several granularities rather than proceed through it in real time [8]), *video* (e.g., VideoManga, which generates pictorial video summaries [236], and a video browser with two indexes for navigation [70]), *artifact* (e.g., Classroom 2000, which combines multimedia artifacts for revisits of lectures [2], and Teamspace, an integrated space to access all meeting materials [77]), and *discourse* (e.g., Rough 'n' Ready, which supports named entity search of meeting transcripts [47], and an interactive document generated from captured media [239]). More recently, researchers have begun to explore richer and more curated media types, such as dashboards, including MeetingVis [216] and EMODASH [68], which visualize meeting dynamics to encourage retrospective awareness of emotions.

These systems commonly help users access meeting information through automatically generated or manually added indexes. Tucker and Whittaker summarize the expected use case as *index-centric random access* of meeting information [234]. This model supports preservation and recall, but it is limited if meetings extend beyond *a single point in time* [154]. Framing meeting artifacts through the transfer/translation/transformation lens reviewed in Chapter 2 [69] suggests a broader design goal than recall alone. Yet little work has explored how people need to mold and transform meeting information to fit different remote-work contexts. One exception is a study of how multimedia artifacts are created, used, and shared in distributed meetings [151]; this work validates users' need to revisit and share media after a meeting and identifies challenges,

including recording useful information and locating scattered information, but solutions remain under-explored.

Only a few systems explicitly support users' continued engagement after meetings beyond accessing preserved information. Examples include HyperMeeting, which links videos from follow-up meetings [79]; Video Threads, which supports users in receiving video updates for and contributing further to a conversation from an initial meeting [20]; and a post-meeting dashboard that displays meeting information and collects users' feedback for automated meeting understanding [63]. A closely related precedent outside the meeting setting is ReBoard, which automatically captured whiteboard content and made it reachable through user-centered access tools. Across a seven-week workplace field study, that content became more valuable for collaboration not because recall grew easier, but because its availability beyond the whiteboard's own time and place enabled workflows that had not previously been possible [28]. Winder makes a related move for asynchronous design collaboration by binding recorded speech to the visual objects it refers to, so that a comment left for a collaborator carries its referent with it rather than requiring the recipient to reconstruct what was being pointed at [114]. These systems treat capture as support for *transitions in work* rather than only as preservation, but they tend to focus on individual techniques rather than generalizable design patterns.

This gap matters because, in remote work, meetings and the work of keeping track of meeting content are themselves forms of articulation work. They help manage the "overhead" of coordinating interdependent work and engaging with artifacts from different group members [67, 193]. As remote work becomes more temporally flexible, complex, and interdependent, this articulation work can consume an increasing share of workers' time and divert them from their primary responsibilities [197]. Digital artifacts can help, as shown by work on the *transfer*, *translation*, *transformation*, and *holding* of knowledge in dispersed virtual teams [69]. Studies of shared editing tools show the same dynamic at a finer grain: in design collaboration conducted through Google

Docs, the document functions less as a container for a finished result than as a site of ongoing disclosure through which collaborators keep track of one another's thinking [103]. Yet despite the abundance of multimodal data generated by remote video meetings, it remains unclear how meeting artifacts might support the specific articulation challenges of catching up, reconnecting, and maintaining continuity across ongoing collaboration.

Building on this prior work, we investigate users' needs for meeting information in post-meeting follow-up activities. To explore concrete designs, we conducted co-design studies on information artifacts that could bridge between remote meetings and asynchronous follow-up activities. The results point to design principles for digital artifacts that can preserve information from remote meetings while also translating and transforming it to fulfill users' subsequent needs.

3.3 Methods

Our interviews and survey of information workers paint a picture of the use of online meeting information in post-meeting activities and the challenges of putting such information to good use. Initially, we interviewed 13 participants who frequently participate in remote meetings to assess their current practices, goals, and challenges when using information after meetings. Preliminary interview themes then informed the design of our survey, which included 198 participants. This broader and more diverse sample confirmed our early impressions and helped us generalize our findings. Both studies were exempt from the Institutional Review Board at the University of Washington.

In both our interview and survey discussions, we scoped our questions to small group meetings that occur online via video conferencing tools (e.g., project update meetings, feedback sessions, brainstorming meetings, 1-on-1 check-in meetings). We focus on small group meetings because they are more likely to be part of a running collaboration that involves follow-on communication,

unlike large lectures or presentations.

3.3.1 Interviews

Participants. We recruited participants online through a screening questionnaire and through word-of-mouth. For online recruitment, we posted the study recruitment message on X (formerly Twitter) and on a 2000-member Slack channel at a large research university in the U.S. We include the screening questionnaire in our supplementary materials.

We began by interviewing local Ph.D. students to quickly gather initial data before branching out to a more diverse pool. Although we strove to capture greater diversity of practices around using meeting information by prioritizing respondents who work in different industries, play different roles on teams (e.g., junior, senior), and use varied strategies to capture meeting information based on their answers to our screening questionnaire, our final sample skewed toward highly educated individuals in research and the technology industry. We stopped recruiting new participants when in-progress analysis suggested we had reached data saturation.

In total, we conducted 13 interviews (represented by I1–I13). Participants include 6 Ph.D. students of different stages, 2 designers, 2 research scientists, 1 program manager, 1 professor, and 1 open-source project leader. Detailed participant demographic information and the interview protocol are included in our supplementary materials.

Procedure. Interviews were conducted remotely over Zoom and lasted 52 minutes on average, ranging from 25–66 minutes. All interviews were led by the first author. We started the interviews by asking participants to identify the different scenarios where they needed to keep track of information from remote meetings for potential future use. For each kind of artifact (e.g., notes and recordings) they mentioned, we asked follow-up questions about their experiences using it. When asking about post-meeting experiences, we focused on specific information participants

found useful from meetings, how they wanted to use the artifact post-meeting, and the ways the artifact succeeded or failed to support their collaboration needs.

Analysis. During data collection, the first author debriefed with other authors after each session and kept a running memo to document emerging themes. All authors met weekly to discuss emerging themes. All interview data were transcribed and analyzed via collaborative qualitative coding. The first two authors coded the interview data following guidance from a reflexive thematic analysis approach [30] to clarify the practices and challenges of capturing and using such information.

In the first round of coding, the first two authors conducted "open coding" on all interview data. Over 500 open codes were generated. All authors then discussed and iteratively clustered the data into common practices and challenges when capturing and using meeting information. Examples of clusters include "*Create to-do list after meeting from notes and recordings,*" "*Use recording to track comprehensive information,*" and "*Challenges of locating information from missed meeting.*" Based on the list of practices and challenges identified, all authors iteratively discussed and derived roles that information from meetings could play post-meeting and challenges in creating, consuming, and sharing meeting information.

3.3.2 Survey

Participants. We recruited survey participants via Prolific.[1] We limited our pool to people based in the U.S., between the ages of 19 and 65, who are employed part-time or full-time, and who have some experience with remote meetings. Our survey took 12.8 minutes to complete on average. Each participant was paid $6.04 regardless of their completion time. The final survey sample consists of 198 responses; five responses were removed for failing an attention check. Our sample

[1] https://www.prolific.co

is balanced in gender, with a 50/50 breakdown, but skewed slightly toward a younger population. It represents participants across a variety of educational backgrounds, industries, company types, and job roles. Slightly over half of the participants reported having management experience. We include more specific demographic information in the supplementary materials.

Survey Design. Our survey included multiple-choice questions, free-response questions, and 5-point Likert scales. The design was iterated upon and discussed within the research team. We also piloted the survey with 39 students and 3 initial Prolific participants to receive feedback before deploying the survey. We include the survey in our supplementary materials.

We started the survey by asking participants to indicate the information artifacts that they have used to keep track of information in meetings (e.g., using recordings, notes, screenshots). As practices around these different artifacts may vary greatly, and given our survey length constraints, we then asked focused questions on the practices of using *notes* and *recordings*, two widely accessible artifacts that represent distinct media formats and often involve different practices and norms.

Specifically, for participants using recordings or notes in meetings, we inquired about their frequency of producing and referring to recordings or notes and their likelihood of sharing the produced artifacts with others. We also asked them to mark the goals and strategies they use from a multiple-choice list. We iterated on this list by combining preliminary findings from the interview study, multiple brainstorming sessions with all authors, and multiple pilot tests.

Finally, for all participants, regardless of whether they used recordings or notes in meetings, we asked what could prevent them from using each, as well as their perceptions of the difficulty of completing common tasks using each. We also collected data on participants' preferences for the type of information they want to capture in remote meetings in different scenarios. At the end of the survey, participants were asked to reflect on their remote meeting experiences in general, regardless of the strategies they selected.

Analysis. For multiple choice and Likert scale questions, we analyzed the responses via descriptive analysis. To analyze the open-ended questions, we used inductive coding and clustering to surface common themes in the responses.

3.4 Current Practices and Challenges

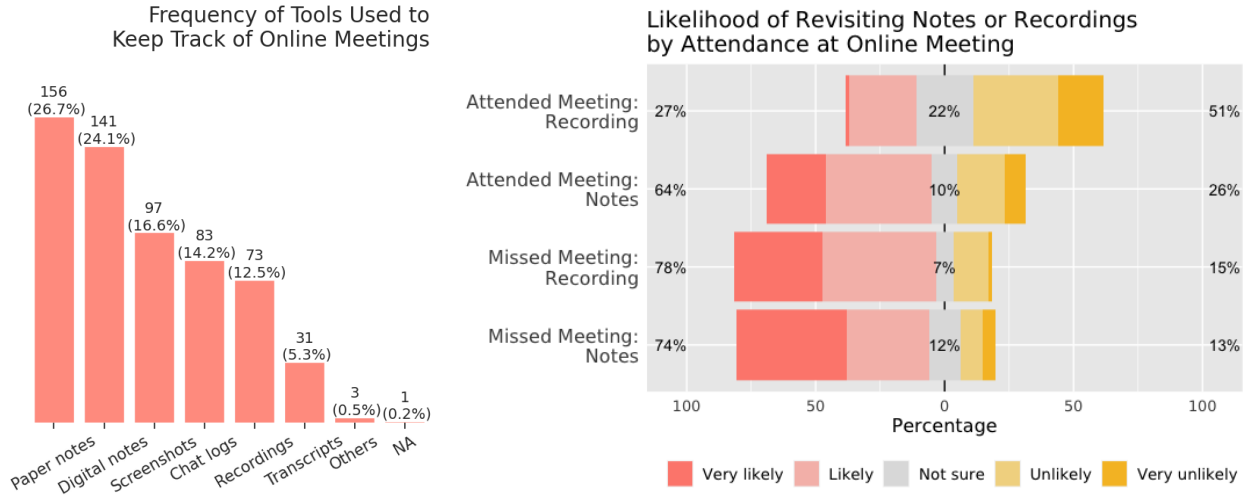
3.4.1 Current Practices

We now describe current practices that interviewees and survey respondents use to capture and revisit information from remote meetings. This includes the type of information they capture, the frequency of using that information in post-meeting activities, and the distinct ways of using the information after meetings.

Type of Information Captured.

We found that participants capture information from remote meetings using a variety of artifacts (see Figure 3.2a). 156 survey respondents (79%) indicated using paper notes, and 141 (71%) indicated using digital notes to keep track of meeting information. In contrast, recordings were used less frequently, with 36.9% of survey respondents stating that they keep track of information from remote meetings using video or audio recordings.

For participants who indicated the use of notes and recordings, we additionally asked about the frequency of use for each artifact. Of those who use notes, 63.8% report using notes in most or almost all meetings they attend. In comparison, only 28.8% report using recordings in most or almost all meetings. In interviews, participants also mentioned not making frequent use of recordings; those who record meetings usually work regularly in teams that maintain such conventions (I10, I11, I12). For others, the decision to record is case-by-case, either when someone



(a) (b)

Figure 3.2: (a) Survey results show that people use a variety of artifacts to keep track of meeting information. Paper and digital notes are used most widely among participants. (b) Participants reported being likely to revisit notes regardless of whether they attended the meeting, but likely to revisit recordings only when they missed the meeting.

requests it or when the meeting itself is critical and dense with information (I1, I5, I6, I8, I9, I13).

Frequency of Information Use After Meetings.

We found that while note takers refer back to their notes, those who record meetings rarely do. Most survey participants (80%) return to at least *some notes* taken either by themselves or others, but only 39% return to at least *some recordings*. Indeed, most survey respondents return to only a few recordings (51%) or do not bother to return to them at all (9.6%). This suggests that many recordings taken for synchronous meetings are *never* actually used.

We also asked about the circumstances in which participants would refer to recordings or notes after meetings (Figure 3.2b). Over 70% of survey respondents said they would refer to *notes* if they missed a meeting. When they attended the meeting, 64% still indicated that they were likely to refer to notes. In contrast, only 27% indicated that they were likely to revisit *recordings* when they

attended the meeting, but 78% did so when they missed the meeting. When revisiting recordings, survey respondents reported a more expansive list of goals compared to their goals when revisiting notes, including more archival goals, such as: "Absorb all major and minor information" and "Find the exact quotes of certain statements that were said" (Figure 3.3).

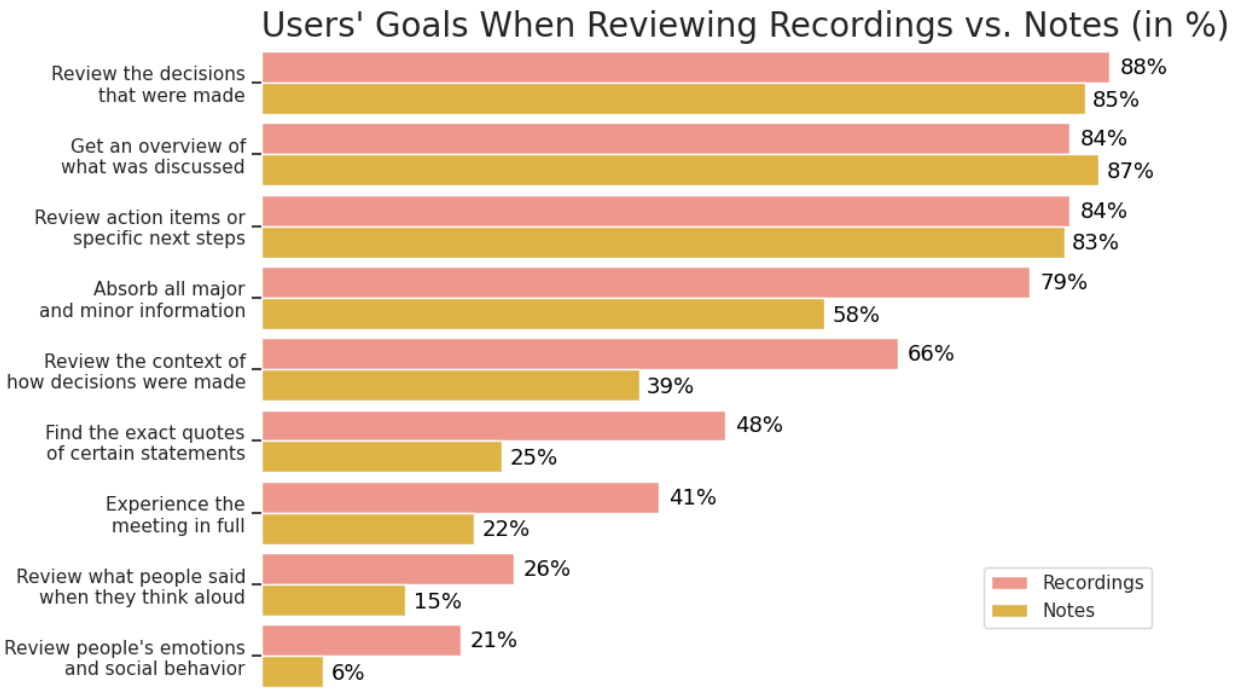


Figure 3.3: When revisiting recordings, participants reported a more expansive list of goals compared to their goals when revisiting notes.

We also queried how frequently participants would share information captured from meetings with others. 73% of survey participants reported sharing notes at least sometimes, and 53% reported sharing recordings at least sometimes after meetings.

In general, our results show that a variety of information is captured during online work meetings. People frequently use captured information by revisiting or sharing it after meetings end. In particular, information captured in meeting notes continues to support post-meeting asynchronous activities. However, not all captured information is actively used; recordings are

sometimes never seen again after being captured.

Post-Meeting Uses for Captured Information.

Our interview results capture a nuanced picture of how information gleaned from meetings is used to support post-meeting activities. We synthesize five main roles that interviewees described for such use.

Personal or Group Archive. Most commonly, meeting attendees want to keep a record of meeting information to help them personally or collectively revisit it asynchronously when needed. They expect that the archive could provide a truthful representation of meeting events and decisions. For example, I8 described notes as "*more true than my real memory*" as notes are captured in closer proximity to the events. Many users find archived information especially useful when the meeting involves dense, complex information that is difficult to remember or digest in real time. For example, as a non-native English speaker, I13 revisited recordings because they had difficulty following English content in technical discussions but felt embarrassed to ask clarifying questions. For others, an archive of meetings helps fill in lapses of memory. When captured for archiving, information is accessed only on an as-needed basis. For instance, I6 recorded meetings in which they received feedback on their work but many times they "*didn't feel a need to [go back to watch it]*" because "*the feedback I got was actually not that much and very manageable*." Indeed, many participants referred to recordings and notes as a "*backup*" (I4, I8), "*fall-back mechanisms*" (I13), or "*safety nets*" (I13). However, this role is important, as I13 states: "*It [the need to go back to recordings or notes] doesn't happen a lot, but it's very critical.*"

Participants also commented on the trade-offs between different artifacts. They often prioritized revisiting notes because notes were easier to search, and referred to recordings only when notes were insufficient. However, they preferred recordings when they wanted to access nuance (e.g., verbatim quotations) and minor information, revisit visual aids, or understand the context of

decisions that were made. For example, I10 watched meeting recordings to capture "*the insights that maybe the notes didn't catch.*" Often, users had a sense of whether recordings were needed before meetings occurred based on the meeting's importance, expected information density, and expected level of activity during the meeting. For example, I6 said: "*I only really personally record meetings if I'm getting feedback on something and I feel like I can't type the feedback quick....to make it useful.*" Interviewees also clarified the distinction between personal and group meeting archives. Many mentioned that they would separate personal notes from shared group notes and treat them differently. For example, I2 described how they use specific terminology that makes sense only to them in their personal notes, but use commonly defined vocabulary in shared notes.

Personal or Group Task Reminders. Information captured in meetings can also be used as a reminder to guide post-meeting tasks. Here, interviewees generally talked exclusively about notes. Several mentioned notes being "*action-oriented.*" For example, I3 described how he would document to-do items in group notes to keep track of tasks: "*Most of the notes are oriented around providing enough information for the people who are going to do those things to go back and remember what they're supposed to do.*" To-do items also get prioritized when people revisit notes for the purpose of reminders. For example, I12 prioritized the following: "*What are some of the things that I have to do until we meet next time or whatever the deadline is? What are some of the things that I need from other people? What are some of the things that I need to do for other people?* " Unlike meeting archives, interviewees preferred reminders to be documented in ways that could be made visible to people assigned to tasks. For example, I6 took notes in Google Docs because they could tag people with specific tasks. I6 also liked when people sent a post-meeting follow-up email containing a to-do list.

Supporting Group Onboarding and Inclusion. Often in group collaborations, a member could miss a meeting, or a new member could join after the start of the project, or the outcome of the meeting must be shared with a broader audience. Information captured during meetings

helped bring these members on board. Interviewees stated that richer formats especially help when someone missed a meeting. For example, I9's team, which does not regularly record team meetings, recorded them when "*someone really wanted to be at that meeting because they wanted to hear about it but are not able to be there.*" However, those seeking to catch up generally wanted to focus on the main takeaways or decisions derived from the meetings, not on comprehensive details. Especially for meetings deemed relatively unimportant, such as project meetings where they play a supporting role, interviewees found that "*there's not a need for keeping track of everything.* (I7)" Some missing content is not necessarily a problem because it could serve as an invitation for follow-on communication, which we describe in Section 3.4.1. For example, I11 mentioned that they ask clarifying questions when notes or recording cannot answer all their questions.

Additional Sensemaking. Information captured in meetings can serve as raw material for further sensemaking by meeting attendees. One way is by allowing people to make new connections between the information captured and additional information from outside the meeting. For example, I6 usually kept track of their to-do items on a Trello board during project meetings. However, since they were involved in multiple projects, they needed to assess the priority of each item in the context of all projects: "*[the Trello board] gives that bird's eye view of seeing everything at once... so it really helps me make that kind of assessment when I have them all out in front of me.*" I11 also explained that looking across several weekly meetings together lets them not only get a general sense of "*what has been most recently happening,*" but ask more interpretive questions, such as "*the timeline of how did a feature get made*" in their open source project.

Additional sensemaking also occurred when participants referred to and reorganized information first captured during a meeting. I7, who is involved in a complex research project, often reorganized their notes of project meetings as they revisited them, and this process helped them further digest the discussion: "*I'll have written like really shorthand something [during the meeting]*

like more citations about X. And then maybe I'll go back [to the notes] and specific links or flesh out why we needed to do that [after the meeting]." I10, a user researcher, shared that when returning to recordings of meetings with customers, they looked for "*their interpretation of what they've heard, or [a] colleague's interpretation*" to make sense of what occurred in concert with others.

Starting Point for Follow-on Collaboration. Information captured during meetings also motivates or structures follow-on collaborations. For teams with recurring meetings, captured information provides a structure for subsequent meetings. Several participants (I2, I3, I5, I6, I7) mentioned using dedicated Google Docs to take notes for recurring project meetings. For example, I6 shared that: "*I can look back and be like, oh this is what we talked about last week or these were the action items out of last week. And so it gives you that starting point for the next meeting.*"

Follow-on collaboration can also include asynchronous discussions since "*a ton of stuff could happen between two weekly meetings*" (I11). For people who attended the meeting, asynchronous conversations centered around updates from topics discussed during the meeting. For example, I7 posted updates to to-do items from weekly meetings during the week and solicited additional feedback: "*I'll provide an update on Slack [for the to-do items]. For example, when I make a graph, [the other collaborators] will say oh this is pretty good, but actually, we should reorganize or change these colors or make other smaller adjustments. And then I'll update again in Slack later.*" I12 noted that they would add questions that they were unable to clarify on the spot to a shared feedback document for people to discuss asynchronously.

For those not present at the meeting, information captured during the meeting offered an entry point for them to contribute to the conversation asynchronously. For example, when I2 went through notes from meetings they missed, they saw the process as a "*simulation of the meeting by imagining that they were there*" and considered any disagreements with the decisions made. If there was a decision they did not understand where "*[their] input would have changed the decision or added more background or context to a piece of decision,*" they would initiate a conversation with

meeting attendees to revisit the decision. Similarly, I11 found it challenging to participate in every meeting synchronously, but they found that reviewing meeting notes lets them participate in meetings asynchronously: "*Someone had a meeting last week that I wasn't in, but they wanted my opinion on something. So they linked me to the meeting notes and Slacked me...The question was very easy...so I just gave them my answer.*" However, when questions became too complicated to discuss asynchronously, interviewees would defer them to the next synchronous meeting. I12 observed how looking back to meeting notes from a missed meeting helped them more easily continue the conversation with the team in a follow-on meeting.

3.4.2 Challenges

Despite the benefits of meeting information in supporting post-meeting collaboration, we observed challenges for those trying to create and use such information with current tooling. We separate challenges according to three stages of the meeting information lifecycle: *capturing* information during a meeting, *consuming* information after a meeting, and *sharing* information with others.

Challenges with Capturing Information during Meetings

Lack of tools to capture rich data in real time. Participants in our survey and interviews reported being unable to write or type quickly enough to follow meeting progress, which resulted in only partial notes during meetings. In the survey, 74% reported that they *often* or *always* have to *take notes in short phrases/incomplete sentences during meetings*. I3 explained the challenge this way: "*I just capture random words that people say... not always coherent to other people [who read the notes].*" I13 echoed that their notes often include only partial information, such as the outcome of a discussion, not the nuances, such as "*justification or trajectory that led to decisions*."

The tools used to capture information also constrain what they can keep track of. For example,

I12 found it challenging to keep track of digital links when taking notes on paper. However, I13 took notes on paper or tablet instead of digital documents so they could draw diagrams to represent ideas non-linearly. To address this limitation, some participants used multiple tools for information capture, such as recording content as backup so they can "*relax a little bit in my note-taking*" (I10). Participants referred back to the recording to complete a more detailed version of their notes or confirm the notes' information. I11 said: "*The meeting minutes are great, [but] it's likely missing context. There are nuances to the discussion that aren't contained...*"

Capturing information affects meeting engagement. Of the respondents who took notes, 82.9% reported doing so themselves in some or many of their meetings. I8 took notes by trying "*...to type as the other person talking [sic] or as I'm talking.*" However, note-taking can be "*cognitively demanding,*" as indicated by I5 and many others, because it usually involves a lot of "*context-switching...not only the tools' context but also my mind's context*" (I5). As a result, note-taking often distracted attendees from actively participating in meetings.

While recordings capture rich data and make fewer cognitive demands than note-taking, they still present challenges for meeting participants. 83 survey respondents indicated that "Participants might be hesitant about speaking up if it's recorded." A similar concern was echoed by our interviewees. For instance, I12 observed that "*People become more conservative about the comments that they make because they know it's being recorded...*" I13 explained that "*People could be suspicious about why you are recording. Like are you gonna use this recording against me in the future, you know, when something bad happens?*" Participants also mentioned that getting explicit consent from all attendees could be awkward given that individual preference for being recorded or documented might not be explicit during meetings.

Challenges with Consuming Information after Meetings

Lacking the necessary context for understanding. Despite being prevalent, notes often

lack the context needed for post-meeting use. For example, I3 found it challenging to catch up on missed meetings by reviewing notes taken by other people because they lacked sufficient context to follow others' thought processes. Note-takers also often used abbreviations or notations that others might not understand, or that even they themselves might forget after a period of time. Even meeting transcripts, which capture conversations more comprehensively, can be insufficient in providing context for meetings that involve sharing of multimedia information. For instance, though revisiting meeting transcripts after meetings worked well for I6, it proved ineffective for design critique meetings, which involve extensive discussions of visual elements: "*People use words like 'it's this' or 'move that there,' but I don't necessarily remember the context here and I have no idea what they're talking about.*" I13 tried to revisit transcripts after meetings but found them unhelpful because the automated transcript incorrectly transcribed foreign names or technical terms. Several participants referred to information captured in multiple media formats when they found an individual media format to be insufficient. For instance, I2 spent extra time returning to more raw sources, such as the recordings, to recover context not captured in notes.

Difficulty locating desired information. Raw information captured from meetings can be noisy, and people need to filter it to find useful content. However, many found that this process is burdensome. Per I2: "*The meeting takes an hour, but it takes almost an hour to go back through [the] recording.*" I8 agreed, especially for transcripts that "*...are too long and hard to read.*" I9's team used to record every meeting for archival purposes but soon abandoned the practice since she found that people rarely return to the recordings, and it became "*more than anybody can handle.*" I3 found it hard to quickly catch up with missed meetings using recordings because lab meetings often involve "*random shoot the breeze kind of talking.*" I1 preferred notes to quickly get a sense of what happened during meetings because they "*filter out irrelevant information.*"

When seeking specific information, participants reported challenges in locating it. For example, I11 found it difficult to search for information in video or audio recordings: "*I wish I could jump to*

a point in the city council meeting where a bill was being discussed. The city council meetings are two hours long, and I don't want to sit there and scramble through the timeline and find that point." Even when functions such as search via keywords are available, I13 still found locating specific information to be challenging: "*Maybe I don't know the terms to search for [desired information]... Maybe I remember a scene in a video but I don't know how to search for the scene. Like I need to know the meta information to be able to find it.*"

Challenges with Sharing Information with Others

Reluctance to share personal notes or to trust notes taken by others. Some participants avoid sharing notes because of the personal decisions they made about format and content while note-taking. One survey respondent said: "*I generally do not share my informal notes from meetings, as those are structured for my own purposes.*" Indeed, we discovered that participants had highly personalized note-taking behaviors, including different forms of organization and structure. They also often prioritized what to capture based on their personal needs for revisiting meeting content. One respondent thought that others would not be interested in their notes: "*Nothing seemed important for everyone to know. Sometimes it's just things I want to keep to myself to help with my work.*" Another expressed a lack of confidence in the quality of their notes.

Even when shared, some participants expressed hesitation about trusting notes taken by others, instead preferring to take their own. For example, I2 worried about the mismatch in their judgment of relevancy: "*they might write things that I don't care about or they might not write things that I care about.*" Similarly, I10 said: "*When I have a colleague help me take notes...they're not as invested in that conversation as I am because this is going to be informing my work. It's just a trust issue and I don't trust that they're going to catch the things that I need to catch.*" Others expressed concerns about being misled by biased interpretations in team members' notes. For instance, I8 often used notes as an archive for meeting information, but "*A lot of people put their*

thoughts in their notes, which for me, compromises [notes] as a source of memory." I10 echoed the concern: "*People who create action items [in notes] might have their own bias on what's important and what's not important... I won't be able to form my own opinions.*"

Reluctance to share recordings due to privacy concerns. Privacy of documented information is another common concern that prevents the sharing of meeting content, particularly for recorded data. Only 53.4% of survey respondents indicated having shared their recordings with others at least occasionally. When asked open-ended questions about why, respondents expressed concerns regarding privacy, namely, that it would be difficult to edit out sensitive information before sharing. This was also reflected in survey question results probing the difficulty of performing certain tasks when using recordings; the task rated most difficult was "Share only selected portions of a meeting with other people," with 38% of respondents finding it difficult or very difficult.

3.5 Meeting Bridges: Designing information artifacts for use after meetings

Our interviews and surveys highlight that a variety of information is being captured in the form of recordings and notes during remote meetings. This information has the potential to facilitate post-meeting activities. However, much of it is uncurated or poorly curated, making it difficult to use. This presents a significant challenge in forming a critical bridge from remote meetings to post-meeting collaboration. To address this challenge, we take inspiration from prior work on boundary objects and articulation work in CSCW [12, 69, 227, 229, 124, 27] that inform the design and use of information artifacts for providing a shared structure and space for team members to negotiate and coordinate with each other. We explore the potential of designing enhanced

information artifacts captured from remote meetings that could more effectively serve the need for asynchronous follow-up work after meetings.

We refer to curated information artifacts that are explicitly designed to support bridging from meetings to post-meeting collaboration as *meeting bridges*. **A meeting bridge is a type of information artifact created in a synchronous meeting with the explicit purpose of facilitating or bridging to asynchronous interaction and collaboration after the meeting is over.** For example, a video recording of an online meeting is a common type of meeting bridge since it is created during a synchronous meeting with the intent of being useful for interaction and collaboration post-meeting. Our distinction of which artifacts constitute meeting bridges rests on the *intent* being about supporting post-meeting activity. Thus, throwaway information for facilitating ongoing synchronous interaction, such as a chat backchannel or sketches on a whiteboard, is not intended for bridging; however, this content could get *repurposed* into an artifact that does seek to serve post-meeting bridging aims; e.g., live captions of the dialogue during a meeting could also be incorporated into a meeting transcript for people to read and review afterward.

We call attention to the design concept of *meeting bridges* to highlight opportunities for designing information artifacts that better support bridging. However, the concept also raises new questions. In particular, how can designers concretely approach the design concept? What kind of formal structure should a meeting bridge embody to reduce redundant articulation work? To explore the design space of meeting bridges, we conducted a co-design study with five groups of 16 total participants to elicit design ideas. We first describe our study methods (Section 3.5.1) and then discuss the design principles we synthesized from the study (Sections 3.5.2 to 3.5.2).

3.5.1 Co-design Study Methods

We conducted co-design sessions to elicit design ideas from participants for *meeting bridges*. These sessions also sought to elicit our participants' desires for different forms of meeting bridges. We conducted 16 individual study sessions where we asked participants to sketch desired designs based on real or realistic synchronous meetings that they participated in. To help them explore design ideas more concretely, we collected and made available to them raw information (e.g., recordings, notes, chat logs) captured from synchronous meetings during sketching activities. The co-design study is also exempt from the Institutional Review Board at the University of Washington.

Participants. Given that our co-design activity uses content produced from real or realistic meetings our participants attended, we recruited participants through word of mouth to reduce privacy concerns. Participants were recruited in groups of people who knew each other so that they could share a more natural experience in a remote synchronous meeting. We recruited a mix of designers and lay users working in different occupations. In total, 16 participants were recruited and divided into five groups to participate in real or realistic meetings. We include a table of more specific demographic information in our supplementary materials.

Preparatory online synchronous meetings. Our study design necessitated access to raw materials from a remote meeting. For groups with a scheduled existing meeting, we asked them to record one session of the meeting and share the recording and any notes with us. For groups without an existing meeting, we set up a mock meeting and captured recordings and notes. Of the five groups, 2 recorded existing meetings, and 3 recorded mock meetings. All meetings were conducted over Zoom except for one real meeting, which took place on Google Meet.

For groups that had a mock meeting, we tasked them with coming up with proposals for a fund that is dedicated to improving an aspect of their work environment, adapted to each team's

specific context. We designed the task so participants could have open-ended discussions on a general topic relevant to their work. We also selected topics that were information-dense, so participants could hypothetically imagine using the artifacts after the meeting. All task prompts are provided in the supplemental material.

We collected as many forms of rich information artifacts as we could from the mock meetings. We video-recorded the meetings and also asked participants to take notes so they could continue the discussion in a hypothetical follow-up meeting and so a hypothetical team member who missed the meeting could catch up afterward. Two paper coauthors also attended the meetings; during the ensuing discussion, the coauthors were muted and had their video turned off while making detailed logs of notable events (e.g., someone sharing their screen). For groups that recorded real meetings, we asked one participant to take group notes and obtained consent from every meeting participant before the meeting to capture a recording. We then produced a log of events by watching the recording obtained from the group.

Co-design sessions. We conducted 16 individual co-design sessions with each participant 1–3 weeks after the meeting. The sessions were conducted in person to make it easier for participants to sketch and share their ideas. The median length of a session was 90 minutes, with a range of 45–95 minutes. There were two components of the co-design session: (1) an *interview* about their previous experiences with meeting information artifacts, followed by (2) *sketching* ideas for meeting bridges from the preparatory meeting. The interview covered questions similar to those posed in Study 1 interviews and was designed to stimulate participant thinking about challenges in their experiences and potential solutions. We include our interview questions in the supplementary materials.

The main co-design activity was a series of sketching exercises that aimed to elicit participants' ideas about the desired characteristics of meeting bridges. We introduced 5 scenarios that covered different ways in which information artifacts from meetings could be used. We also adapted the

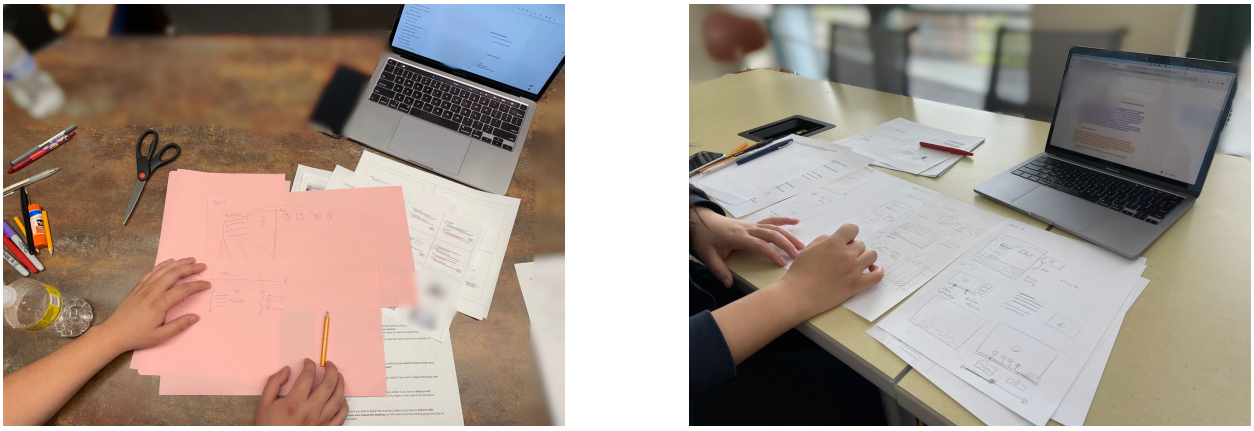


Figure 3.4: Co-design session setup. Parts of the images are blurred for anonymity. Participants sketched ideas for meeting bridges based on a real or realistic meeting they had attended. We provided existing information artifacts from the meeting, including notes, recordings, and transcripts, as design materials.

scenarios for each group to be relevant to their specific discussion context. Specifically, participants were asked to sketch designs of artifacts that could be (1) used for personal revisiting, (2) shared with team members who also attended the meeting, (3) shared with lightly involved team members who missed the meeting, (4) shared with heavily involved team members who missed the meeting, and (5) shared broadly.

To prepare for the sketching activity, we organized the participant's personal notes, any meeting scribe's notes, our log of events, transcripts of the dialogue, the chat log, and the video recording into one Google Drive folder. We also printed out all artifacts that are text documents as well as screenshots from the video recording. The prepared materials served as prompts for users to consider how their desired artifacts were similar or different from existing ones. These elements could also form "building blocks" for participants to consider how to recombine them into a new design. If participants wanted to use any of the printouts, we provided scissors, glue, markers, and highlighters. See Figure 3.4 for pictures of our setup.

We asked participants to sketch on paper and explain their thinking out loud throughout the

session. In a few sessions, the coauthors sketched on behalf of participants when they expressed being more comfortable describing their ideas in words; in these cases, we confirmed that the sketches aligned with what they imagined. When participants finished sketching for each scenario, we also asked them to describe how they imagined the artifacts being used. This helped us understand users' needs for meeting bridges in context. Participants could work on any of the five scenarios in no particular order. In some sessions, participants decided that they wanted similar designs for multiple scenarios. We include example sketches in our supplementary materials.

Analysis. All co-design sessions were audio and video recorded and then transcribed into textual format. We coded all data using a collaborative coding app. The first two coauthors coded the transcripts along with the sketches following guidance from a reflexive thematic analysis approach [30] to identify design ideas for artifacts as well as users' needs. Our initial round of coding and clustering generated 12 groups of design patterns. We further discussed these design patterns among all authors and consolidated them into broader themes about desired meeting bridge characteristics.

3.5.2 Five Design Principles for Meeting Bridges

We organize our results around five high-level design principles that arose from our analysis. We do not claim that these design principles are comprehensive, given that our co-design participants skew toward knowledge-intensive work. However, they provide examples of how we might design artifacts with features that explicitly support bridging.

#1: Incorporation of multiple linked data types and rich media. Most participants wanted meeting bridges to incorporate multiple data types and media formats (Figure 3.5), as this could help users revisit or build on meetings from multiple perspectives. For example, C7 wanted meeting bridges to "*combine different forms of media to be as comprehensive as possible.*" When asked

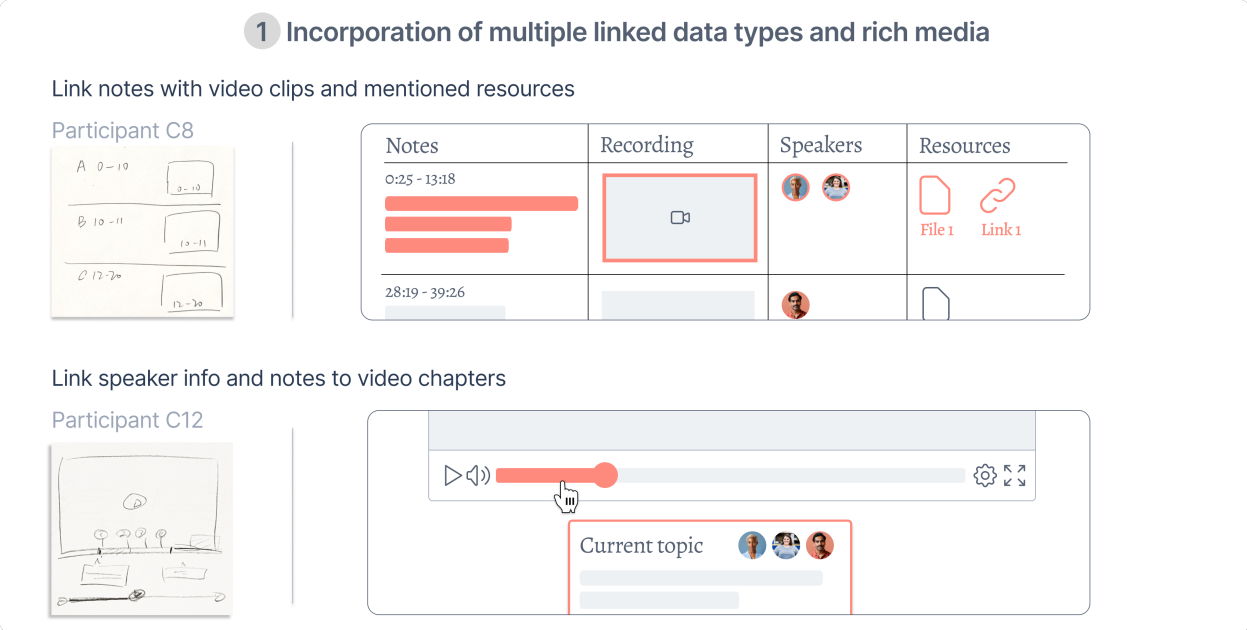


Figure 3.5: Design Principle #1: Incorporation of multiple data types and rich media. We show two design examples: linking notes with video clips and resources (top), and linking speaker information and note snippets to video chapters (bottom). For each example, we show original sketches collected in co-design sessions and a synthesized mockup based on the sketches.

about the kind of information to include in the artifact, C14 said "*every piece of information here,*" referring to all materials we provided. Participants additionally wanted links shared, attribution of information sources (e.g., speakers) and meta-information (e.g., information about meeting participants). Meeting bridges with multiple data types offered a more comprehensive, nuanced, and accurate view of the meetings. For example, per C9, text notes sufficed for understanding meeting outcomes at a high level, but recordings were necessary to grasp the discussions leading to the outcomes. Incorporating various data types also enabled individuals to access and make sense of the information using multiple senses, improving accessibility for C4, C7, and C11. However, there were some use cases when comprehensive information was less desirable. For example, C13 noted that meeting bridges, when serving as task reminders, should provide just enough information for people to recognize the content without going into comprehensive detail: "*The purpose [of the meeting bridge] is more of a reminder, so if people just see one sentence and they can remember what happened in the meeting, that's enough. If they do forget, they can just go back to see the full log.*"

Participants also wanted to connect different media formats so they could understand each piece of information in the context of others. For instance, almost all participants suggested linking notes to specific time stamps on video recordings. C12 wanted to see version histories of notes aligned with a video timeline for the context of how the text notes were created. For C15, recordings also helped him re-experience the meeting: "*Sometimes people write down decisions they made, but not the rationale [in notes]. If I don't [initially] like this decision, but I was probably convinced for some reason [during the meeting], but I don't remember why. I kind of want to go back. I want to know how I was convinced.*"

Another commonly desired linkage was the attribution of ideas and opinions to specific participants. C15 thought that documenting the originator and supporter of ideas discussed in meetings lets him "*put the rationale into context.*" This information is especially desirable for

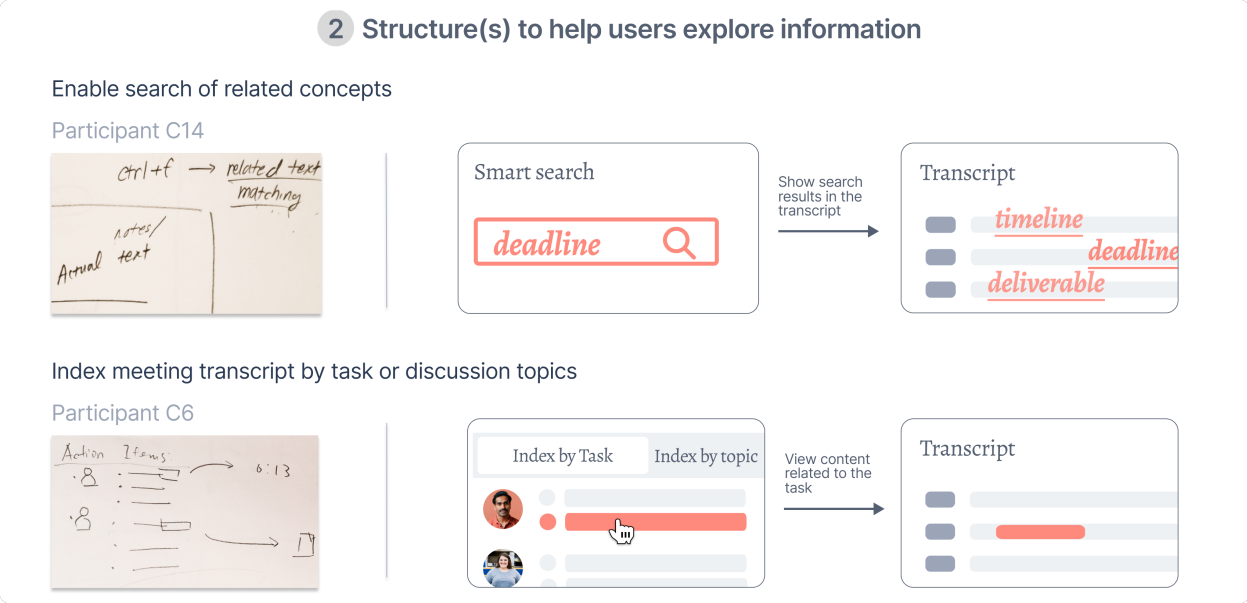


Figure 3.6: Design Principle #2: Structures to help users explore information. We show two design examples: enabling search of related concepts (top) and indexing meeting transcripts by task (bottom).

brainstorming meetings, where people could hold vastly different opinions. C11 and C16 similarly thought that knowing who said what in meetings is helpful to understand the "*weight*" of opinions, especially when making decisions. C3 added that linking people to ideas helps teams follow up with the corresponding person after the meeting. However, participants also raised concerns that attributing ideas to specific people could negatively affect discussion dynamics. For example, C15 explained that it is important to leave "*gray ideas*" in the design. Assigning people to ideas that are then placed on opposing sides could force people into separate 'camps' when they might have been open to either side. Similarly, C16 thought that the linking need not be one-to-one but instead could be a many-to-many connection that showed a high level aggregate sense of who is interested in what.

#2: Structure(s) to help users explore information. Though participants wanted richness, presenting meeting information in its raw form is overwhelming. Several (C4, C11) mentioned that they did not want unedited video recordings or raw notes from synchronous meetings. Instead, they wanted meeting bridges to be "*skimmable*" (C14) so they could "*glean from it*" (C4).

Participants proposed several ways of making artifacts more easily consumable. Some mentioned adding a summary to help the audience make sense of the rich information. C13 proposed that "*someone should write a summary [of the meeting] on the document in the form of bullet points.*" C7, C9, and C14 also proposed one-line or paragraph summaries for the main ideas in the meeting. These summaries could serve as a "*table of contents*" (C7) for users to explore the content using "*their preferred way of navigating [information]*" (C13). Participants also wanted to curate and organize information using more explicit structures. While temporal order is often the default structure for displaying information, and C6 stated that a temporal index helps him skip to relevant parts, C2 found it not useful as conversations can progress non-linearly. Relevant ideas and comments might not be in temporal proximity, which often led C2 to "*go through the whole recording [to find what I want].*" Even for notes, C15 found it hard to consume and locate information in a temporal order because "*it's not a reflection of how [he] thinks about the project.*"

Several participants proposed indexing meeting information as a way to create structure that aligns with their goals and mental models. Participants identified a variety of indices, including person, topic, tasks, resources, and action items. For example, C16, working on an academic research project, proposed using the academic papers mentioned in the discussion as an index. C6 echoed the idea: "*I really like to have the thoughts linked to shared paper, video or links. I think this kind of structure would be helpful...quotes are no longer organized by [temporal] order. People are no longer an important part of it.* " C1 and C3 further proposed to visualize information in the form of a mind map, making it easier to skim.

Finally, some participants proposed searchable artifacts that go beyond the structure enabled

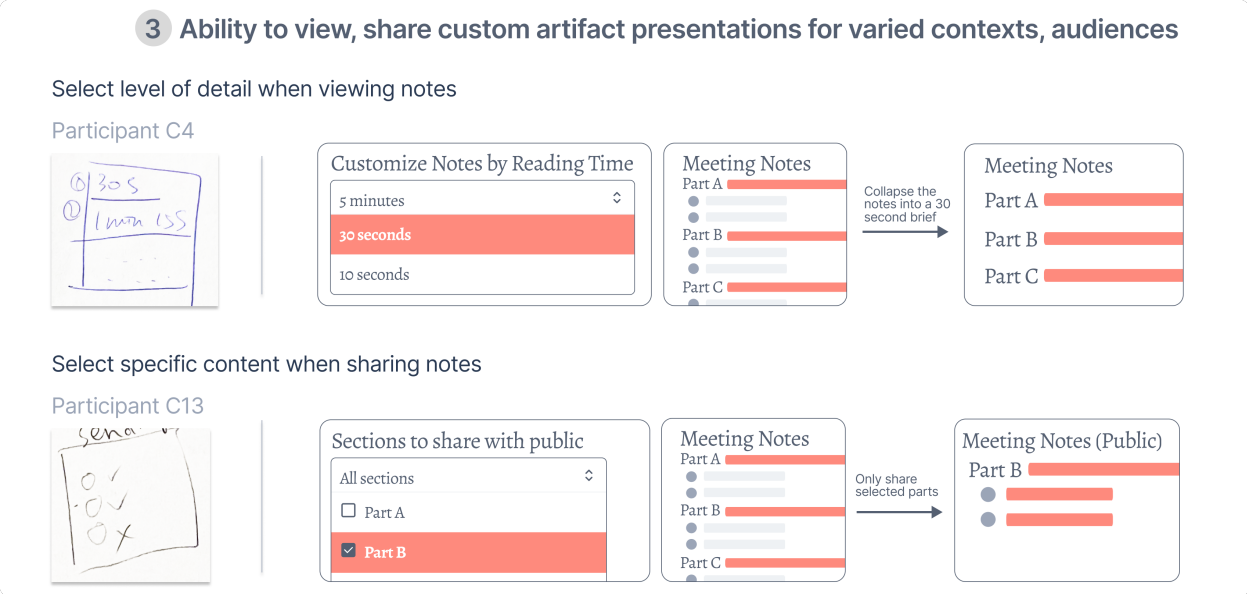


Figure 3.7: Design Principle #3: Ability to view and share custom artifact presentations for varied contexts and audiences. We show two design examples: selecting the level of detail when viewing notes (top) and selecting specific content when sharing notes (bottom).

by an index. C4 wanted searchable components (e.g., transcripts) so she could search for relevant discussions using keywords: "*If I attended the meeting, I always have these little sound nuggets in my head—I know we were talking about something and this person said the word 'back and forth'... I attach a lot of my own social-emotional meaning to certain types of words. So in a full transcript...even if it's one little word, I would be able to find that part of the meeting.*" Some participants desired even more flexibility in search (C6, C14). For example, C14 proposed searching not only for exact text matches but also for semantically related concepts or discussions (Figure 3.6): "*if I were to type 'writing,' 'reading,' something that's semantically related would populate there. That's really important for my skimmability.*"

#3: Ability to view and share custom artifact presentations for varied contexts and audiences. Participants showed strong interest in customizing and personalizing artifacts

for different use cases since not all information is necessary for all people, and different team members may play different roles and differ in their information needs. Participants wanted the ability to customize the content provided, level of detail, and meeting bridge structure, both as authors sharing the artifact and as viewers receiving it.

When sharing meeting bridges with others, some participants wanted to customize the shared content for privacy reasons. For example, C16 wanted to take private notes for himself to explain some terms discussed in the meeting, which he felt was unnecessary for others in the team. Participants such as C13 also wanted to customize the presentation of meeting bridges according to the platform on which it is shared: "*...after the user clicks share, there will be different icons for different platforms. If you share via email, they [the receiver] will only be able to see the text...If you are sharing in Zoom or Slack or other platforms, the receiver will receive the whole [artifact].*" Participants also sought flexibility in selecting the level of information detail based on the audience's backgrounds or interests. For example, C11 noted that the artifact needs minimal detail when shown to meeting attendees but more detail for those who missed the meeting. C16 felt that information could be tailored according to roles and responsibilities. Finally, participants wanted customization to personalize their experience. C10 preferred to view meeting content in chronological order if they missed the meeting but preferred a topic-based organization for meetings they attended.

When it came to designs, C3 proposed an expandable, collapsible design: "*This document with everything collapsed is like a TLDR. This is what we discussed but people can choose to go to whatever point. For example...[another meeting participant]...is new and she may want to see specific things about what is [company name]...So we have that link and people can go to [it]...So now they get to choose like this is when I need more information.*" Others (C4, C9, C14) proposed a design where users could adjust a slider that sets the artifact's level of detail. For example, C4 suggested a slider design that filters based on the amount of time the viewer wants to spend (Figure 3.7). To achieve

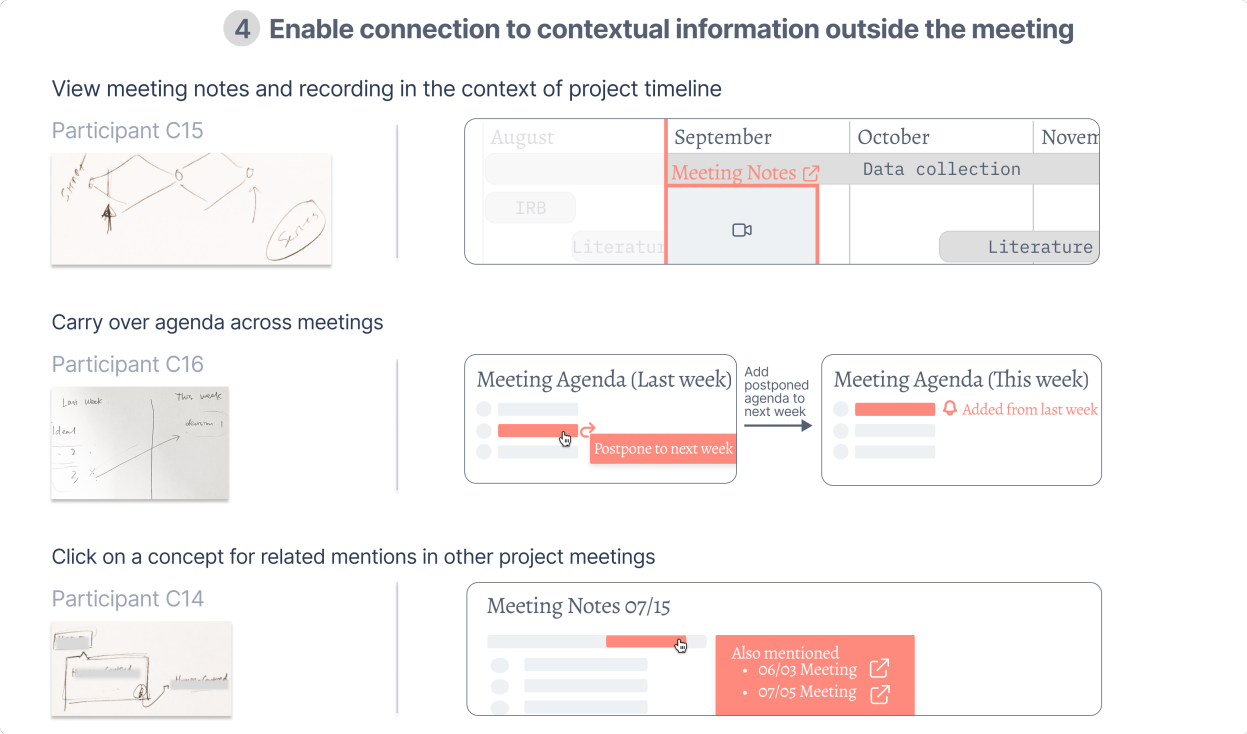


Figure 3.8: Design Principle #4: Connection to contextual information outside the meeting. We show three design examples: viewing meeting materials in the context of the project timeline (top), carrying over agenda items across meetings (middle), and clicking on a concept for related mentions in other project meetings (bottom).

structural flexibility, many participants suggested providing multiple indices in meeting bridges for users to choose from. For example, C9 proposed an interactive interface that could switch between topic- and time-based content organizations (Figure 3.7).

#4: Connection to contextual information outside the meeting. Meetings do not exist in a vacuum; rather, they are often situated in ongoing team collaborations and address issues related to a broader context (e.g., for a larger project or recurring meetings). As a result, participants wanted to contextualize artifact information for sensemaking and follow-on collaboration.

Several participants thought visualizing the relationship between the current and other project meetings would be helpful. C16 proposed providing a project timeline overview that could help

the team understand each meeting's goals based on the stages of the current project (Figure 3.8). C7 and C15 each proposed a design where users could see multiple weeks of meeting bridges at a glance. C15 thought a multi-week design could help him keep track of items raised in previous discussions that were not followed up on (Figure 3.8). For C14, when discussion on one topic spans multiple meetings, viewing all meeting bridges together would let them make needed connections. Participants also expressed a desire to situate artifacts in their personal context. C11 proposed a personal multi-meeting dashboard so she could prioritize tasks across different meetings.

Another type of context mentioned by participants is common but implicit knowledge shared by the team, such as team-specific jargon. Here, C16 proposed linking terms and concepts to previous instances where they were discussed (Figure 3.8). He mentioned a case where meeting attendees used the term *persona* in a meeting to describe a specific paper they discussed, not in its usual sense. Similarly, C14 suggested a shared team dictionary to define team-specific terms.

Finally, participants also wanted to contextualize meeting bridges in their established workflow. Per C13: "*During the meeting [the artifact] serves as a note-taking sheet, but after the meeting it's a reminder for people who want to go back to the meeting content and follow up.*" When artifacts are used as reminders of the next steps, participants wanted to integrate them into their calendar or task management system.

#5: Ability to evolve after the meeting. A team's collaboration typically does not end when the remote synchronous meeting ends. As a result, information captured from a meeting may need to change after the meeting concludes. For this reason, participants wanted artifacts to be updatable to reflect the latest information, especially when they are used to facilitate further conversations and collaboration among team members. For example, per C9: "*Sometimes I come back [to the meeting notes] after a week and I found that at some point we already made a decision on something, [but it's not reflected in the meeting notes].* " C13 wanted status updates in artifacts



Figure 3.9: Design Principle #5: Ability to evolve after the meeting. We show design examples for creating and tracking tasks after meetings and for Q&A with the team after meetings.

from a particular meeting; specifically, he wanted the artifact to reflect whether team members reviewed the information captured from the meeting, which helps him set his expectations for the next team meeting.

Participants also suggested that users who did not attend the meeting may have questions when reviewing the artifact. Many therefore wanted a Q&A section in the artifact. Several participants also wanted to include a bot to help answer such questions. C9 offered that if the bot could not answer a particular question, it would notify meeting participants to solicit help (Figure 3.9). Beyond questions, meeting attendees could also have new ideas related to the meeting topic after the meeting ends. Some participants wanted to collect these new thoughts and feedback in the artifact as well. For instance, C13 designed their meeting bridge with a separate section for post-meeting thoughts. Collaborators who access the meeting bridge from various platforms (e.g., via email, Slack, or a collaborative doc) could all contribute to the section, which would then be summarized.

3.6 Discussion

We reflect on the concept of meeting bridges by examining their role in supporting remote collaboration, discussing the feasibility of proposed design ideas, and describing tensions designers might experience when creating meeting bridges given current technology.

3.6.1 Designing Meeting Bridges to Support Remote Team Collaboration

Our findings indicate that bridging the transition from remote meetings to asynchronous follow-up activities offers a promising starting point for going beyond meeting representations as just "*a point in time*" [154] toward representing the broader context of an ongoing collaboration involving many touchpoints. The challenges we observe in using notes and recordings for asynchronous collaboration echo prior empirical work on the high cognitive effort of taking notes in meetings [187, 189, 257], the difficulty of browsing videos [181, 36], and privacy concerns with sharing meeting records [36, 246, 71]. Many have tried to address these challenges by using in-situ interactions such as annotation [159, 262] or automated methods [165] to add structures that help people make sense of information [39, 271]. Our work reframes these prior issues and designs to consider them in the context of building *bridges* to support asynchronous collaboration with a concluded synchronous event. By strengthening such bridges, we argue that we can de-emphasize participation in synchronous meetings, alleviating meeting fatigue for team members while avoiding creating more asynchronous work or causing collaboration breakdowns. With this new frame, we demonstrate how existing systems can embody aspects of our design principles for meeting bridges in Table 3.1. We note that many current systems address principles that support users in presenting and accessing rich media information (DP #1, DP #2), but few systems support users in customizing, sharing, and updating meeting information after meetings (DP #3, DP #5), or connecting meeting content with external information (DP #4).

Design principles	Systems that exemplify the principle
#1: Incorporation of multiple linked data types and rich media	Integrated multimedia platforms such as Classroom 2000 [2], Teamspace [78]; dashboards such as EMODASH [68], Ferret [52]; interactive documents generated from captured media [239]
#2: Structure(s) to help users explore information	Auto-generated playback plans such as HyperMeeting [80]; visual summaries such as MeetingVis [216], Video Manga [236]; interactive video summaries such as ElasticPlay [100]; searchable meeting materials such as Rough 'n' Ready [47]; user-generated indexes such as Hotspots [105] and Filochat [245]
#3: Ability to view and share custom artifact presentations for varied contexts and audiences	A system to curate videos to share with heterogeneous participants [36]; A system to share multimedia screen content that is viewable from any source [35]
#4: Connection to contextual information outside the meeting	Ability to create links across meetings, such as HyperMeeting [80] and Time Travel Proxy [232]
#5: Ability to evolve after the meeting	Asynchronous video sharing systems such as Video Threads [20]; interfaces that collect users' post-meeting feedback for automated meeting understanding [63]

Table 3.1: Example systems that illustrate each design principle for meeting bridges.

To demonstrate how designers can use our principles to design more effective bridges, we apply relevant design principles to each asynchronous use of meeting information that we uncovered in our first study, deriving concrete ideas (Table 3.2). For example, to design a meeting bridge to onboard new members who join an ongoing project, our design principles suggest linked multimedia artifacts that help new members absorb information more effectively, links from team-specific jargon to prior documents where terms are defined or discussed, and a feature for new members to ask follow-up questions based on meeting artifacts.

3.6.2 Technical Considerations in Implementing Meeting Bridges

Technical challenges can arise when building artifacts that meet users' expressed needs. For example, indices that provide structure for existing meeting bridges are often created or extracted manually, adding a burden for users. Similarly, customized views of meeting bridges are often defined manually by users or system developers, which can limit the extent to which the meeting

Asynchronous Uses	Design Ideas and Relevant Design Principles
Personal or group archive	As an archive, meeting bridges should support users in accessing information on demand, with specific objectives guiding their search. The bridges could incorporate advanced search and indexing functions **(DP #2)** and integrate multiple interconnected data types **(DP #1)** to support user-driven exploration. For example, customizable smart indexes could enable users to filter through multimedia content in ways that resonate with their personal mental model of the meeting.
Personal or group task reminder	As task reminders, meeting bridges should be action-oriented and updated with the latest information. They should integrate reminders into the overall project workflow **(DP #4)** and enable users to update task statuses easily **(DP #5)**. A possible design idea is to show a brief task description on calendars or group workspaces (e.g., Slack), with clear deadlines, relevant team members, and completion status that are visible to all members.
Group onboarding/ inclusion	To support onboarding, meeting bridges should highlight key takeaways and decisions and allow users to explore details or seek clarifications based on their interests or the perceived importance of information. For example, to help people catch up after missing a meeting, meeting bridges could offer a default overview highlighting main themes using short video snippets of key moments from the meeting **(DP #1)**, helping users gain a quick grasp of the main ideas. Users could then dive deeper based on their preferences via features like personalized Q&A **(DP #3)**. To help onboard new members joining the group, meeting bridges could clarify unfamiliar terms by referencing relevant segments from past meetings **(DP #4)**.
Additional Sensemaking	To support additional sensemaking, meeting bridges should enable users to link captured information across several meetings and with broader contextual details, as well as collaboratively refer to and restructure this information. Drawing on the design principles, a possible design could highlight common themes from recent meetings or related projects **(DP #4)**. It could also offer flexible customization options (e.g., different views) to support users in organizing information in ways that best support their understanding and analysis **(DP #3)**.
Starting point for follow-on collaboration	To support collaboration beyond the meeting, meeting bridges should provide structure for subsequent synchronous meetings or asynchronous discussions and ensure users can access the information in different contexts. A possible design could share video snippets and notes from unresolved discussions to dedicated asynchronous discussion spaces (e.g., Slack) to encourage continued discussion outside live meetings **(DP #4)**. Updates from these discussions would then be reflected in the meeting bridges to maintain a continuous view of progress **(DP #5)**.

Table 3.2: Example design ideas generated by applying design principles to each asynchronous use of meeting information.

bridge fulfills personal needs.

The rapidly improving capabilities of emerging technologies, however, introduce new opportunities to mitigate existing challenges. For example, with the growing capabilities of large language models (LLMs) to recognize patterns and generate content based on multimodal data, one can imagine using LLMs to identify latent patterns in meetings and create indices or summaries to help users navigate meeting bridges based on their personal preferences. Currently, multimedia data captured in remote meetings are mostly text- or video-based. As sensors of different types become more ubiquitous, one can imagine including emotion or gaze data that enriches meeting bridges, given appropriate privacy guardrails. There is also potential to combine the strengths of human and machine intelligence to produce more refined meeting bridges with less effort [100, 247].

Enabling information to flow between collaboration and productivity tools is also an essential prerequisite for implementing meeting bridges that connect meeting software to other collaboration platforms. However, closed platforms may create barriers to this. The increasing popularity of open APIs and tools such as Zapier, which provide no-code or automated solutions to connect across platforms, can reduce the cost of manually transferring and transforming information from meeting platforms to other platforms where follow-up activities occur.

3.6.3 Limitations and Future Work

This paper focuses on bridging from synchronous meetings to asynchronous interaction. The other side of the bridge, artifacts for bridging asynchronous communication to synchronous interactions, should also be considered. For example, team participants in a remote synchronous meeting may want to connect to topics discussed in older chat conversations, documents, and other artifacts generated in an asynchronous setting. While this could present new opportunities to improve live meeting conversations, we chose the synchronous-to-asynchronous bridge because our formative

study demonstrated that this was an area of greater need for participants.

This work also focuses on artifacts that people typically create from remote synchronous video and audio calls. However, people also engage synchronously in collaborative documents (e.g., Google Docs), shared canvases (e.g., Miro, Figma), or virtual spaces (e.g., Gather.town), often while also engaging in synchronous conversations via a video or audio call. Even within video conferencing software such as Zoom, there are additional ways to synchronously collaborate, such as a whiteboard or polls. Future work could consider how to incorporate information created from these other kinds of shared spaces into a meeting bridge. Some ideas participants expressed in our co-design sessions could pair with these artifacts. For instance, in a mind map of topics discussed in a video call, map nodes could directly integrate with the contents of a shared canvas being edited synchronously by members of the call.

A wide variety of meetings for diverse purposes occur in the wild. Although we collected survey data from participants with diverse backgrounds, our interview and co-design data were collected from a population that is highly educated and involved in knowledge-intensive work. In addition, we focused on remote meetings that involve discussion and require follow-up work, and followed this criterion both in the design of the study material and the recruitment of study participants. However, there may be nuanced behavioral differences across meeting types, organizational cultures, attendee roles, and other factors. Given the scope of our data collection, we do not have data to examine such nuances in depth. We acknowledge that our findings are best suited for interpreting remote meetings that are recurrent in professional contexts with long-term and familiar members, involving brainstorming and discussion. We encourage future work to collect more in-depth empirical evidence across different meeting setups. Similarly, we acknowledge that our study participants are limited to the U.S. Given that work culture differs across geographical areas, some of our findings may not generalize to other regions.

Finally, our co-design study led to five design principles that can guide design ideas for meeting

bridges. However, they should not be used as a comprehensive checklist, and future research should evaluate these principles quantitatively to understand their effectiveness in facilitating follow-up activities after remote meetings.

3.7 Conclusion

We explored the current practices, roles, and challenges affecting post-meeting use of information captured from remote meetings. We identified five roles information plays in this regard: archive, task reminder, support for onboarding and group inclusion, additional sensemaking, and a starting point for follow-on collaboration. We observed that current challenges in capturing, consuming, and sharing information artifacts prevent them from realizing their full potential in supporting these roles. Given the potential uses and challenges we encountered in our survey and interviews, we proposed the concept of *meeting bridges*, information artifacts that bridge from remote meetings to asynchronous collaboration. We clarified this concept by identifying five design principles that meeting bridge designers should consider. These principles can guide the future design of information artifacts that help users access and engage with post-meeting information toward reducing meeting and collaboration overload.

Chapter 4

Social-RAG

4.1 Introduction

AI-enabled agents are increasingly being deployed in online social environments, including group conversational and collaborative spaces. They serve as assistants, facilitators, or collaborators to support varied tasks such as conversation summarization [263], brainstorming [194, 218], and conflict resolution [81, 233]. These agents often leverage AI to insert automated messages into group spaces where human-human conversation and interaction naturally occur; for instance, they can post messages to facilitate collaborative learning in group chat channels [33] or post cues in brainstorming meetings to inspire new directions for discussion [194].

While these interjections may offer benefits such as providing useful information to groups or fostering prosocial group dynamics [57, 56], they can easily become unhelpful or even annoying to group members, who use the same space to communicate with each other [33, 16]. These systems often abide by rigid message templates and rules defined by system designers, limiting their ability to adapt their content to align with evolving group preferences and specific social contexts [58, 113, 141]. When they fail, existing systems often do not provide easy ways for group

members to give feedback in the moment [16]. As a result, without proper integration into a group's social environment, AI-generated messages can be perceived as irrelevant [264] or intrusive [185], eventually leading users to ignore the messages or even abandon the system [16, 194].

Recent advances in large language models (LLMs) can generate more flexible and human-like text than templated messages [250]. However, these systems often lack the social knowledge required to align generated content with group preferences. *Retrieval-augmented generation* (RAG) techniques augment LLMs by incorporating external knowledge that may not be part of the training data [131]. We find that most RAG applications curate and retrieve *factual knowledge* from external sources (e.g., web content, domain-specific documents) to produce fact-based or domain-specific outputs [109, 127, 110]. Little work to date explores how to augment LLMs with *social knowledge* to generate contextually relevant and socially aligned output in multi-user environments.

In this work, we propose **Social-RAG**, a workflow for LLM-based AI agents that takes social signals (e.g., topical preferences) gleaned from prior interactions between group members and between members and the AI agent to contextualize and adapt the agent's generation (Figure 4.1). Much like traditional RAG pipelines use factual knowledge bases to improve factual grounding, Social-RAG retrieves and processes "social facts" from a social knowledge base to improve an agent's *social* grounding. As such knowledge can be difficult to ascertain without tedious user input [3], we instead leverage the rich history of social interactions between group members commonly stored in group spaces and devise strategies for parsing and retrieving meaningful social information from this history. Group spaces also have many affordances for members to give each other social feedback (e.g., reactions and replies) that can be repurposed for members to give feedback to the agent.

To test Social-RAG in a real-world social setting, we built **PAPERPING**, an AI agent that proactively posts scholarly paper recommendations and LLM-generated explanations into group

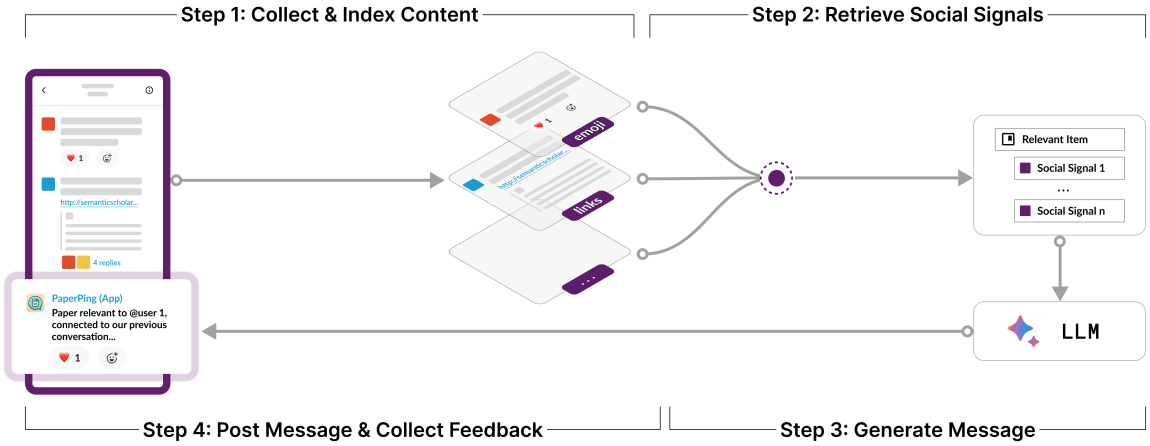


Figure 4.1: We introduce Social-RAG, a workflow for LLM-based AI agents to generate content aligned with group interests and norms. Step 1 collects and indexes the group's conversation history into a social knowledge base. When tasked with making a post, relevant item suggestions (e.g., papers) and signals (e.g., the group's topical interests) are retrieved and ranked (Step 2). Social-RAG then feeds these signals as context into an LLM to generate a socially grounded message (Step 3). The message is posted to group channels, where feedback can also be indexed (Step 4).

chat spaces dedicated to research labs or projects. We chose this test bed due to the collaborative nature of scientific research [119, 101], where exchanging and discussing scholarly articles is common practice and important for conducting high-quality research [101]; further, rapid growth in publications motivates researchers' need to identify relevant papers. Indeed, in two formative studies with 39 researchers, we found that researchers frequently recommend papers in group chats by targeting a specific member or relating them to prior conversations; they also leverage affordances such as emoji reactions and threaded replies to react to each other's recommendations. In addition to creating a rich repository for learning people's interests, this existing social behavior also informs PaperPing's design and how it selects relevant social signals.

PaperPing was deployed for three months in 18 pre-existing Slack channels with a range of social dynamics (e.g., number of participants and frequency of messages), with total exposure

to over 500 researchers. We find that PAPERPING effectively learns researchers' individual and group preferences with minimal upfront effort from users. Users considered its LLM-synthesized messages to be contextually relevant, outperforming generic paper summaries in explaining a paper recommendation to their group. PAPERPING also fits into existing group practices without disrupting group dynamics and supports users in reaching common ground. Our deployment also surfaced areas needing additional investigation, such as how to balance group versus individual preferences. We conclude by discussing the broader implications of Social-RAG for designing more adaptive and socially aware AI systems in diverse collaborative settings as well as how to apply our strategies for social grounding in additional contexts.

In summary, our work contributes:

- A novel workflow, Social-RAG, for augmenting LLM-based AI agents with social knowledge extracted from past and ongoing group interactions, enabling agents to generate content aligned with the group's interests and norms.

- PAPERPING, an AI agent that uses the Social-RAG workflow to proactively post relevant scholarly paper recommendations to a group chat channel with LLM-generated explanations that are contextualized to the group.

- Formative studies with 39 researchers that provide input on how researchers socially interact when discussing academic papers as well as their preferences for an AI agent that recommends papers, which we use to inform the design of PAPERPING.

- A field deployment of PAPERPING on 18 Slack channels reaching over 500 researchers, demonstrating the effectiveness and limitations of Social-RAG in real-world contexts.

4.2 Related Work

From the thesis-wide review in Chapter 2, this chapter especially builds on three themes. First, AI agents in shared group spaces can support productivity and interpersonal dynamics, but ungrounded interjections are often disruptive [16, 273], motivating socially aware adaptation. Second, aligning agents to groups via explicit preference elicitation is costly and can itself interrupt interaction, while learning from activity traces is promising yet rarely validated continuously in the wild [6, 121]. Third, RAG can ground LLM generation in external knowledge, and personalization RAG has mostly targeted individuals with explicit text cues rather than groups whose social knowledge is implicit in interaction traces [131, 93, 143].

Social-RAG takes the next step suggested by that review: it treats group interaction history as a *social* knowledge base, curates *social facts* from past and ongoing interactions, and retrieves them to generate messages that are not only topically relevant but socially appropriate for a specific group channel. The remainder of this chapter develops that workflow and evaluates it through PaperPing.

4.2.1 Direct Elicitation of Group Preferences

A straightforward method to customize AI agents to a group's context is to allow users to explicitly indicate their preferences to agents. For example, Agarwal et al. developed an agent that polls group members' opinions before sending facilitation messages to the group [6]. Conversational group recommendation systems also explicitly learn about users' goals through a multi-turn dialog [240, 167, 169, 168]. Studies that use this method show that continuous and interactive feedback mechanisms lead to better alignment with user expectations [84, 50].

However, explicit feedback often requires users to invest substantial upfront effort solely to inform systems about their preferences. As a result, users see the benefits of the system only after

that effort, creating a 'cold start problem' that can hinder adoption [169]. Frequent interactions in group spaces that are only for steering the system can also disrupt group dynamics, as with an overly frequent poll prompt [6].

4.2.2 Learning Group Preferences from Activity Traces

A unique asset of real-world groups is their rich group communication history, which captures the process of individual and team activities. Researchers have explored using this history to make inferences about individual and group preferences [212] and to inform interactive systems that facilitate team processes, such as peer evaluation [215] and information summarization [262]. Researchers have also explored the design of AI systems that use human social interaction as input. For instance, Krishna et al. showed that collective activity traces of whether users respond to agents' questions can teach an AI agent to ask socially appropriate questions [121]. Others have explored providing agents with contextual information about a group gathered from documents and ongoing conversations to facilitate social awareness [178, 33, 42]. Recent benchmarks make the inference itself the object of evaluation, asking whether a model can recover a user's context and preferences from a history of interactions rather than from stated instructions [115]. These target individual users, and the group case is harder in a specific way: what must be inferred is not one person's preferences but a set of expectations held jointly, which no member states outright and which no single member's history reveals on its own.

Despite their potential, many current strategies for contextualizing agents were validated only in controlled lab environments [57, 15, 16] or using a Wizard-of-Oz approach [200, 194]. Researchers have yet to explore how AI agents can continuously monitor group interactions to adapt their messages to suit the preferences of diverse teams in the wild.

4.2.3 Designing AI Agent Behavior to Fit Social Norms

Many existing agents are developed with rule-based algorithms or predefined language templates that aim to apply across deployment contexts rather than align with the social norms of a specific context [57, 15, 16]. As a result, agents can struggle to convince users of their relevance and utility [141]. The premise that one configuration can serve every context is questionable on its own terms. Searching the space of team structures with a multi-armed bandit, Zhou et al. found no universally effective arrangement: the structure that made one team productive did not transfer to the next, and an algorithm that adapted to the particular team outperformed any fixed assignment [267]. If the arrangement that suits a group is not general, there is little reason to expect the way of addressing one to be. The capabilities of large language models (LLMs) for flexibly generating human-understandable text [179] create new possibilities for agents to integrate more naturally into multi-user shared spaces. Researchers have explored using LLMs to enable personalized interactions that cater to different individuals. For example, Ha et al. [85] created personalized agent personas with distinct expertise to increase the relevance of the information shared with users. Lian et al. [135] explored the potential for LLMs to act as personalized recommenders and tailor explanations to individual user needs.

Despite this promise, integrating LLM agents into practical multi-user systems remains a challenge. Traditional LLM agents are trained or fine-tuned on datasets representing general world knowledge, which are not necessarily tailored to the preferences and backgrounds of specific groups. Current LLMs predominantly focus on informational content instead of social factors [96] and thus show limited abilities to reason about social knowledge (e.g., speaker intention, social proximity of speaker and receiver) [**?** 134]. Natural language processing (NLP) research has also shown that LLMs often lack social intelligence (i.e., cognitive intelligence, situational intelligence, and behavioral intelligence) [250]. These factors limit their capability to understand and interpret

the dynamics of a group's conversation and subsequently generate responses that are natural to the group's context.

4.2.4 Retrieval-Augmented Generation (RAG)

Retrieval-augmented generation (RAG) [131] augments LLM generation with knowledge not in the training data. RAG was originally proposed to address the issue of LLMs lacking access to a constantly expanding and changing body of knowledge, which can lead to false information ("hallucinations") in NLP tasks such as question answering [219, 182, 149, 241, 131] and fact checking [268, 259]. Most RAG systems are built using web data [268, 241, 76], pre-compiled static document corpora, or knowledge bases such as Wikipedia [219] and domain-specific corpora [259, 112, 255, 109, 110]. Using information retrieval (IR) techniques such as BM25 [201] or semantic embeddings [252, 131, 126, 183], these systems retrieve relevant documents to include in their prompts. A growing body of human-computer interaction (HCI) research uses RAG to build interactive systems that support knowledge-intensive tasks, such as sensemaking [140], decision-making [112, 255], or brainstorming [109, 110]. However, little research has explored how RAG can help address the sociotechnical gap.

More closely related to this dissertation, research has used RAG to personalize text generation by retrieving data from personal interaction histories [93, 143, 192, 261, 207]. However, these applications have been limited to individual users and often require explicit, text-based cues from past interactions to infer user needs [192, 143, 179]. In contrast, in group contexts, social knowledge is often implicitly embedded in observable social interactions (e.g., emoji reactions to messages) because explicit preference elicitation can be disruptive in shared spaces [6]. As a result, additional inference is needed to interpret such implicit signals, and existing RAG techniques, such as embedding textual messages, retrieving based on textual similarity, or directly injecting

group messages into prompt contexts, cannot capture and apply social knowledge effectively.

4.3 Social-RAG: Socially Grounding AI Generation via Retrieval from Group Interactions

In the following, we describe *design goals* for our workflow that are drawn from prior work and through building, iterating on, and using a prototype AI agent (described in more detail in Section 4.4). We then describe the *steps of our workflow*, including the curation of a social knowledge base gleaned from group interactions and the retrieval pipeline for real-time zero-shot learning for message generation.

4.3.1 Design Goals

Our design goals are informed by prior literature as well as a *research through design* (RtD) approach [272], where we iteratively designed and built a prototype LLM-based AI agent over the course of 18 months that we deployed to solicit user feedback in our own social spaces. In Section 4.4, we provide details of our final system and evaluation. Though we built the system for a specific task and social context, the process of iteratively working through design requirements helped us gain insights into the broader question of how to best leverage existing group social information and infrastructure to align an AI agent to a group.

DG1: Contextualize suggestions in the group's social history. Research has shown that tailored explanations that convey the relevance of AI suggestions [137, 64] and citations to past social interactions [51] can improve users' trust in and acceptance of AI agents. With access to a group's social history, we can use the same approach to contextualize AI suggestions to the group.

DG2: Integrate AI content into the group without disturbing existing group social

practices. Although the AI agent will be interacting with a group in its shared social space, it should not crowd out existing productive human-human interactions (e.g., by posting overly long messages). Poor integration could result in an undesirable group outcome, such as discouraging discussion or making users leave for other channels.

DG3: Minimize upfront user effort for specifying preferences to the AI and also enable continued improvement. Many existing AI agents require explicit preference choices from users (e.g., by polling users' opinions [6]), which can be tedious. We can learn from prior group interaction data to determine preferences, reducing user effort needed to bootstrap the agent. We can also lower barriers to user feedback for additional adaptation by leveraging the existing affordances of the space and existing common behaviors of users.

DG4: Promote transparency of information shared between humans and the agent to maintain common ground in the group. Maintaining common ground is essential for productive asynchronous collaboration [174]. Thus, agent communication to the group, even when targeted to specific members, should occur in the shared group space. Users' feedback to the agent should also be visible to others to ensure transparency and help group members learn and respect each other's preferences.

4.3.2 Social-RAG Workflow

We now describe the steps of the Social-RAG workflow (Figure 4.1) in detail and explain how they instantiate our four design goals.

Step 1: Collect and index content from group interactions to create a social knowledge base

The first step involves determining how to leverage a group's pre-existing interaction history to populate a social knowledge base, minimizing upfront user effort (DG3). Depending on the specific goal of the AI agent's interjection and technical feasibility, the scope of relevant items in the history could vary. For instance, given a bot that recommends the latest news items to a group that likes to discuss the news, prior messages discussing news-related topics and news articles posted by group members could be collected as an indication of group members' interests. In addition, group members' reactions and engagement around these messages could signal the preferences of more group members. For example, if a group chat had an active discussion about an article, then it is possible that another article on a similar topic could inspire future conversations.

After collecting relevant posts, they must be parsed to extract indexable items for an algorithm to retrieve since we found that simply providing snippets of raw conversation history as context to an LLM performed poorly. Items to be extracted could include any relevant information for suggestions, such as an article URL for a bot that recommends articles. How to index an item depends on what aspects of the item should be used to determine relevance. For instance, news articles could be indexed by the semantic embedding of the article title.

Next, each item is associated with additional information. Each item should initially be associated with the **posts** that mention it. Additional context, such as the sentiment of the message or reactions to it, could also be stored to better identify the **direction of preference**, as opposed to simply interest. Additional **metadata** about each item could be gathered from external data sources or through additional analysis (e.g., the publisher of an article or its main topics). Indexed items can also be associated with the **group members** that mention or interact with the entity so that the agent knows who in the group would be most interested.

This index can now be used to improve the underlying suggestions that the AI agent provides to better target the group. For instance, when recommending a new article to the group, a news recommendation bot could query the index to determine whether the article might be interesting to the group (i.e., if the group has discussed similar articles) and recommend only those articles that return the most results.

Step 2: Retrieve social signals from the knowledge base and contextualize AI explanations

After the agent decides on a suggestion to post, it can use the social knowledge base to help explain the relevance of the suggestion to the group (DG1). We present three categories of social signals that can be retrieved and how those signals contextualize AI suggestions in different ways:

Cite relevant prior posts in group history. Given an index of items previously mentioned and the posts from which they came, the agent can retrieve a **prior post** discussing similar items. The agent could cite this prior message and its associated thread of discussion when suggesting a new item to explain its relevance. The citation could then link to the prior post, enabling members to investigate the connection and establish trust in the agent's suggestions; this is akin to how traditional RAG systems cite the sources from which they retrieved information in their generation. The ability to see older relevant discussions also reminds group members of what was already discussed to further establish common ground (DG4) for continuing discussion. For instance, an AI agent that makes automated *pull requests* (PRs) in a code repository with suggested code edits could link to prior PRs that recently edited the same piece of code or attempted to fix the same problem.

Highlight relevant metadata in an explanation. Each item in our index can be associated with one or more pieces of **metadata** gathered from external data sources. From this information, the agent can determine what metadata is potentially interesting to the group by storing and retrieving counts of how often an item with that metadata was mentioned. For instance, if group

members in the past have shared many news articles from a particular writer or publisher, they might be more interested in a new article if they know that it is also from that writer or publisher. Thus, when a new item is set to be suggested to the group, the agent can also determine whether any particular metadata about it should be mentioned when explaining the suggestion.

Mention relevant group members. Finally, each item in our index can be associated with one or more specific **group members** who posted about the item or reacted or responded to a post about that item. When a new item is set to be suggested, the agent can determine as a signal of interest which items it is most similar to and which group members interacted with those items. Additionally, it could use external data sources with other information about group members to determine connections to specific group members. Then, the AI agent can @-mention specific group members as another explanation of relevance when posting to the group. Alerting all group members of specific members' interests could potentially encourage them to interact with the agent's message in the channel and make them aware of each other's interests (DG4).

Step 3: Rank social signals and generate a succinct message for the AI agent to post

In this step, we gather all relevant social signals associated with a suggestion and combine them into a message for the agent to post. Our goal is to craft messages that naturally suit the existing human-human conversational cadence in the social space (DG2). To achieve this goal, we use LLMs (1) to flexibly generate a message in natural language mimicking existing group dialogue and (2) to summarize social signals in a succinct way to avoid overwhelming group members. We take the social signals gathered in the last step and the original suggested item and provide them as context to an LLM, along with instructions and examples of how the AI agent should provide its suggestion and explanation. For instance, the prompt could ask for a specific format, length, or tone. If the prior step resulted in many social signals retrieved, we can use heuristics to select one or a few most relevant signals to provide as context or simply provide all information to the LLM

and let it determine what to retain to form a short explanation.

Step 4: Post generated message to social channel and collect and index group reactions

Finally, with an LLM generation in hand, the AI agent can post the message in the group channel for all members to see (DG4). At this point, as with any other human-generated post, members can interact with the post, such as by reacting or replying to it. These interactions can also be collected and indexed (similar to how group member posts are collected and indexed in Step 1). Enabling group reactions to the post helps the agent further learn about other group members' preferences (DG4). By using the existing affordances of the social space, we also provide an easy and more natural way to give feedback to the AI agent for additional adaptation (DG3).

4.4 The PaperPing System

We implement the Social-RAG workflow in a system called PaperPing, which automates academic paper recommendations in a scholarly group chat context. We chose this as our test setting due to the highly collaborative nature of scientific research, which often involves regular, information-dense communication among collaborators [120, 176, 119, 101], much of which occurs today using digital tools. We also chose to focus on paper recommendations since academic papers play an important role in facilitating knowledge sharing, fostering collaboration, and informing decision making in scientific team communication. The process of discovering and searching through papers is often open ended and opportunistic [164]. Finally, paper recommendations often come from social networks [222], and researchers commonly form groups for the purpose of sharing and exchanging relevant research literature.

4.4.1 Formative Studies

We first describe two formative studies we conducted with scholars to identify (1) if and how they use social channels to discuss and recommend papers to each other, and (2) their preferences for an automated paper recommendation bot. Findings from these studies motivate and inform the system design of PaperPing, including the specific social signals (e.g., paper metadata) we choose to capture about each paper and the design of automated messages (e.g., format, tone). Our study was reviewed and deemed exempt by our university's Institutional Review Board (IRB).

Procedure and Participants

In our first study, we **interviewed** 13 researchers who regularly discuss research in group chat channels. The participants were part of research project and interest groups, such as a lab discussion channel, focused mainly in the areas of HCI and NLP. The interviews lasted 19 minutes to 1 hour, and we asked participants about their motivations and considerations when sharing scientific literature and their experiences with literature recommendations. We analyzed the interview transcripts using inductive coding and clustered the codes into common themes.

In the second study, we conducted a **survey** with 26 participants from two active research discussion channels on Slack, consisting of a mix of research scientists, professors, PhD students, and undergraduate students. In the survey, we asked participants to compare different styles of Slack messages that recommend academic papers in order to surface people's preferred message presentation in an academic Slack setting. These styles varied across several dimensions, including tone (neutral vs. enthusiastic, promotional language, humor), use of emojis, and formatting (single block of text vs. bulleted format, use of text highlighting such as bold). Finally, we asked participants to indicate their preferences for pairs of messages showcasing different styles.

Results

Researchers regularly share papers with each other in group chat. In project-specific channels, interviewees share papers that directly support specific goals, such as answering research method questions or illustrating ideas to team members. These shared papers play a crucial role in shaping the research project by providing potential citations, serving as model papers, and enhancing collaboration by ensuring consensus of perspective. In interest-based channels, papers are shared for their general relevance to the group's common interests; although not shared in response to specific questions, these papers spark new conversations and facilitate relationship building among like-minded researchers.

When sharing papers, researchers attempt to connect the papers with group context. Interviewees who post paper recommendations mentioned that they like to contextualize the recommendation in the discussion. For example, P1 often highlights specific quotes or explains reasons for sharing the paper. They also use @-mentions to draw the attention of relevant individuals to specific papers. Interviewees looked at prior papers that were shared and group engagement about them to determine what kinds of papers to share, how to share, and the frequency of sharing. However, some said it currently takes too much effort to craft a message explaining a paper recommendation, calling it "*free labor.*" They are occasionally hesitant to write their own paper summaries since they often recommend papers without having fully read them and worry about providing an inaccurate summary. Some junior scholars also worry about interrupting group dynamics or taking too much space with papers that may be perceived to be low quality or irrelevant to their groups.

Researchers reading paper recommendations prefer short and neutral explanations. Survey participants overwhelmingly preferred straightforward and neutral Slack messages when introducing academic papers in their channels. Messages with minimal formatting and without

text decoration or humor were favored, as seen in the strong preference for neutral tone (92.3%) and simple formatting (96.2%). Promotional language and humor were particularly unpopular, with only 15.4% and 7.7% of participants supporting those styles, respectively. While there was some division on emoji usage, undergraduates were more receptive (60%) compared to PhD-level participants (43%). The overall trend leaned toward simplicity and clarity.

Researchers prefer quick ways to acknowledge paper recommendations. When responding to papers that others shared in chat, interviewees commonly use emojis and quick comments to indicate their interest as a way to show appreciation and encourage others to continue to share. Emojis are liked because they are "*unobtrusive and don't take space*."

Summary

Our findings motivate the utility of a paper recommendation bot that posts in group chat since researchers already perform this activity but occasionally find it difficult to do so easily and succinctly. In terms of bot design, interviewees provide examples of ways they connect their recommendation to the group; these inform the social signals we implement in PaperPing from the Social-RAG workflow. In addition, our findings guide our effort to design bot interjections that are short and unobtrusive, with feedback mechanisms that are quick and easy.

4.4.2 PaperPing Example User Scenario

Figure 4.2 demonstrates how users interact with PaperPing. As an example scenario, imagine that Kara leads an interdisciplinary research group consisting of PhD students, postdocs, and undergraduate researchers. The group also shares common interests. To help the group keep current with the latest research, Kara encourages everyone to share and discuss relevant papers in their group chat on Slack. Upon learning about PaperPing, Kara introduces it to her group to

help them stay informed about the latest literature without adding to their workload. She set the recommendation frequency to be once every other day.

One day, PAPERPING notifies the group of a new paper recommendation. The message includes a brief explanation that highlights the connection between the recommended paper and their recent chat, citing a previous discussion thread for reference. The message is also personalized with an @-mention to her postdoc Ryan since he has previously reacted positively to similar papers. Ryan views the linked previous discussion to gain more context and has some new thoughts about a question that confused the group in previous discussions. He shares his thoughts under the latest paper recommendation while @-mentioning other relevant students and up-voting the bot's post. Kara sees the discussion and contributes another paper that she recently encountered and thought relevant. The team continues to interact with PAPERPING and with each other as recommendations arrive. Over time, the positive feedback and ongoing interactions from the group help PAPERPING refine its knowledge of the team's preferences, recommending more relevant papers and providing more targeted explanations.

4.4.3 PAPERPING Process and Features

Building on findings from our formative interviews and survey, we describe in more detail PAPERPING's process and features specific to our academic paper recommendation use case (Figure 4.3).

Step 1

To create a social knowledge base from prior chat history, PAPERPING parses prior messages to extract links to academic papers shared by others. It converts these links into an embedding for each paper to store in a vector database, along with the messages mentioning that paper, message authors, and when messages were posted. In addition, it collects users' emoji reactions to the

message, noting which emojis have a positive versus negative sentiment (e.g., thumbs up as a positive reaction) and any comments in the thread under the message.

PaperPing also links this paper database with external data sources that provide additional relevant metadata. In particular, it collects author names on the paper, their affiliations, the papers they cite and that cite them, and the venue in which the paper is published. Finally, PaperPing optionally allows users to link their group chat account to their academic publication record, letting it know the user's affiliations and publication history.

When PaperPing prepares to make a recommendation, it sends to an off-the-shelf paper recommender system recent items from this paper database that have (1) received positive emoji reactions, or (2) one or more comments in response from a group member.

Step 2

After accessing a paper, PaperPing next retrieves social signals related to it. It focuses on three groups of social signals: (1) Connections from the new paper to previously shared papers in the chat channel, looking for potential topic overlaps, citation relationships, or common authorships. (2) Metadata items about the paper, such as authors or publisher, that have some connection to prior interactions in the chat channel; for example, if an author of the new paper was previously mentioned in the group chat history, it will identify the author's name as a social signal. (3) Team members who might be interested in the paper based on their previous interactions in the chat channel. These align with the three types of social signals discussed in Section 4.3.2. If it retrieves many social signals from this step, PaperPing uses heuristics developed over several months of piloting to rank and select only a few that it deems the most important. See the Appendix for a detailed list of retrievable social signals implemented in PaperPing.

Step 3

As Figure 4.4 shows, PAPERPING uses chains of LLM prompts to turn the retrieved information into a natural language explanation. The prompt chains are dynamically constructed based on the availability of social signals from Step 2. PAPERPING uses one prompt for each group of social signals before using one final prompt to succinctly summarize the information and implement a friendly and informative tone. This approach lets the output address meaningful connections between the recommended paper and the channel context while reducing hallucinations. We iteratively developed these prompts by piloting our prototype in two channels we participate in that are not part of the deployment study. The Appendix provides the specific LLM prompts we used.

Index	Channel Type	Channel Size	Frequency of Recommendation	Study Duration (weeks)	Participants
1	Lab channel	31-100	every-other-day	14	P1, P2, P4
2	Common interest	2-10	every-other-day	14	P3, P5
3	Lab channel	100+	weekly	21	P12, P17
4	Lab channel	31-100	weekly	18	P10, P15
5	Project	2-10	every-other-day	24	P8, P23
6	Project	2-10	every-other-day	24	P8, P23
7	Lab channel	10-30	every-other-day	24	P8, P23
8	Project	2-10	every-other-day	24	P8, P23
9	Lab channel	2-10	every-other-day	24	P14, P20
10	Project	2-10	daily	23	P11
11	Seminar	31-100	weekly	16	P16, P9
12	Seminar	31-100	every-other-day	23	P13, P18, P19
13	Common interest	10-30	weekly	10	P24
14	Lab channel	10-30	weekly	15	P24
15	Lab channel	10-30	daily	2	P6, P7
16	Common interest	10-30	weekly	23	P21, P25*
17	Common interest	10-30	weekly	24	P26*, P27*
18	Common interest	10-30	every-other-day->weekly	22	P22, P28*

Table 4.1: **Participant information by channel type and recommendation frequency.** Participants marked with an asterisk (*) participated asynchronously in the offline evaluation task without joining the exit interview. Duration counts the time from the installation of PAPERPING until all participants in the channel had completed the study. While not counted in the study duration, 11 channels continued to use PAPERPING voluntarily for more than six months after the study concluded.

Our LLM prompt set instructs the generation to take the following format. The generation first introduces the paper using a contextualized explanation based on the preceding social signals. Whenever the explanation mentions a user, it will also @-mention the user, unless it has already done so recently. When there is a relevant previously mentioned paper, the explanation also cites the previous thread that mentions the paper; this citation is linked, allowing group members to easily view the previous thread. Finally, the explanation also provides the paper's meta information, such as title, author name, venue, etc., in a structured format separate from the contextualized explanation. The text design came from feedback from formative study participants and pilot users. For example, users indicated that they prefer to see keywords bolded in the explanation to help them skim through the text.

Step 4

The message gets posted in the group channel, after which group members can provide an emoji reaction or reply to the message to give feedback. Group members can adjust the frequency of recommendations to daily, once every other day, or weekly based on their individual preference.

4.4.4 PaperPing System Implementation

PaperPing is implemented as a Slack application, with recommendations arriving in the group conversation as chat messages from the PaperPing bot account. It is built with the Slack Web API. The backend server is built in Node.js and interfaces with a PostgreSQL database. PaperPing leverages the Semantic Scholar Recommendation API[1] to select paper recommendations based on a group's interests and prior interactions. To retrieve the paper's metadata, SPECTER embeddings for each paper [46], and citations from the citation graph, PaperPing uses the public Semantic

[1] `https://api.semanticscholar.org/api-docs/recommendations`

Scholar Academic Graph API. Finally, PAPERPING uses `GPT4-turbo-preview` through the OpenAI API for all text generation with an LLM.

4.5 Field Deployment

We conducted a **three-month field study** where we deployed PAPERPING in 18 Slack channels, collectively reaching over 500 channel members. From these channels, we collected in-depth feedback from 28 members via **exit interviews and questionnaires** and an **offline evaluation task**. Three participants (P8, P23, P24) were members of multiple channels that installed PAPERPING, while all others were in only one participating channel. All but one channel installed PAPERPING for at least 3 months; members of that channel, who decided to discontinue the study after one week, were still invited to debrief about their experience, and their feedback is also reported below. Eleven channels continued to use PAPERPING voluntarily for more than six months after the study concluded. The study was reviewed and deemed exempt by our university's IRB.

4.5.1 Research Questions

Our research questions are based on our design goals for Social-RAG (Section 3.1). We focus on identifying how channel members perceive and interact with PAPERPING's suggestions and how PAPERPING affects group dynamics. In particular, our research questions include:

- RQ1: How effective and effortful are PAPERPING's feedback mechanisms in eliciting individual and group interests? How relevant are PAPERPING's suggestions? (DG3)
- RQ2: How effective are the social explanations in contextualizing PAPERPING's suggestions? (DG1)
- RQ3: How does PAPERPING affect existing social practices in the channels? (DG2)

- RQ4: What impact do PaperPing's suggestions have on the establishment of common ground in channels? (DG4)

4.5.2 Study Participants and Procedure

Participants

Via social media advertisement and word-of-mouth, we recruited a total of 28 participants from 18 channels (Table 4.1) to participate in interviews for our deployment study, ranging from junior PhD students to professors across various domains (e.g., AI, HCI, NLP, Programming Languages). The channels included group research labs, reading groups, project teams, and course channels, with sizes ranging from 2 to over 200 members each. These channels had varying levels of paper-sharing activities prior to installing the bot, ranging from 0 to over 5 papers per week. We paid participants who participated in both the interview and offline evaluation (N=24) $50 Tango gift cards and those who participated in only the offline evaluation (N=4) $25 Tango gift cards.

Eighteen of our participants were interviewed before launching the tool in their channel. No participants were required to interact with PaperPing during the length of the study. We additionally recruited 10 participants for interviews after the tool had been installed in their channel for a period of time by sending recruitment messages to the channel. These participants included those who indicated that they had not thus far used PaperPing much.

Installing PaperPing in channels

We invited participants who volunteered to be their channel's contact person to an onboarding session, where we explained PaperPing's key features and guided them through installation and setup. To set up, participants selected a recommendation frequency suited to their channel's norms (e.g., once a week). We also invited participants to share several papers in the channels to

seed PAPERPING if the channel did not have much of a conversation history. Upon installation, PAPERPING introduced itself to the group with a brief message, highlighting core features such as link sharing and emoji reactions. During the study, we sent check-in surveys every few weeks to all paid participants. These surveys captured recent memorable positive and negative experiences, which we used in post-study interviews to help participants recall their experiences.

Exit interview and questionnaire

After using PAPERPING for at least 3 months, we invited our 28 participants to reflect on their experience in an exit interview lasting 30–75 minutes. During the session, they first completed the offline evaluation task (below), after which they discussed their experience in a semi-structured interview focusing on their perceptions of PaperPing's alignment with their individual and group preferences and its impact on group dynamics.

Finally, they completed a questionnaire containing Likert scale questions about their overall satisfaction in and the usability of PAPERPING. We developed survey questions by reviewing prior work on the evaluation of conversational AI tools [15].

Offline evaluation task

We designed an offline comparative experiment to examine the effect of different variations of messages on participants' experiences. In the task, we presented four variations of explanations for papers (see Figure 4.5). Specifically, Condition 4 (LLM synthesis of social grounding information and paper summary) shows PAPERPING's messages posted in the Slack channel. Condition 1 (TLDR), Condition 2 (social grounding information presented in template form), and Condition 3 (social grounding information and paper summary information presented in template form) each explore a different component of a generated message and its effects on users' perception. Each participant was asked to evaluate five papers that were randomly selected from the recommendations that

they had received in the channel. Each paper was shown in four variations (Figure 4.5), presented in random order. For each variation, participants rated Likert scale questions, such as whether the explanation was concise, interesting, or helped them understand the relevance to the group. After seeing all four variations, they selected the variations that best suited their own and group preferences, respectively.

4.5.3 Data Collection and Analysis

Qualitative coding for exit interviews

We took detailed memos during the interview process and discussed emerging themes in regular project meetings. We used Dovetail[2] to transcribe the interviews, which produced transcriptions that were linked to audio and video recordings. During the coding process, we referred back to the original video recordings when transcripts were ambiguous or inaccurate. We analyzed the data following the guidance of reflective thematic analysis [29], wherein we first developed codes based on the research questions and emerging themes in the memos and added new codes as they emerged in the analysis process. The final codebook contained 35 codes focused on: alignment of recommendations with participants' preferences, the effectiveness of the social signals, and PaperPing's impact on social dynamics; examples include: *relevance to individual preference*, *conciseness of explanations*, and *individual willingness to engage with PaperPing*.

Statistical analysis of offline evaluation task ratings

PaperPing made 832 recommendations in 18 channels during the deployment study. We randomly selected 136 of these papers and collected 540 observations from 28 participants in the offline evaluation task. Participants' ratings for this task were treated as ordinal variables and analyzed

[2]https://dovetail.com/

using a mixed ordinal logistic regression, where variations were fixed effects and participants were treated as random effects. We calculated the pairwise differences between the four conditions and applied Tukey correction. We also applied a Holm correction for testing multiple hypotheses.

Usage logs

To gain insight into how people interact with PAPERPING in real-world settings, we collected and analyzed paper-related activities (e.g., reactions and comments to shared papers) within each channel, until all participants in the channel had completed the offline evaluation tasks and exit interviews. Notably, while we captured activities around papers shared via common academic platforms (e.g., ACM Digital Library, Semantic Scholar, arXiv), we were unable to track papers shared through social media links (e.g., X, LinkedIn posts).

4.5.4 Results

PAPERPING effectively learns individual and group preferences with minimal user effort (DG1, DG3)

PAPERPING recommends papers that suit individual and group preferences and that directly benefit ongoing research PAPERPING's messages were considered relevant and aligned well with researchers' personal and group preferences. In the offline evaluation, 59.6% of the 136 reviewed recommendations were marked as relevant to participants' individual preferences, and 75.7% were marked as relevant to groups' preferences (Figure 4.6). Moreover, 82% of participants agreed that PAPERPING helped them discover papers they would not have found otherwise. Over half (54%) of the participants thought PAPERPING's suggestions aligned with their interests, which is comparable to those who consider recommendations from other channel members to be relevant. PAPERPING's suggestions were also able to directly benefit researchers' ongoing research.

Participants considered benchmark papers suggested by PaperPing in their research (P8, P11): "*We definitely use the papers that are suggested.*" Several participants (P5, P15, P16) appreciated PaperPing's ability to surface relevant papers in fast-moving fields. Compared to general-purpose literature discovery tools, PaperPing was considered more helpful since it is "*seeded*" and "*ha[s] a good sense of the topics and research communities that I want to read papers from*" (P3). P11 echoed: "*With PaperPing we can focus our time and energy on our own paper.*"

Indicating one's preference to PaperPing involves minimal effort While PaperPing suggests relevant content that brings practical benefits, researchers agreed that indicating their interests to PaperPing was a natural process without being effortful. In the post-study questionnaire, 88% of participants agreed that using PaperPing required little effort. Sharing paper links in the channel was the most frequently used method to communicate interests to PaperPing. Participants appreciated that these feedback mechanisms fit into their existing workflows. P3 described reading and sharing papers in channels to be a "*natural part of my day-to-day interaction of being on Slack.*" P4 also thought that reacting to papers in group chats is a natural interaction: "*it feels social...other people also are interested in this, and they also react to this.*" P20 described group members' emoji reactions and comments as a form of "*collective annotation.*"

Limitations of existing feedback mechanisms Despite promising results, we observed several limitations of PaperPing's existing feedback mechanisms. In less active groups, PaperPing struggled to gather sufficient signals, leading to less accurate recommendations. P10 mentioned: "*there are definitely more interests that we have, but we just didn't put in [these in] the channel, so PaperPing wasn't able to pick up the signal.*" P5 and P13 both wished their group members had reacted to papers more frequently to better inform PaperPing about their groups' interests. Diverse group norms about using emojis could also introduce ambiguity in the interpretation of the

feedback. For example, P8 noted that in their channel, "*the reaction was more for acknowledgment as opposed to indicating that the paper is very useful.*" Further, although PAPERPING considers all papers shared in the past to have more data available, it struggled to reflect interest shifts in a timely manner: "*It's hard to branch out into super new papers...it feels like the types of papers that [it] recommend[s] don't change too much over time*" (P1).

For groups with diverse interests, the tension between individual and group interests also poses challenges. P4 observed: "*even paper shared by my lab mates I don't relate to most of the time.*" As a result, PAPERPING sometimes "*overfits*" to certain channel members' interests (P5). The misalignment could further turn into barriers for users with unique interests to share papers. P17 felt uncomfortable sharing papers that diverged from the group's primary focus, which represented the interests of only active members in the channel.

Despite recognizing PAPERPING's usefulness, participants did not see it as a replacement for paper recommendations from other human channel members. As P3 describes: "*PAPERPING is not a replacement. It's an addition to the practice of people sharing with other people.*" Participants noted that PAPERPING did complement human recommendations in its timeliness, broad coverage, and ability to tie back to previous discussions. P8 shared: "*there are some papers that [PAPERPING] suggested that I think people just missed or they didn't suggest it in the group. [PAPERPING] helps to have a better picture of the ongoing work related to the project.*" P4 observed that PAPERPING offered more diverse and more timely recommendations than they would typically receive from human group members. However, participants continue to place more trust in human recommendations and find they want to verify recommendations from PAPERPING. For example, P8's comparison of papers from PAPERPING, arXiv, and other human members further illustrated this sentiment: "*PAPERPING is more helpful than browsing arXiv since the topics are more focused, but I still feel like I have to briefly judge whether I wanna actually read it or not. Whereas if a person*

who is familiar with our project suggested it, I would definitely read it without having to consider it again."

Social explanations contextualize suggestions in group context (DG1)

Social signals in explanations help contextualize the recommendation in social context

To adapt AI suggestions to groups' social contexts, PaperPing explained each paper recommendation by summarizing paper content and its relevance to the group. These explanations helped participants assess paper relevance (75% agree) and were perceived as sufficient by most participants (75% agree). In contrast, only 29% of participants' explanations provided by other channel members were sufficient (Figure 4.6). In the offline evaluation, explanations with social signals (PaperPing's LLM synthesis and Template+TLDR) were rated as being significantly more helpful for conveying relevance to the group compared to the TLDRs, which only summarized paper content ($p < 0.001$ for both comparisons; see Figure 4.7).

In the interviews, participants shared how social signals helped them determine a paper's relevance. For example, P10 and P11 mentioned inferring paper relevance by using the importance of the linked paper (i.e., similar or cited by the recommended paper) and the presence of shared authors highlighted in the explanation. Others also mentioned using the channel engagement of the linked paper highlighted in the explanation to determine paper importance: "*three thumbs up versus one neutral reaction makes a difference*" (P4). P17 noted that social signals could be more helpful than paper content summaries in determining relevance: "*I only need to glance at the people who are mentioned; I don't even need to go through the summary of the paper to know how relevant this paper is to me.*" This can be especially helpful for junior scholars like P21, who have less expertise on the topic compared to other channel members: "*I don't have as much background on the main topic of the channel, so social signals were helpful for me in understanding why it'd be relevant for me to read.*" As P21 summarized: "*social signals provided a more contextual way of*

understanding how the recommended paper related to a past paper."

Social signals also foster group members' trust in PAPERPING's suggestions. P8 offered that social signals provided transparency into PAPERPING's actions: "*the fact that there are correct links to who reacted to or who suggested what helps you to trust that...the information is being considered in an accurate or relevant way.*" The association with specific group members highlighted in the explanations made P17 more receptive to PAPERPING's suggestions not immediately relevant to them: "*if PAPERPING mentioned a person who's not in my small group or who I don't work with, I know this paper is not relevant to me but is relevant to the larger group.*"

PAPERPING's LLM synthesis is informative and natural but could lead to unreliable interpretation Participants thought PAPERPING's LLM-synthesized explanations could introduce paper content in personalized and contextualized ways and surface connections that might otherwise be missed. P3 appreciated the LLM explanations that highlighted the novelty of the recommended paper beyond its findings. P5 and P13 valued the explanations and connections with their interests that they had not previously considered: "*That made me decide that I could take a look at the recommended paper*" (P13). However, some participants found LLM explanations to be too speculative or ambiguous. P10 noted that LLM synthesis could overly compress information, making the explanations unclear. P12 also found LLM explanations to be too generic: "*It highlighted the similarities between two papers but can be too broad, like 'boosting computer visual models.' That's what every single paper is about.*" Indeed, the usefulness of social signals relies on the prerequisite of having a clear summary of the paper's content. This is evident in our offline evaluation, where the template condition (Condition 2) was consistently rated lower than both PAPERPING's LLM synthesis (Condition 4) and Template with TLDR (Condition 3) across all evaluation questions except conciseness ($p < 0.001$ for all comparisons; see Figure 4.7). It was also rated lower than TLDR (Condition 1) on all questions ($p < 0.001$) but conciseness and

helpfulness in assessing relevance to the group. For many researchers, a clear summary of the paper is considered to be a "*more reliable source of information*" (P7) compared to inferred social signals (P1, P2, P7, P15).

Participants also noted that PaperPing's awareness of individual and group context could still be insufficient. For example, P10 found that explanations occasionally highlighted connections that were misaligned with their interests: "*it has its own understanding of why and sometimes that leads to inaccuracy or irrelevance.*" P17 noted that PaperPing could miss group context cues and offer redundant information: "*Everyone in the group already knows [popular machine learning model] and has been following the different versions. It's just like announcing a new iPhone; you don't really need to introduce the iPhone 14, iPhone 13 before that. You can just say, 'Here's the new iPhone 15.'*" Participants P6 and P18 observed that the LLM's interpretation of emoji reactions in explanations could be awkward and inaccurate: "*It feels like it's trying to derive the conclusions when not having enough data points.*" (P1); they would prefer to see the exact emoji to better understand the nuances of the reactions, especially given the diverse norms of emoji usage across groups. Due to these uncertainties in LLM interpretations, participants, including P10, preferred the TLDR+template explanations (Condition 3) because "*It felt a little bit more objective.*" P14 also preferred forming their own interpretations after reading a paper instead of relying on PaperPing's interpretation.

Challenges involved in translating social explanations to personal relevance Though social signals can help contextualize a paper's relevance to the group, participants P1 and P19 noted the challenge of translating them to their personal interests. As P19 explained: "*If I decided to open up that paper, those reasons are all related to myself.*" Seeing PaperPing's social explanations related to other group members could help them infer the paper's relevance to themselves, but P1 considers it challenging to do the translation: "*We have very diverse interests, and people have different standards for liking. I don't know how they relate to my own standards.*" Indeed, the utility

of social signals in explanations varies across individual members in groups. P17 and P22 found references to past papers confusing if they had joined the channel later or forgotten the shared context: "*To me that shared context was lost*" (P17). P22 highlighted that "*shared context from working in the same team towards the same goal*" was essential for them to find information about relevance to specific members to be helpful: "*if I had not been there in person or if I didn't know these people, then I don't know how much it would help.*"

For most groups, PAPERPING does not disrupt existing social practices (DG2)

PAPERPING generally fits in well with a group's dynamics, as more than half of the participants believe (54% strongly agree or agree). 86% of participants indicated they would recommend PAPERPING to others, and 75% expressed a desire to continue using PAPERPING in their channels. In fact, 11 out of the 18 participating channels are still voluntarily using PAPERPING as of this writing, more than six months after the conclusion of our study. As P4 described: "*PAPERPING feels natural to me. It doesn't disrupt the natural flow of the conversation in the channel.*" P1 described PAPERPING as "*an invited robot guest in the group's space, adapting to groups' norms rather than trying to make the group adapt to it.*" They elaborated: "*It doesn't really feel out of place. It recommends things in a message and links back to previous threads just like any other member.*" P23 echoed the sentiment: "*PAPERPING's behavior didn't interfere with anything. It didn't disrupt the natural conversation.*"

P4 appreciated the pace of posting: "*The frequency is nice. I didn't feel like I was annoyed by the bot or anything.*" P8 and P11 noted that PAPERPING's contextualized, natural language explanations made it feel more human. Participants including P16 and P21 also liked the concise and stylized summary with key information bolded: "*I like bolded text in the explanations. [It is] especially helpful for Slack messages, allowing me to skim it first and then...decide if you want to read it.*" (P21) However, other participants found PAPERPING's explanations could be verbose: "*If there's a huge chunk of text in Slack, I tend to glance at the first sentence or maybe word. If it's not important, then I*

just dismiss it." (P15). The chunk of text could potentially bury human conversations if not posted at an ideal time (P23).

PaperPing fosters common ground within groups (DG4)

Social signals help facilitate the establishment of common ground PaperPing not only suggests useful and relevant information without disrupting channels but also plays a social-facilitator role by exposing group members to each other's interests and fostering common ground within groups. Participants noted that social signals, such as tagging relevant members and referencing prior discussions and engagements, could lead to potential follow-up interactions and foster group cohesion. As P14 reflected, "*The social signal is the strongest aspect because it shows relevance to me or to my group, which helps me understand my group pattern to some extent.*" Participants appreciated how PaperPing's social signals facilitated group members learning more about each other's interests. As P3 shared, "*The recommendations and the social signals helped me realize my group members were interested in this type of research.*" Integrated into existing social spaces and practices, PaperPing encourages researchers to develop further interactions among each other. As P14 mentioned, "*PaperPing is a part of a chat group with other members. I get to understand the interests of other people. It's a part of a practice that I already have in real life, and it has a lot of potential to assist and support in connecting better with [each] other.*" P10, a member in a large research lab channel, also sees PaperPing's explanations as opportunities to develop long-term collaborations: "*Just having it [PaperPing] in the group is helpful for me to know what other people's interest is. Even if the interest is slightly different, seeing some new things from related fields is helpful for me to learn what other people are working on, common things that we work on, and things that we could be chatting about.*"

PAPERPING reinvigorates activity in inactive channels but does not necessarily establish new norms Several participating channels were lab groups that wanted to encourage paper-sharing but have not to date been successful. In these inactive channels where there was little conversation among members, PAPERPING helped revive activity by posting recommendations and tagging members to prompt engagement. Participants felt that PAPERPING "*Keep[s] the channel alive*" (P3, P14). As P14 described: "*Whenever there was a new message, I would look at it.*" P10 observed that PAPERPING prompted more contributions from channel members: "*The channel hasn't been active at all for half a year or more. PAPERPING definitely prompts some other people to post. At least I'm seeing two or three other people who have posted on this channel.*" P8 also thought that PAPERPING served as a reminder for other human members to engage: "*It reminded people to send paper recommendations. We have always been encouraged to do that, but people tend[ed] to forget previously. Seeing PAPERPING's recommendation every two days in the channel reminded people.*" (Figure 4.8(a)). P13 added that PAPERPING incentivized them to contribute more: "*It made me feel my reactions or activities in the channel are being seen and are going somewhere, which gets me excited to contribute. It will be captured by PAPERPING, and sometime later it is going to give me a reward.*" Participants also noted that tagging specific channel members is an effective way to keep people engaged; P5 usually skims through PAPERPING's posts, but "*If my name is tagged, I would read a little more into it.*"

However, participants also noted that PAPERPING did not establish new norms, such as having discussions about papers in groups, that were not previously established practices. Only 25% of the participants self-reported often engaging with PAPERPING's recommendations, compared to 50% of the participants who self-reported often engaging with other human recommendations (Figure 4.6). P15 observed that although PAPERPING was helpful in suggesting relevant papers, this did not lead to more paper discussions in the channel. Several other channels also

show a similar engagement pattern with Channel #4, to which P15 belongs (see Figure 4.8(c)). These channels often have little social history or established norms, with several created right before the study or inactive for an extended period of time. Instead of fostering a social space, these groups used PaperPing to create a feed of paper recommendations, which members checked regularly but only engaged with passively. The lack of extrinsic incentives could be one reason for less engagement with PaperPing's recommendation. Although we disclosed PaperPing's feedback mechanism to the whole group upon installation, P8 noted that some group members could be unaware that their engagement could inform future recommendations and thus were not incentivized to engage. Even when aware of the mechanism, participants could still be hesitant to engage because of the lack of potential follow-up interactions with PaperPing: "*I know PaperPing won't respond.*" Participants also agreed that respected human members were more effective at encouraging engagement than PaperPing alone. P4 observed that their channel became more receptive to PaperPing's recommendations and had more engagement after their advisor reacted to one of the posts.

4.6 Discussion

Our evaluations of PaperPing showed that an agent that is augmented with Social-RAG can deliver contextually relevant and socially aligned messages. PaperPing's strategies to convey relevance to groups, such as referencing prior discussions and highlighting group members' interests, effectively contextualized the message to suit group social norms (DG1). Its ability to learn interests based on past paper-sharing history and incorporate ongoing feedback via emoji reactions, comments, and shared links let users provide continuous feedback with minimal effort (DG3). Users' interactions with PaperPing (e.g., receiving messages in group chats) naturally suited their existing social practices (DG2) and helped them understand other group members'

interests, occasionally even reviving group activity (DG4).

Nonetheless, participants expressed interest in even more nuanced levels of social awareness, such as improvements in distinguishing between group and individual preferences (DG1), tailoring the verbosity and tone of messages based on group norms (DG2), and enabling more group-specific feedback mechanisms (DG3). Based on our findings, we discuss the implications for future systems that plan to support socially nuanced text generation.

4.6.1 Bootstrapping Social Grounding with Past and Ongoing Group Interactions

The use of past and ongoing group interactions as a basis for social grounding is a core advantage of Social-RAG. In contrast to explicit preference elicitation, which requires significant user effort [169] and risks disrupting group dynamics [6], Social-RAG infers user preferences from past and ongoing group social interactions without requiring additional user labor. From collected user preferences, or *social facts*, Social-RAG generates contextually relevant messages with an LLM. For this, we developed techniques to curate and maintain a social knowledge base from past and ongoing group interactions. For example, the PAPERPING system preserves the connection between extracted items (e.g., shared papers) and their social contexts (e.g., reactions and comments from others) when indexing group conversation history into the social knowledge base. These preserved connections enable further social inference in the Social-RAG workflow. We also proposed complementing the off-the-shelf retrieval methods typically used in traditional RAG systems (e.g., semantic embeddings [131]) with empirically validated heuristics to guide the retrieval and interpretation of social knowledge. This hybrid approach, which combines heuristics with off-the-shelf embedding methods, is also starting to show promise in recent RAG applications in other contexts that require more complex reasoning (e.g., UI linting [145]).

4.6.2 Costs and Benefits of Different Levels of Group Social Grounding

Grounding systems in their social contexts has been highlighted as a major design requirement for both classic social computing systems [3] and recent LLM-powered systems [238]. However, realizing and implementing social grounding effectively remains an open challenge [238]. We contribute to this ongoing discussion by reflecting on the different levels of social grounding of generated text in a group setting based on our experience of developing Social-RAG and implementing PaperPing. Different levels may require different amounts of resources and considerations and could potentially lead to different benefits.

No social grounding. At the lowest level, a one-size-fits-all system could apply uniform behavior across all groups, relying on generic templates or heuristics (e.g., generate template text that is likely acceptable across all settings). This level of grounding is resource friendly and predictable, but it lacks the ability to adapt to specific social contexts.

Category-level social grounding. A system could predefine more detailed rule-based heuristics to generate content for groups based on their broad characteristics, informed by empirical observations. For example, in the deployment of PaperPing, we observed that groups working on active research projects would prefer exhaustively exploring closely related papers on certain topics, but interest-based reading groups would prefer exploring diverse sets of papers. However, it would be challenging for systems at this level to adapt to nuanced differences in similar categories.

Group-level social grounding. Further, a system could tailor its output to group-specific preferences and norms. In Social-RAG and PaperPing, we demonstrate that past and ongoing conversations in a group could effectively inform AI systems about the group's implicit topical preferences. A system at this level could also consider adapting the cadence and tone of generated messages to make them less intrusive to groups.

Individual-level grounding. Finally, a system could target and tailor its messages to specific

group members (e.g., via mentions). While individual-level grounding could increase engagement and encourage understanding among group members, a system at this level needs to carefully design the balance between individual preferences and group preferences as well as carefully navigate power dynamics to behave in a socially appropriate way.

To more concretely illustrate the above, we can position PAPERPING (i.e., Condition 4 in the offline evaluation) within this framework: selecting which paper to recommend is grounded at the group level by pooling all previously shared papers as the input to a recommender system. This gives group members access to papers of both common and individual interest. The generated explanations were tailored at both the group and individual level, with relevant members being mentioned. This promoted engagement with PAPERPING as well as awareness of specific members' interests among all group members.

4.6.3 The Tension of Group vs. Individual Preferences

As previously noted, a key design consideration for a socially grounded system is how to balance group- and individual-level personalization. Empirical observations from our 3-month deployment showed that within the same group, individuals might have different preferences for explanations relative to each other or even prefer different messages. For example, members who are more familiar with a recommended paper might prefer explanations that are more in-depth compared to other members.

One way to address this tension is by aligning the generation with an "average group" preference. Future systems could consider experimenting with different ways to aggregate individual preferences that optimize various goals. Another option is to scaffold the explanation, e.g., starting with a more generic explanation tailored to all members and following with in-depth explanations tailored to individual members. However, if such modifications are made, the potential risk of

diminishing common ground among members must be carefully considered (DG4).

4.6.4 The Impact of Proactive Agents on Group Social Dynamics

Another key design goal of a socially grounded system is ensuring that AI agents enhance rather than crowd out desirable human-human interactions. In our deployment of PaperPing, we observed varied impacts on group dynamics. In some cases (Figure 4.8(a) and (b)), PaperPing had a positive influence. Channel members became more active in sharing and reacting to papers or offloading paper-sharing to PaperPing while continuing to engage through reactions or comments. This suggests that AI agents can play a beneficial role in group spaces by taking on tedious tasks while preserving human engagement. However, as seen in Figure 4.8(c), the presence of PaperPing alone did not automatically foster new social norms (e.g., active paper discussion), particularly when human members primarily valued receiving recommendations over engaging in conversations. In some cases (Figure 4.8(d)), a mismatch between PaperPing's scheduled posts and the natural pace of group interactions could overwhelm the channel, leading to disengagement and abandonment of the space. Overall, these observed patterns point to a promising future of AI agents taking on pro-social roles in groups but highlight the importance of designing agents that can flexibly adapt their content and behavior (e.g., pacing and timing of posting) to different group dynamics.

To complement the interview and questionnaire findings from the field deployment, we conducted an extended usage-log analysis of PaperPing interactions across research channels (December 2023–March 2025). The analysis distinguishes clicks, emoji reactions, and replies as a spectrum from passive interest to active discussion, and compares engagement with PaperPing recommendations versus human-shared papers. In brief, PaperPing often functioned as a discovery feed: clicks were common, while visible reactions and replies remained rarer than for human

posts. Deeper engagement concentrated when recommendations aligned with a group's active research interests. Timing analyses also show that attention is not limited to the first day after a post: median matched click time was nearly eight days, with a later uptick in the 60–90 day range. Full methods, dataset summary, and design implications appear in **??**.

4.7 Limitations

While our participants found PAPERPING useful in real-world online chat groups, our field deployment also highlighted limitations of our current implementation of Social-RAG that can inform future work. Specifically, observable social interactions in conversation history may not fully capture group dynamics or offline interactions, potentially leading to biased or inaccurate interpretations of preferences and norms. For example, in less active groups, PAPERPING had fewer social signals to work with (e.g., shared papers, user reactions, and comments) and produced lower quality recommendations and explanations. Similarly, as group dynamics evolve over long periods of time, the relevance of past interactions may become stale or diminish even for active groups. Some participants reported finding it hard to shift to new topics, and some even created a new channel to indicate new interests to PAPERPING.

4.8 Future Work

While this work focused on leveraging social signals to tailor the right information with the most relevant social context, future work could also consider tailoring the tone of explanations, level of detail, group posting cadence, and other aspects of delivery. Similarly, the current implementation of PAPERPING focuses on aligning with a group's topical preferences, but one could also extend the workflow to account for social hierarchies. The challenge would be to identify relevant social

signals from the group's interaction history and use them effectively as context for LLMs to generate socially appropriate messages.

Another area of future work would be to explore more flexible ways to adapt to a group's communication norms. For example, we currently hard-coded how the system interprets emoji feedback from users (e.g., a thumbs-up emoji indicates positive interest in the paper), a method informed by the norms observed in the formative interviews. However, our deployment study showed that a few individuals or groups did not follow this assumption (e.g., using a thumbs-up emoji only as an acknowledgment). Future work could explore how agents could more flexibly interpret social signals based on observation of groups' past social interactions. In sum, the flexibility to ground different aspects of a system at varying levels offers opportunities for creating more socially aware and contextually relevant AI agents. However, it also requires thoughtful design choices and a deep understanding of the social dynamics within target groups.

As with any system designed to be embedded in social spaces, learning from interactions introduces privacy concerns. One way to mitigate these is to limit the scope of what the system is allowed to learn from. For example, we made the conscious decision to allow PaperPing to access user interactions only on messages where a research paper was shared. This is a trade-off between privacy concerns and how well our system can learn from interactions between members. Future work could provide greater user control over which aspects of their interactions are used to inform AI-generated content, potentially through customizable privacy settings or explicit consent mechanisms.

Further, PaperPing does not currently allow users to manually insert into or edit the collected social signals. Future work could explore ways to encourage and facilitate user engagement in less active groups as well as complement Social-RAG with direct elicitation techniques to support scenarios where the group needs to make more drastic changes to its preferences (e.g., when switching focus to a new project).

Finally, Social-RAG could be explored in other collaborative spaces and scenarios that are rich with social interaction histories. For example, in Reddit communities, a misinformation correction bot empowered by Social-RAG could generate explanations suited to each community's norms in order to convey trust. Specifically, when retrieving external documents, the system could learn about the types of sources (e.g., governmental vs. journalistic) that are often shared and trusted by the community, as indicated by the sentiment of replies. When generating posts, the system could learn from successful debunking strategies that were used in the past by members. Another example of a Social-RAG system could be a meeting summary tool that can generate summaries and tag the most relevant team members for awareness. The tool could analyze past documents to understand team members' interests based on signals, such as contribution history and viewing frequency, or analyze meeting information, such as speaking frequency in discussions of different topics or speaker roles.

4.9 Conclusion

This work introduced a novel LLM workflow, Social-RAG, that enables AI agents to ground their generation in the social context of group interactions. By retrieving and using "social facts" from past and ongoing group interactions, Social-RAG enables AI agents to dynamically adapt their outputs to group norms and preferences with minimal user effort. We instantiated this workflow in PAPERPING, an AI agent that recommends scholarly papers and generates contextually tailored explanations in research group chats. Our 3-month deployments of PAPERPING in 18 Slack channels demonstrated Social-RAG's ability to adapt to diverse group dynamics, foster trust and common ground, and minimize disruption to existing social practices. Overall, our work demonstrates Social-RAG's potential to address the sociotechnical gap in designing adaptive and socially aware AI systems by introducing strategies to retrieve implicit group dynamics and generate contextually

aligned content. Social-RAG moves us closer to more natural and effective integration of AI agents in collaborative environments.

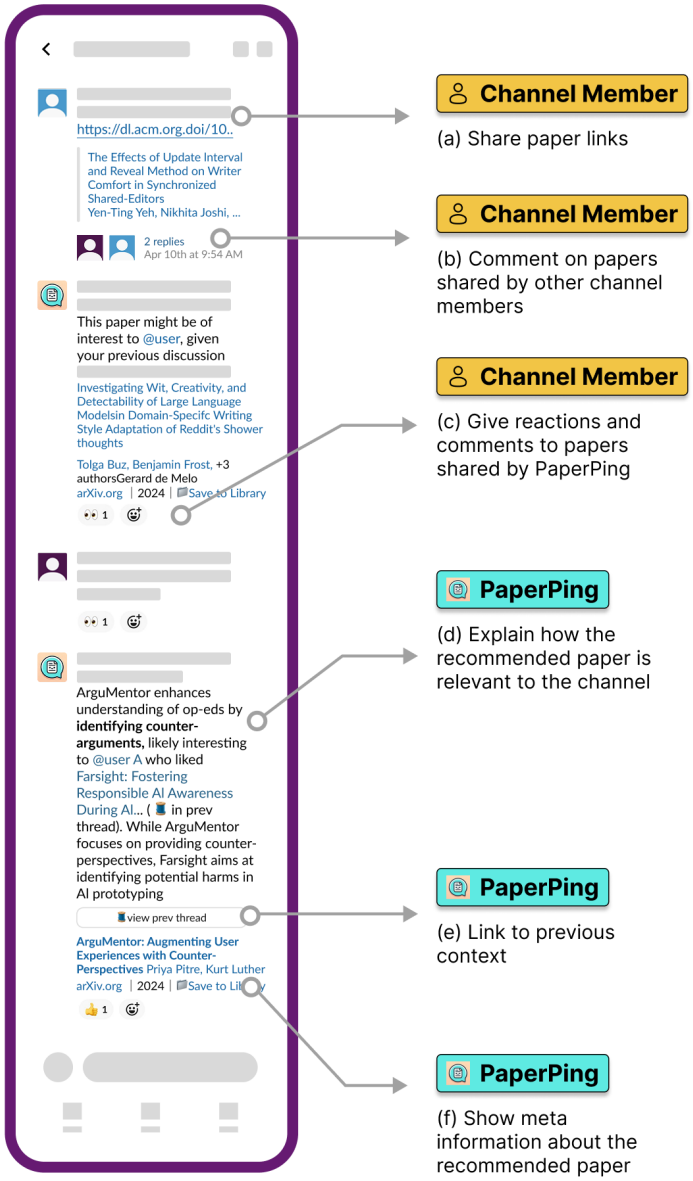


Figure 4.2: **How users interact with PAPERPING.** Channel members share paper links, react to papers shared by other members or PAPERPING, and comment on shared papers. After gathering and processing this information, PAPERPING sends contextually grounded paper recommendations with explanations of how the recommended papers are relevant to the channel. It also provides links to previous related discussions and includes meta-information about the recommended papers.

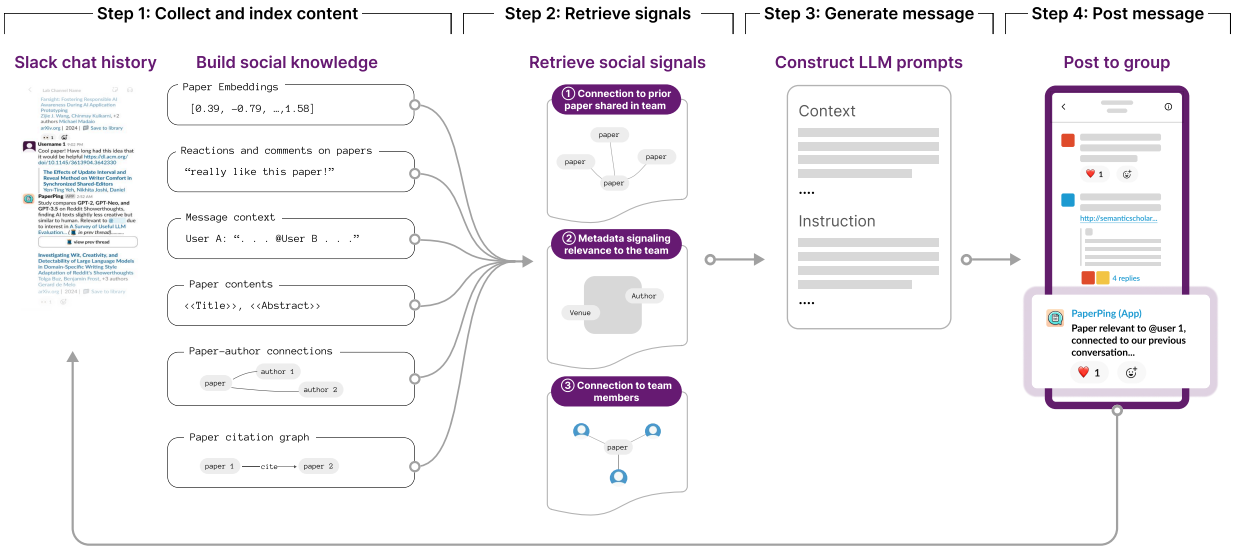


Figure 4.3: **The PaperPing implementation, using the Social-RAG workflow.** PaperPing extracts relevant social information from two data sources: the Semantic Scholar knowledge base, providing paper content, paper-author connections, and citation graphs; and Slack chat history, which includes previously mentioned papers, emoji reactions, comments, and user interactions (Step 1). PaperPing then retrieves relevant social signals from the data using three heuristics, developed over a three-month iterative design process: relevant metadata, the most similar and discussed prior paper in the chat, and the member most likely to be interested (Step 2). Next, PaperPing uses chains of LLM prompts to turn the retrieved social signals into a natural language explanation (Step 3). Finally, PaperPing posts the message to research group channels, where group members can provide emoji reactions or reply to the message to give feedback (Step 4).

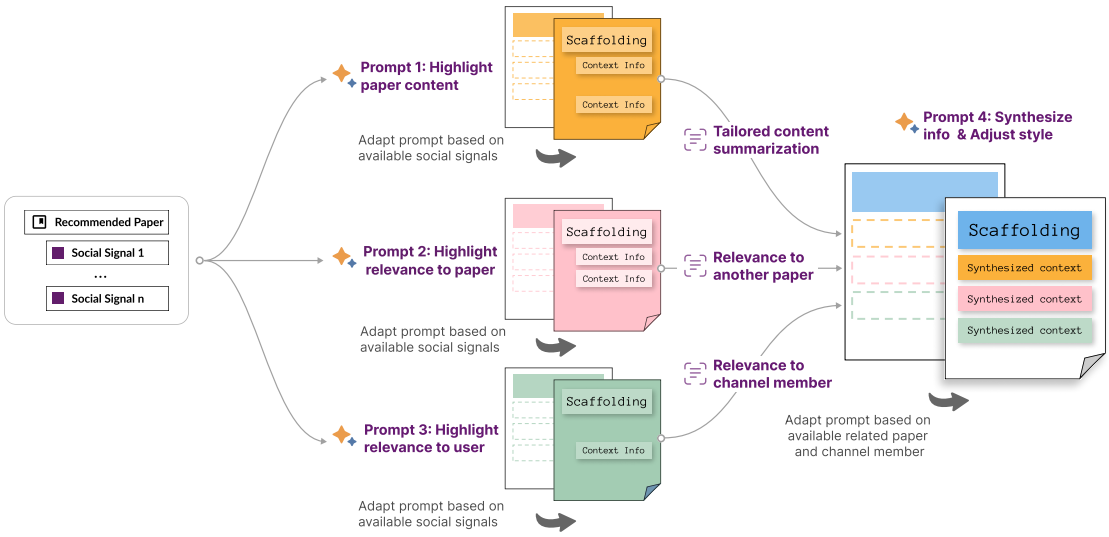


Figure 4.4: **PAPERPING prompt pipeline (Condition 4).** The pipeline starts with social signals leading to three prompts: Prompt 1 highlights paper content; Prompt 2 highlights relevance to a previous paper; and Prompt 3 highlights relevance to a channel member. Finally, outputs from these prompts feed into Prompt 4, which synthesizes the information and adjusts the style, resulting in the final output.

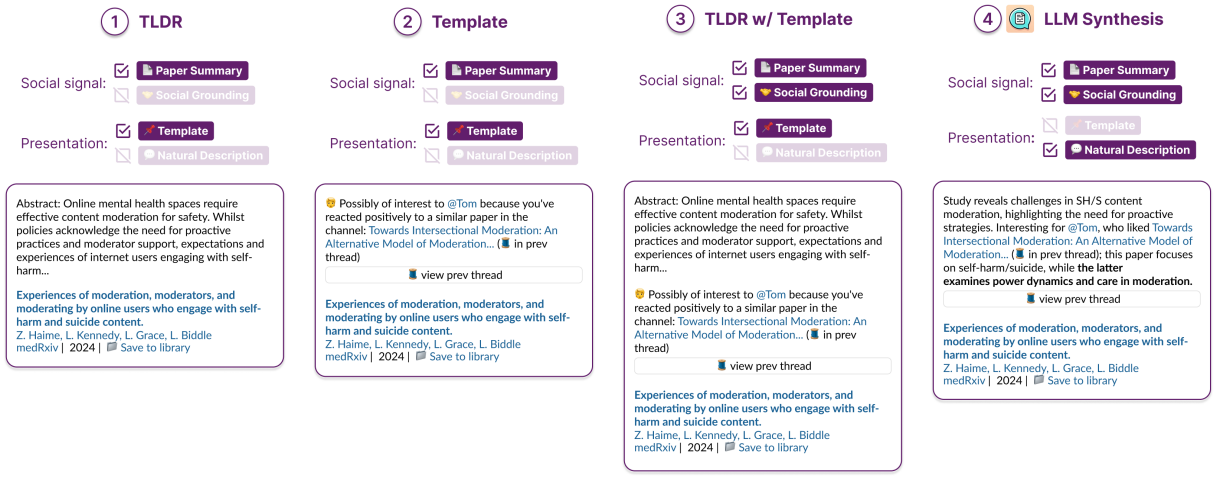


Figure 4.5: **Four PAPERPING conditions shown in the offline evaluation task and example explanations.** Names are anonymized.

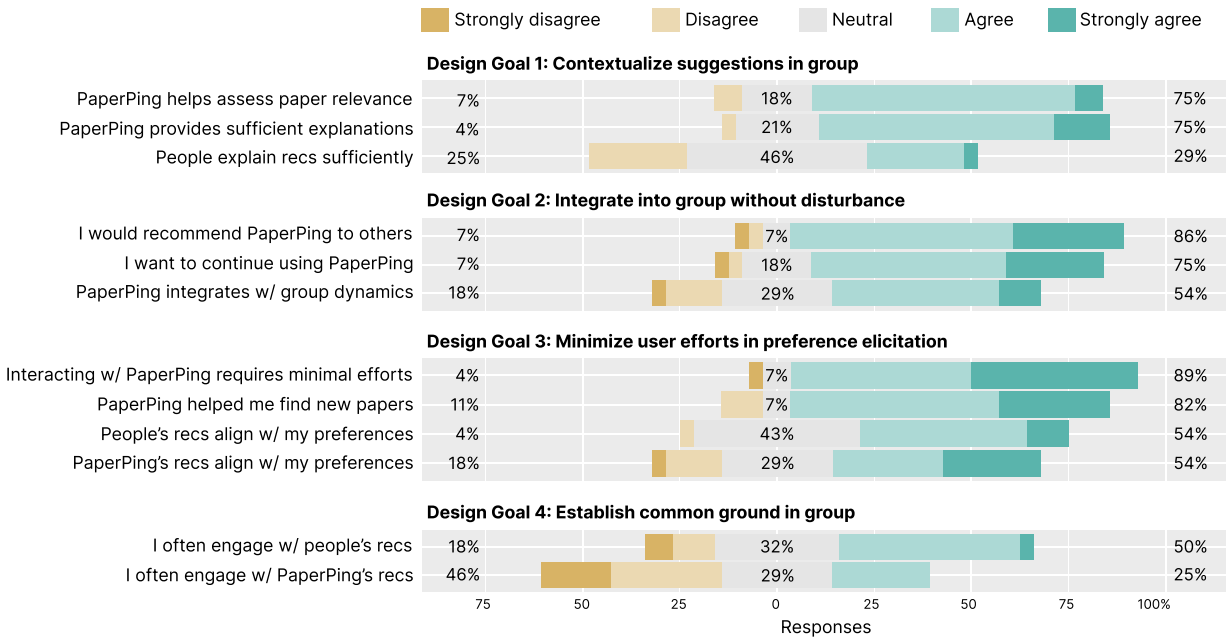


Figure 4.6: **Results from post-study questionnaire.** Responses are grouped based on the four design goals. We present the original questions posed by the questionnaire in the appendix.

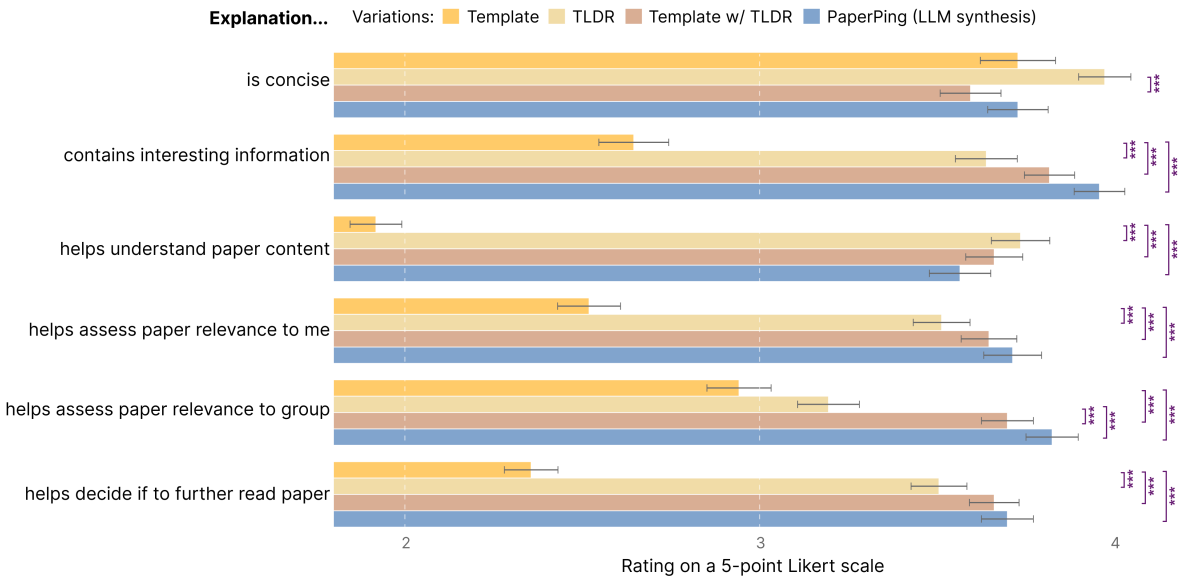


Figure 4.7: **Results of our offline evaluation.** Error bars show standard errors of the mean (SEM). The "template" messages (Condition 2) were consistently rated lower than PAPERPING's LLM synthesis (Condition 4) and Template+TLDR (Condition 3) across all questions (p<0.001), except for conciseness. The "template" messages (Condition 2) were also rated lower than TLDR (Condition 1) across all questions (p<0.001), except for conciseness and "explanation helps assess paper relevance to group." Additionally, explanations with social signals (PAPERPING's LLM synthesis (Condition 4) and Template+TLDR (Condition 3)) were rated significantly more helpful on "explanation helps assess paper relevance to group" compared to the TLDRs, which only summarized paper content (p<0.001 for both comparisons). We observed no statistical significance in other comparisons.

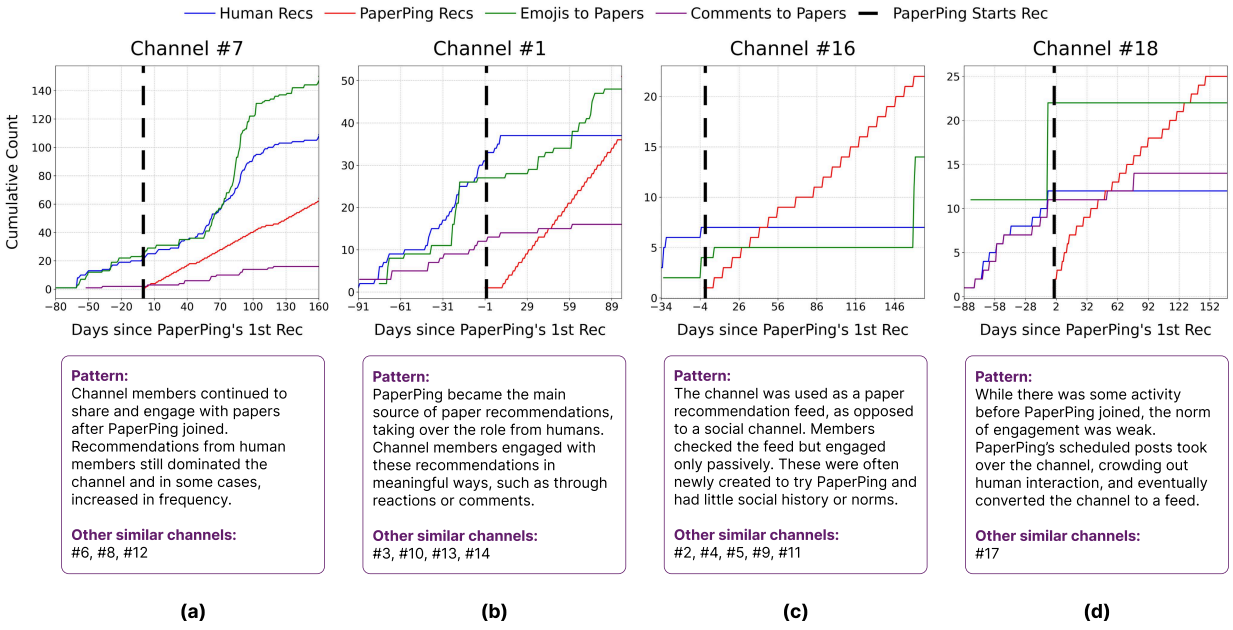


Figure 4.8: **Representative patterns of PaperPing usage during the deployment study.** The y-axis represents the cumulative counts of usage in each channel, and the x-axis represents the days since PaperPing's first recommendation. Only activities around papers shared via common academic platforms (e.g., ACM Digital Library, Semantic Scholar) are captured. We were unable to track papers shared through social media links (e.g., X, LinkedIn posts).

Chapter 5

ProSKI: Measuring Whether AI Models Know When to Contribute Knowledge to a Group Conversation

5.1 Introduction

AI systems have, until recently, spoken only when spoken to: assistants answer only when users ask, and coding agents execute only when users delegate. In these scenarios, users first express intentions through conversational or graphical interfaces; the AI system then acts on its understanding of that intention. More recently, agents have gained the ability to proactively plan and to expand a single request into a sequence of steps, which frees users from spelling out instructions turn by turn [254]. But even so, users still have to decide at the outset whether the agent should act at all. Giving instructions when users interact with agents individually is relatively easy, since the user is already attending to the task. However, it is far more expensive when the agent sits where several people are already interacting: human attention is occupied by

following what others say and holding a position in the exchange, so stepping out to instruct the agent necessarily distracts users from participating. Recent work therefore explores proactive *group conversational agents* that participate without waiting to be asked [256], holding candidate contributions as a conversation proceeds and deciding when one is worth voicing [142], or intervening in multi-party negotiation [144].

Agents' decisions about when to act can directly affect the dynamics of human groups. Like other mixed-initiative systems [92], an agent must weigh the value of acting against the cost of interrupting. In particular, a message posted into a shared channel demands the attention of everyone in it. While providing potentially valuable information, a message arriving too early diverts the discussion, and a message arriving too late makes the contribution irrelevant. Agents in conversation groups thus have to contribute at the right moment to spare a group the work of noticing and asking [256]. Recent work formalizes a model's ability to determine the timing to speak as a speak-or-stay-silent decision at each turn [180, 40, 163]. However, current evaluations typically synthesize conversations around an intended intervention or plant an explicit trigger, so the right moment is a property of the construction procedure and a model scored against them is being asked to recover a planted moment. Since real group conversation carries nuance that simulation does not reproduce [269], assessing timing judgment requires real conversations scored against the perceptions of the people who held them.

In this paper, we focus on one class of contribution for which timing is especially consequential: bringing a group external information relevant to the topic being discussed. Whether the collaboration is professional (a research discussion) or everyday (planning a trip), discussions routinely touch facts that are *nice to have* but that nobody present knows (e.g., what a specific paper found, what a hotel costs). Groups usually absorb such gaps, assuming something plausible and moving on, because stopping to look something up is costly and disrupts participation. An LLM with external knowledge and tools can retrieve such information and offer it in near real time,

filling the group's *external blind spot*. However, such information helps the group conversation only if it is provided at an appropriate time.

We therefore built ProSKI, a dataset of 384 candidate agent contributions drawn from 35 unscripted conversations among real groups discussing research topics and everyday planning. Each contribution was written by a participant in the conversation, who could flag a blind spot as the discussion unfolded or afterward and then compose the corresponding message. Each message was then placed by every member of that group, first independently and then in deliberation. Each annotator named the turn at which the contribution should have appeared and the range of turns over which it would still have been acceptable. Against this dataset we evaluate 15 models in 28 configurations under three task framings differing in how much of the conversation a model sees before committing, and therefore in how closely each approximates deployment: from a turn-by-turn decision with no lookahead to a retrospective choice over the whole conversation. We ask (i) what people expect of good timing and how much they agree; (ii) how well current models' judgments match human expectations; and (iii) when models misjudge, what their justifications weigh at the moment they commit.

Human expectations turn out to form a distribution rather than a point: even after deliberation, annotators land within two turns of one another only about half the time (54%), and only 29.7% of messages have a single moment every annotator independently named. Models place contributions about as well as a human group member does given hindsight, but that competence does not survive deployment conditions: under the realistic framing, accuracy falls to a third of its retrospective value, with 46.4% of predictions firing before any annotator's window has opened against 1.0% arriving after every window has closed, a mean drift of −11.1 turns against a median conversation length of 52. And the reasoning traces show what models weigh when making timing judgments: in 92.8% of firing justifications the model argues only that the content fits at the current turn, while an explicit weighing of whether a later moment would serve better appears in 0.14%.

In summary, we contribute (i) **a benchmark for group social timing in proactive agents** (Sections 5.2.2–5.4) whose timing labels come from the judgments of the people who participated in the conversation; (ii) **an empirical characterization of human timing expectations** (Section 5.4), with a metric design that reflects inherent ambiguity of the judgment (Section 5.3); (iii) **an evaluation of 15 models under three framings** (Section 5.5); and (iv) **an analysis of the failure mechanism** (Section 5.5.1), showing the error is item-specific rather than a global threshold offset and that extended reasoning changes alignment only by moving where the firing threshold sits.

5.2 PROSKI BENCHMARK

PROSKI evaluates how well LLMs align with people on when to contribute external knowledge to a multi-party group conversation.

5.2.1 Task Definition

Whether a moment in a group conversation is right for an AI contribution depends on whether a gap exists at all, whether the agent can find the needed information, and how the message is written. To evaluate models' ability to judge the timing of contributions, we hold the compounding factors constant. Concretely, the model receives a group conversation together with a candidate message already validated as something the group could have used, and its only decision is when, if ever, to surface it. Formally, a **conversation** $C = (u_1, \ldots, u_T)$ is a human-only turn sequence from a real discussion; the agent observes it and the conversation proceeds identically whether or not it contributes, since we evaluate AI's first contribution only. Turns are the unit of timing because the conversations are text-based in this dataset. A **candidate message** m is a concrete agent utterance filling an information blind spot the group had, authored by a participant in that

conversation (Section 5.2.2) and checked by the researchers to be sensible. A **timing judgment** supplies a *best turn* t_i, the single ideal turn to insert the message after, and an *acceptable window* $[s_i, e_i]$ running from the earliest turn at which the message becomes relevant to the last turn at which it is still useful; annotators judge retrospectively, after their conversation has concluded. An annotator may instead mark *do-not-surface*, meaning the message should not be contributed at any point.

5.2.2 Collecting the ProSKI Corpus

Knowing when a contribution belongs is something conversationalists do constantly and can rarely explain [202, 172]. A usable judgment must therefore be anchored to a real conversation and a specific candidate message that could have entered it. The conversation cannot be simulated because real conversations have content and dynamics that simulation cannot mimic [269], and generation smooths away the material the judgment concerns [24, 107]. These considerations lead to our study setup: in a 90-minute collaborative session, a small group produces the conversation, the messages, and timing judgment about them.

Specific steps to collect the dataset. Participants have a 15–25 minute text conversation in groups of 2–4 on a platform we built for data collection (Fig. 5.1). The platform enables participants to complete all study steps (i.e., have conversations, write messages, and annotate messages) with scaffolded AI support and task instructions. The conversation topic is sampled from one of two settings where a group is likely to encounter information blind spots: *open research discussion*, where the useful external knowledge is literature and evidence, and *collaborative planning* of a shared trip or event, where prices, logistics, and options could benefit from external sources. Participants are assigned to groups based on their topic preferences, so they are invested in the discussion and can identify genuine blind spots.

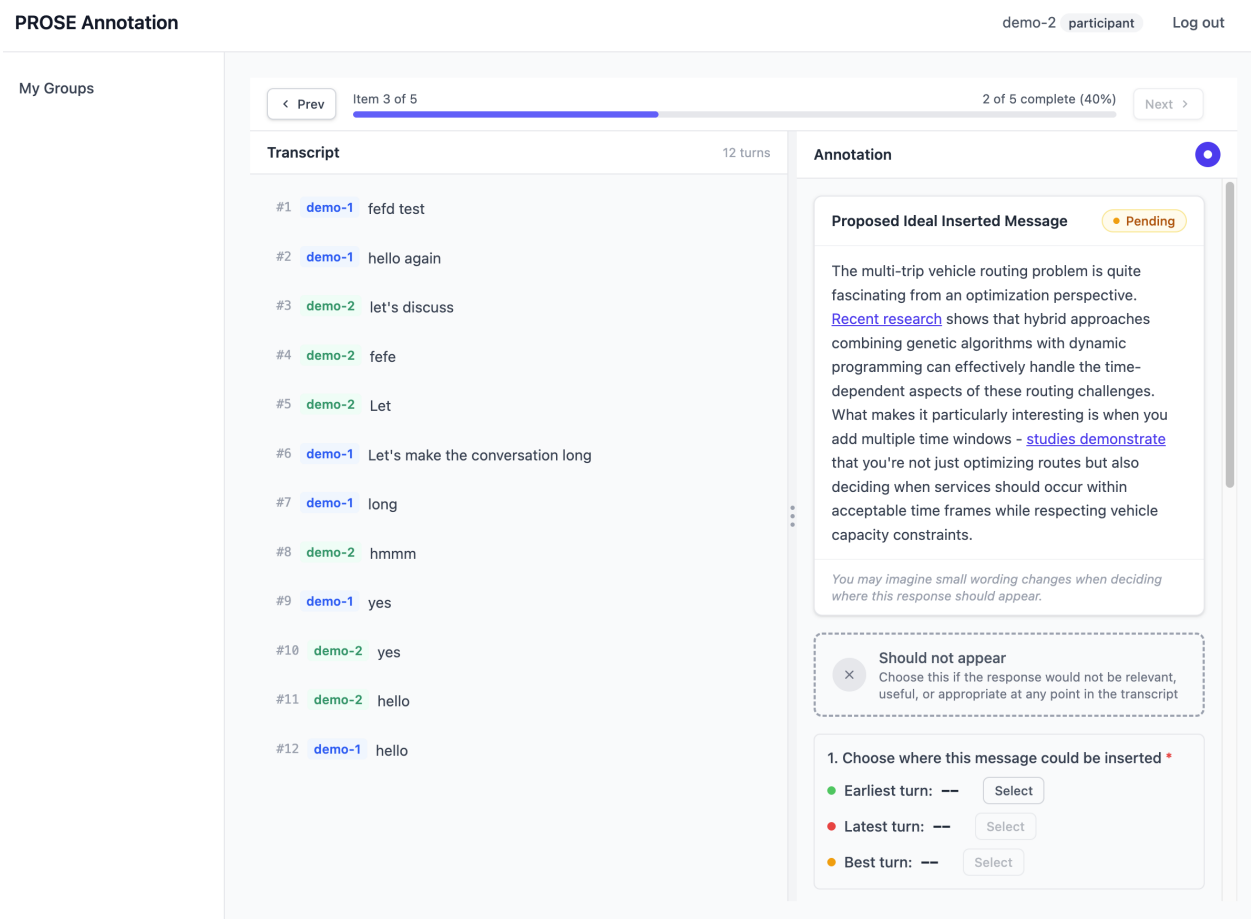


Figure 5.1: Placement interface. The full transcript sits in a scrollable pane on the left and the candidate message and placement controls on the right.

After the conversation, participants review the conversation transcript and write the message their group's blind spot actually called for. The study platform provides search and drafting support so participants can author realistic messages that close the gap and read as if a group might receive them from an agent, within the limited study duration. Each annotator then supplies a best turn and an acceptable window for every message or declines to place it ("not relevant, useful, or appropriate at any point in the transcript"). We observed in pilot sessions that annotators were often confident about roughly where a message belonged and much less confident about exactly when, so we collected both a best-turn judgment and an acceptable window.

Statistic	Value
Conversations / turns	35 / 2,194
Conversation length (mean / median / range)	62.7 / 52 / 16–227
Annotators	96
Candidate messages	384
Timing judgments (one per message–annotator)	1,091 (384 author, 707 peer)
distinct annotators per message	2–4 (median 3, mean 2.8)
placements / do-not-surface	1,047 / 44
Best-turn spread (median / mean / p90)	2 / 6.5 / 19 turns
Messages with a unanimous best turn	114 (30%)
Individual / union / unanimous window (median)	7 / 11 / 3 turns
Models / configurations / API calls / spend	15 / 28 / 77,755 / $507.83

Table 5.1: The ProSKI release. Every message carries the judgments of the people who were in the conversation it came from. A *judgment* is one person's verdict on one message, counted once however many times they revised it. Union window size is the size of the union *set* of endorsed turns, not the span between its extremes.

Finally, every message is placed by everyone in the group that held the conversation, yielding 1091 judgments: 384 author placements made when the message was written, and 707 by the 1–3 peers who were in the conversation. Peers commit independently, then see one another's choices and may revise their own original selection. 578 of the 707 peer judgments (82%) were revisited; 85% of those revisions left the best turn exactly as it was, and the spread that remains is treated as disagreement.

5.2.3 The Resulting Corpus

The resulting dataset comprises 35 conversations and 2,194 turns from 96 participants, yielding 384 candidate messages and 1,091 timing judgments. Each candidate message received a median of 3 judgments and at least 2 judgments (Table 5.1).

5.3 Experiments

To measure how well model judgments align with those of human participants from the conversation group, we define the following metrics and task framings.

5.3.1 Ground truth of human judgments

For a candidate message judged by N annotators, the ground truth of human judgments is the pair

$$\mathcal{B} = \{ t_i \}_{i=1}^{N}, \qquad \mathcal{W} = \bigcup_{i=1}^{N} [s_i, e_i],$$

where $\mathcal{B}$ is the set of turns annotators named as the best turn for the given conversation and message, and $\mathcal{W}$ is the union of turns annotators considered acceptable. We define the ground truth as sets of turns instead of a single point in the conversation to reflect the inherent ambiguity of timing judgment.

5.3.2 Task Framings

We tested 15 models spanning five providers, in standard and, where supported, reasoning configurations, for 28 configurations in total. Each model was tested in all three settings, differing in how much of the conversation is visible before the model commits and thus how realistic each framing is for online deployment settings.

- **Replay**, the most realistic task framing for deployment. The conversation is revealed one turn at a time and the model decides yes or no at each turn. The first "yes" is its committed placement, and it cannot revise.
- **Delayed replay**, an eased version of replay. It is similar to the setup in the replay task, but with a five-turn lookahead at each decision.

- **Pick**, an upper bound rather than a realistic task framing. The whole conversation arrives at once and the model chooses one turn.

5.3.3 Metrics

To quantify model performance in different task framings, we define the following metrics against the ground truth of human judgments.

Metric	Definition
Anyone's Best Turn	The predicted turn is some annotator's best turn. The headline metric.
Near-Miss ($\pm k$)	Within k turns of some annotator's best turn.
Anyone's Window	The predicted turn falls inside some annotator's window.
Unanimous Window	The predicted turn falls inside every annotator's window. Defined on 296 of 384 messages; NA elsewhere.
Timing Drift	Signed turns from the nearest best turn (+ late, − early).

To compare model performance against a human baseline, we also define *endorsement share*, the fraction of the group named that same turn. Endorsement share allows us to compare model performance against a human baseline: model share is $\text{endorsers}(x)/N$ while human share is $(\text{endorsers}(t_i) - 1)/(N - 1)$, the annotator's own vote removed from both sides.

5.3.4 Baselines

For baselines, we define three reference policies: a **random turn** drawn uniformly over the conversation; a **positional prior** placing every message the same swept fraction of the way through its conversation; and **keyword matching** by conversation-local TF-IDF cosine against the candidate message, taking the argmax over the whole conversation offline and, online, firing

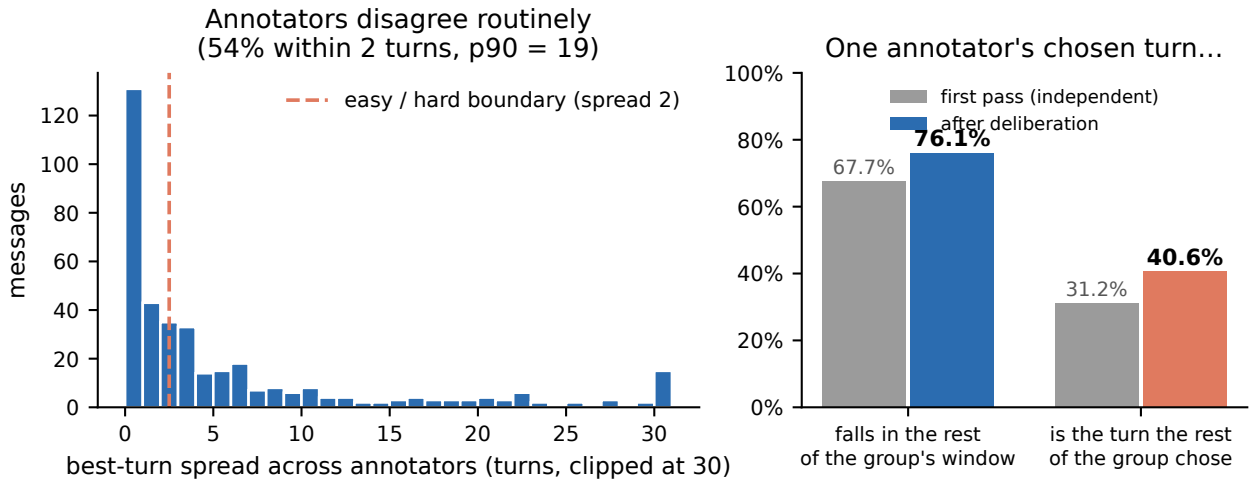


Figure 5.2: Left: how far apart annotators' best turns land, with the easy/hard boundary used throughout. Right: the same annotators scored against the rest of their own group, own vote removed, for both annotation passes. The deliberated bars are the reference standard for every model number in Section 5.5; the independent bars are the comparison for a model that consults nobody.

at the first turn whose similarity clears a swept threshold with the IDF fit causally. Because the sweeps are tuned on the data they are scored against, these are upper bounds on the corresponding shortcut; TF-IDF is a weaker signal than embedding similarity, so the keyword rows bound the shortcut from below.

5.4 RQ1: What People Expect of Good Timing

We first analyze people's expectations of good timing for AI contributions as reflected in PROSKI.

Annotators agree on an acceptable window but disagree on best turn. An individual annotator's acceptable window has a median width of 7 turns, about 13% of a median 52-turn conversation. When it exists, the window that every annotator agrees on is as narrow as 3 turns, under 6% of the median length of conversation (Table 5.1). Pooled across 1,047 placements, people

put these messages in the middle of the conversation, at a mean relative position 0.493, median 0.486, with the middle half between 0.27 and 0.73. There is no rush to the front and no saving for the end. Authors and reviewers also place almost identically, at 0.493 and 0.494.

Annotators disagree routinely on the best turn placement (Appendix Figure 5.2): the best-turn spread, defined as the gap between the earliest and latest best turn, has a median of 2 turns and a p90 of 19. Overall, 54% of messages are tight ($\leq$ 2 turns) and 29% wide ($>$ 5), and only 30% of contributions have a single moment every annotator independently named as the best moment (i.e., unanimous best turn).

To further quantify the disagreement among annotations, we measure each annotator against the rest of their own group (i.e., *endorsement share*). An annotator's chosen turn falls within the rest of the group's acceptable windows 76.1% of the time, but is the exact best turn that the rest of the group chose only 40.6% of the time; it satisfies everyone's window in 63.2% of placements and matches everyone's best turn in 28.6%.

Our analysis indicates that there is a small, widely agreed-upon region where a message belongs, and within it the exact optimum is more a matter of taste.

5.5 RQ2: Model Performance Against Human Judgment

Next, we report results on three task framings (Section 5.3) run on 15 models and 28 configurations.

Models cannot time contributions in the deployable setting. Replay is the task framing that matches deployment. In this task, the model sees the conversation one turn at a time and commits without knowing what follows. Table 5.2 shows that all configurations struggled. Pooled over 28 configurations, predictions name some annotator's best turn on 19.0% of messages and fall inside some annotator's window on 50.5%, against 3.7% and 25.5% for a randomly chosen turn. Only 24.2% land inside every annotator's window. The best configuration (`gemini-3.6-flash` with

Framing	Anyone's Best Turn	Anyone's Window	Unanimous Window†	Median arm	Best arm
Replay *(deployable)*	19.0% [16.5, 21.7]	50.5% [46.8, 54.3]	24.2%	19.0%	32.8%
Delayed replay *(eased)*	22.4% [19.3, 25.5]	52.2% [48.2, 56.1]	27.8%	22.1%	33.9%
Pick *(upper bound)*	57.8% [54.3, 61.2]	92.7% [90.9, 94.3]	73.3%	58.7%	64.8%
Policies with no model in them (each tuned on the scored data; see Section 5.3.4)					
Random turn	3.7%	25.5%	7.6%	—	—
Positional prior *(best fraction)*	6.8%	31.2%	10.1%	—	—
Keyword first-crossing *(online)*	26.8%	47.7%	43.6%	—	—
Keyword argmax *(offline)*	38.5%	75.8%	51.7%	—	—

Table 5.2: Model rows are pooled over 28 configurations, with 95% bootstrap intervals resampling items. Median arm and Best arm are per-configuration values on Anyone's Best Turn; the median arm sits within 0.6 points of the pooled mean in every framing. Best arm is `gemini-3.6-flash` in both replay families and `gpt-5.6-sol` in pick. Baseline rows are point estimates for policies whose sweeps are tuned on the data they are scored against, so they upper-bound the corresponding shortcut. †Unanimous Window is defined on the 296 of 384 messages with a non-empty intersection across annotators and is not comparable to the other columns.

reasoning) reaches 32.8% and the weakest (`gpt-oss-120b` with reasoning) 6.2%.

The failure is specific to deciding online. Unlike in the online task, models perform much better when given the full conversation context in the pick task framing. When models are asked to select a single turn with hindsight, accuracy rises from 19.0% to 57.8% on Anyone's Best Turn and from 50.5% to 92.7% on Anyone's Window. Landing inside *every* annotator's window, which is rare online, becomes much more common (24.2% to 73.3%). Computed per configuration, performance increases on average +38.8 points between the replay task and the pick task framing. Models' performance in delayed replay (where models have five-turn lookahead) improves marginally compared to replay: 22.4% against replay's 19.0% on the best turn and 52.2% against 50.5% on the window. Models' relative performance also shifts across framings. `gemini-3.6-flash` with reasoning is the best replay arm and 26th of 28 in pick, while `gpt-5.6-sol` with reasoning tops pick and sits 8th in replay.

Models match the human reference offline but not online. To contextualize model perfor-

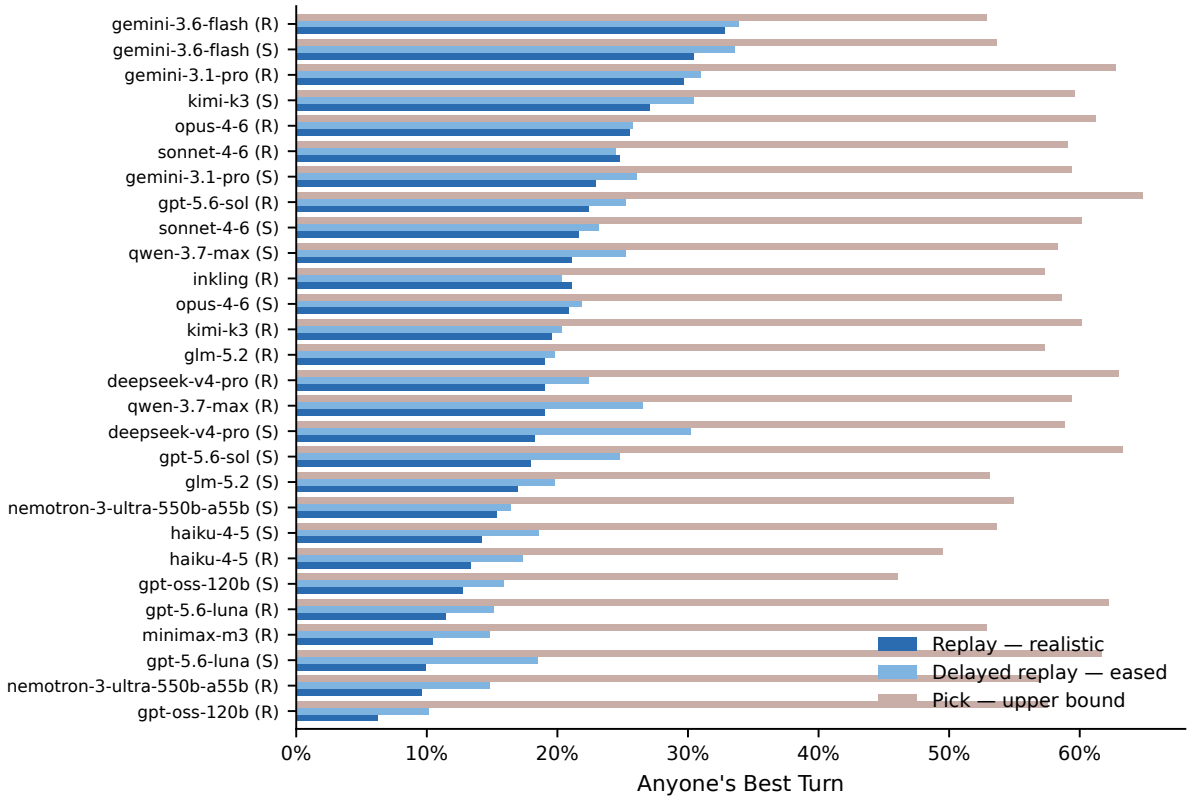


Figure 5.3: Anyone's Best Turn for all 28 configurations. The gap between the pick bar and the replay bar is wider than the entire spread across models within either framing, and it is positive for every configuration. (R) indicates models in native reasoning mode with max effort; (S) indicates models in lowest reasoning effort.

mance relative to human judgment, we compare endorsement share, the fraction of the annotator group that a single placement satisfies. This is the only quantity on which people and models are scored by an identical definition. Results are shown in Table 5.3 and Figures 5.4 and 5.5.

Human annotators do not converge on a single correct moment; this is reflected in their endorsement shares. Even after deliberation, an annotator's placement satisfies 40.6% of the rest of their group on the best-turn criterion and 76.1% on the window criterion, and satisfies the entire group 28.6% and 63.2% of the time. Before deliberation, judging independently, the same annotators reach 31.3% and 67.7%. Models reach this reference performance only in the pick

	Best-turn criterion		Window criterion	
	mean share	satisfied all	mean share	satisfied all
Human annotator, after deliberation	40.6%	28.6%	76.1%	63.2%
Human annotator, first pass	31.3%	18.0%	67.7%	53.2%
Pick *(upper bound)*	37.1%	18.1%	76.8%	56.5%
Delayed replay *(eased)*	14.1%	6.7%	36.2%	21.4%
Replay *(deployable)*	11.8%	5.4%	34.5%	18.7%

Table 5.3: Share of the annotator group a single placement satisfies, the only quantity on which humans and models are scored by an identical definition. Bold marks the two human reference rows. Both human rows exclude the annotator's own vote from numerator and denominator; a model has no vote in the group, which is what makes the rows comparable. The first-pass row is each annotator's independent judgment before seeing anyone else's. Paired differences reported in the text are computed on the n items where every arm produced a placement, so they will not equal differences of these marginal columns.

framing. Pick sits +1.7 points above the deliberated human reference on the window criterion (95% CI $[-0.7, +4.1]$) and −2.8 points below it on the best turn ($[-6.3, +0.7]$). Against the first-pass reference, which is the closer comparison for a model that consults no one, pick exceeds the human rate on both criteria.

In the two realistic online framings, however, models fall significantly short. Replay is −28.1 points below the deliberated human reference on the best turn ($[-32.4, -24.0]$) and −40.6 points below on the window ($[-44.4, -36.7]$): a human placement satisfies everyone's window 63.2% of the time against replay's 18.7%. Delayed replay again improves only marginally, reaching 14.1% and 36.2% against replay's 11.8% and 34.5%. The per-configuration view shows the same pattern across models (Figure 5.5). In pick, three arms perform better than the deliberated human bar on the best turn, and eighteen are better on the window, while in replay and delayed replay none of the 28 arms perform better than human annotators. Overall, models can reproduce human judgment of where a contribution belongs when the whole conversation is available, but retain under a third of the human best-turn share and under half of the window share when the same

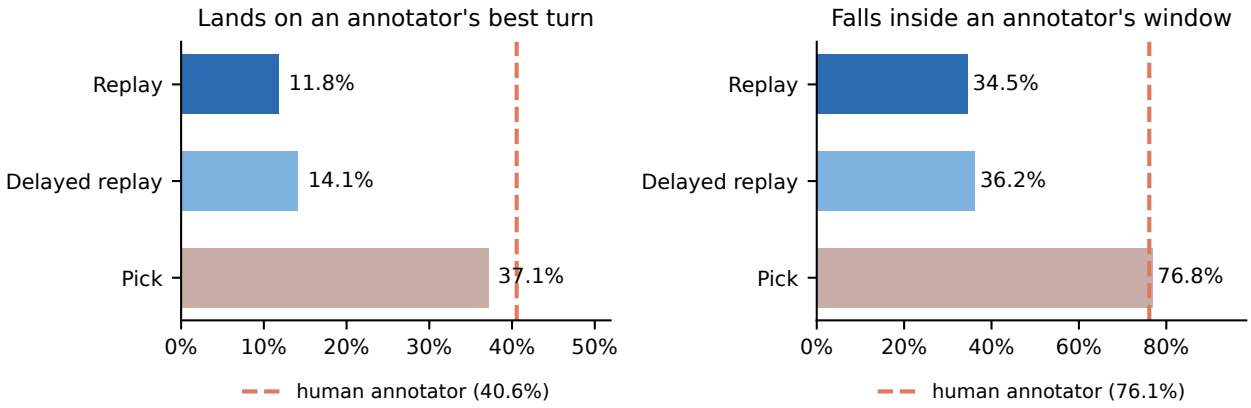


Figure 5.4: Share of the annotator group a single placement satisfies, against the human reference (dashed). Pick reaches the human line; both online framings fall far short of it.

judgment must be made online.

5.5.1 Error Analysis

Errors in the online task framings are systematically premature.

We begin by characterizing where predictions fall relative to the annotators' acceptable windows (Figure 5.6). In replay, 46.4% of predictions are issued before any annotator's window has opened, compared with 1.0% issued after every window has closed and 0.4% falling in the gap between two of the group's windows. This pattern indicates that models tend to place a given message before any of the acceptable windows have begun. In replay, 75.1% of trajectories fire early, with a median drift of −5 turns and a mean of −11.1 (−15.2 conditional on being early) (Figure 5.7). Considered against a median conversation length of 52 turns, the extended tail of this distribution implies that the average online prediction arrives approximately one fifth of a conversation before the point at which the group judged it appropriate.

We then plot the placement positions of model judgments and the annotated windows (Figure 5.8). Annotators place messages at a mean of 0.493 of the way through a conversation, whereas

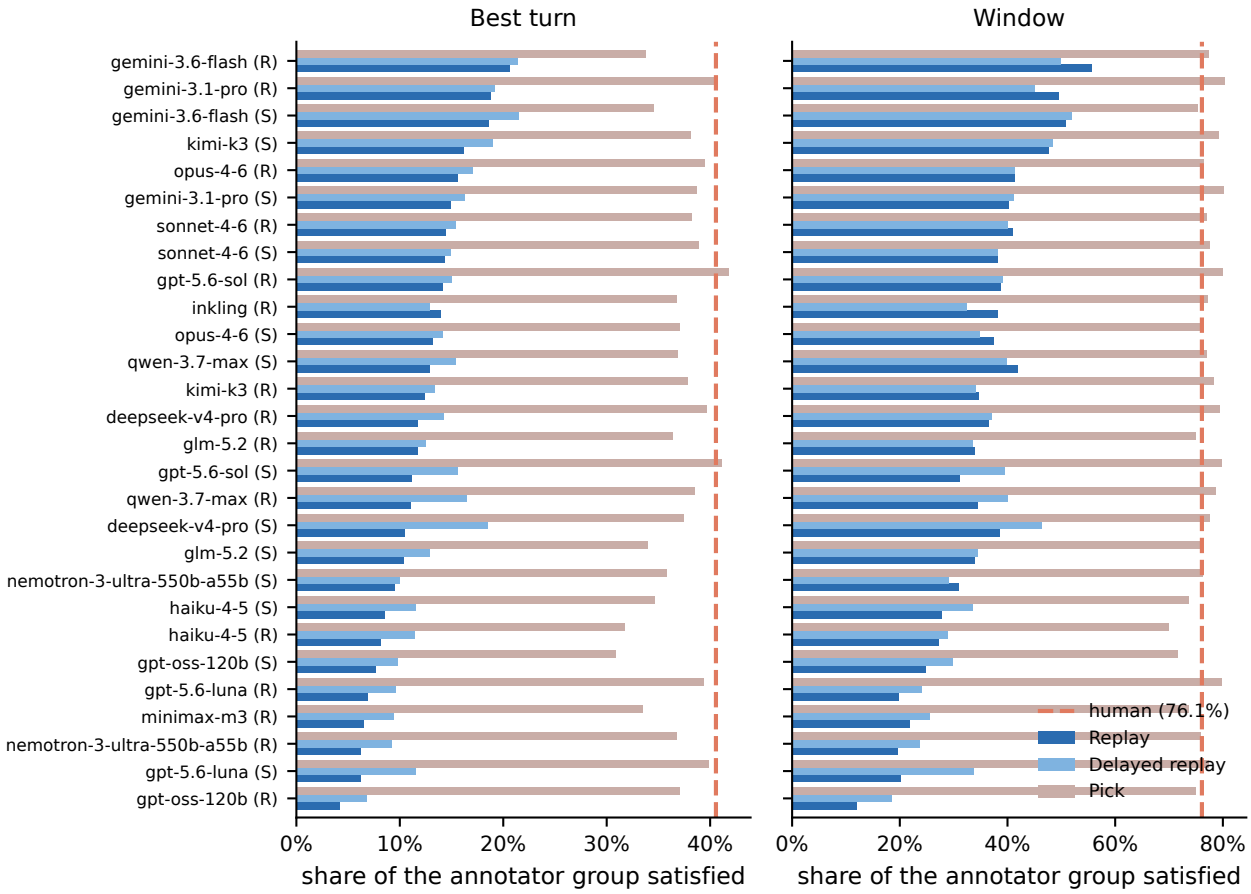


Figure 5.5: Every configuration against the human reference (dashed). In pick, three arms clear the human best-turn bar and eighteen clear the window bar; in both online framings, none do.

replay and delayed replay place at 0.269 and pick at 0.522. Computed pairwise within message, the replay offset is −0.229 of a conversation, and the pick offset is +0.026. The direction of the shift is consistent across the models we evaluated: all 15 place earlier than annotators in replay and later in pick (sign test $p = 6 \times 10^{-5}$), although the magnitude varies considerably, from −18.6 turns for `gpt-oss-120b` to −5.7 for `gemini-3.6-flash`.

Justifications at the point of commitment are most often about relevance, not timing. To examine what models attend to when they commit, we used an LLM judge to classify 4,349 replay trajectories drawn from 13 models, using `deepseek-v4-pro` and withholding both model identity

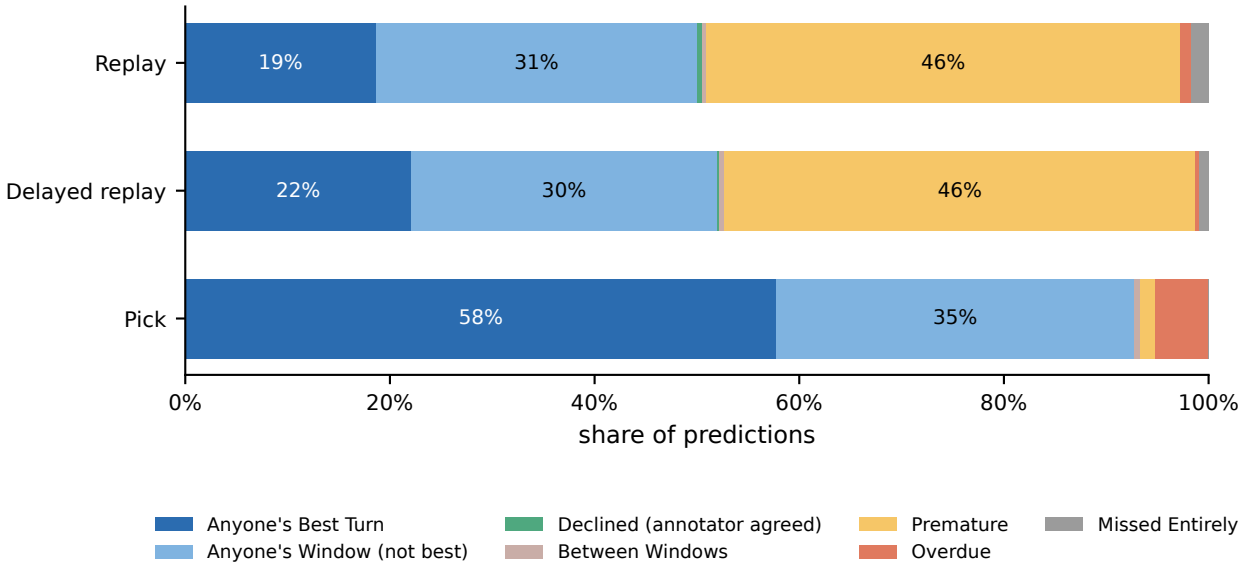


Figure 5.6: Where predictions land. Online, the single dominant outcome is firing before any annotator's window opened; with hindsight it nearly vanishes and the residual error becomes lateness.

and prediction correctness from the judge so that a correctness halo could not form. The judge was shown each trajectory's reasoning at the checkpoint at which it committed, together with the conversation truncated exactly as the model had seen it, and was asked to classify the cues used to justify placement decisions. In replay, 92.8% of justifications argue that the content fits the current point in the conversation, 7.0% that the present moment is the appropriate one, and 0.14% weigh the option of waiting explicitly; in delayed replay these proportions shift to 86.7%, 13.0% and 0.23%. When asked whether a message should be delivered at the present turn, models thus appear to answer a related but distinct question, namely whether the message belongs in the conversation at all. The relevance-oriented justifications are notably uniform, typically citing the participant or the turn to which the message responds, and taken individually they are accurate statements about fit; none, however, constitutes an argument that the present turn is preferable to a later one.

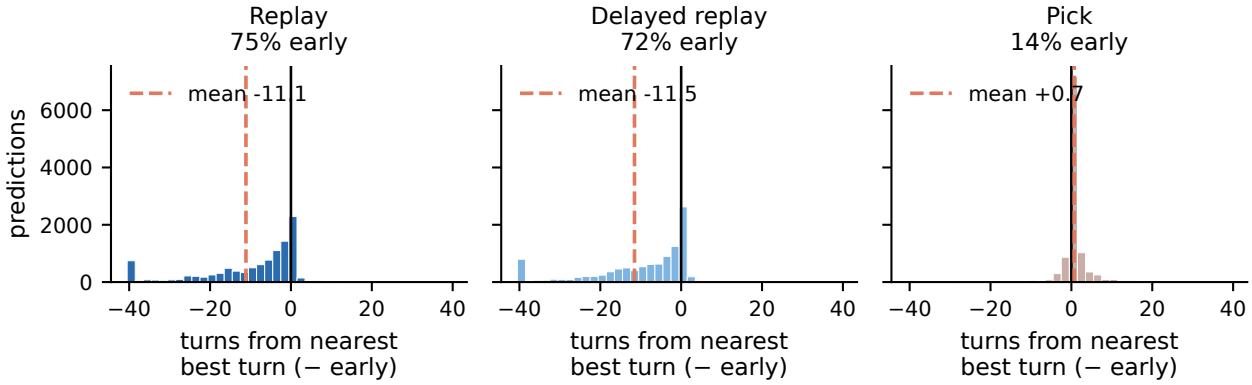


Figure 5.7: Timing drift by framing. Averaging the online and hindsight distributions would report a mild early bias that describes neither, which is why the framings are never pooled.

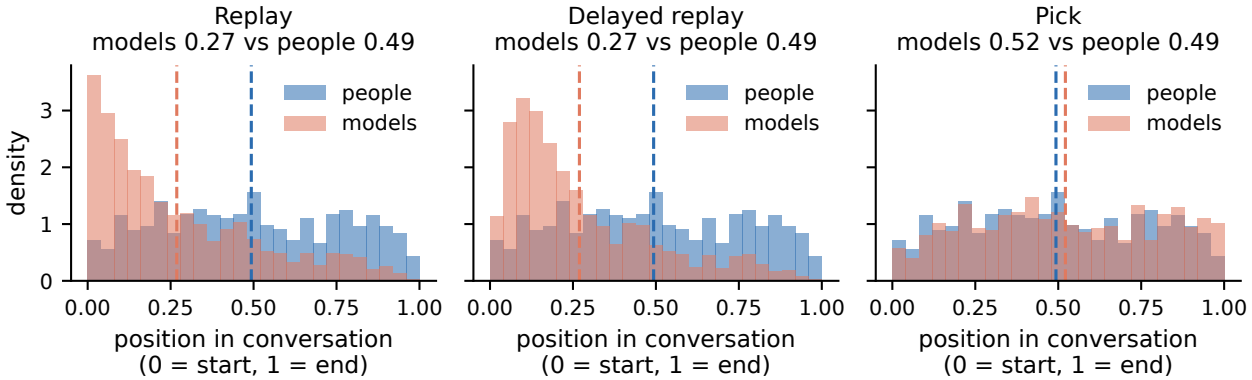


Figure 5.8: Where models place messages against where people do. The distributions barely overlap online and nearly coincide with hindsight.

5.6 Discussion

5.6.1 Timing for proactive agents is a span instead of a point.

Half the messages in the PROSKI corpus have annotators whose ideal placements land more than two turns apart, even after deliberation. For some messages, the timing of placement is multimodal. For example, one annotator placed a snack suggestion at turn 50, where a group first raises the subject of food, and two placed it at turns 181-184, where the group returns to it. Both moments would have been considered correct and aligned with human expectations. This motivates us

to define metrics and ground truth that reflect this spread, instead of forcing a single gold turn, because a system representing timing as a single peak cannot express what these annotators jointly expressed.

Qualitative analysis shows that annotators converge when the conversation contains a specific anchor for the message, such as when a participant asks outright what a bus fare costs. However, they diverge when a message could serve two functions, typically on recurring topics. A contribution about medical evaluation can reopen a thread the group let drop or substantiate a claim a participant stakes out twenty turns later. Deliberation does not resolve these cases because the difference is functional rather than factual.

5.6.2 What Counts as a Fair Standard for a Proactive Agent

The core comparison in this chapter is arguably not symmetrical. Annotators judged timing with the whole conversation before them, and most revised after seeing what their peers had chosen, whereas a model in the replay framing walks the conversation forward one turn at a time, commits at its first yes, and consults no one. A gap between them may therefore be read as unfair to the model rather than as a deficiency in the models.

Despite the asymmetry, I argue that human ground truth defined in the chapter is a desirable target for models used for proactive agents. The goal of the benchmark is not to evaluate whether a model can perform the same task under the same conditions as a human annotator. Instead, the goal is to specify the outcome a deployed proactive agent should achieve. If the agent speaks, the group should later judge that the contribution arrived within a window it considers acceptable and not disruptive.

The comparison between model performance, the human baseline, and endorsement share shows that the target may require better-than-human predictive performance. A deployed agent

has less information than a retrospective annotator, less social standing than a group member, and fewer opportunities to repair a poorly timed interruption. Matching an individual human annotator is therefore not enough. A group member's chosen turn matches the rest of the group's chosen turn just 40.6% of the time, and 31.3% before deliberation; an agent that matched this rate would still misplace many contributions while spending the whole group's attention. A proactive agent contributing to group conversation should aim for the outcome: contribute when the group would accept the timing, and stay silent when no acceptable region exists.

On that outcome, current models remain far from ready. In replay, they land inside an acceptable region on 34.5% of messages against a human 76.1%, and decline on 2.2% of decisions against 4.0%, never in pick. The result suggests that proactive agents need more than relevance detection. They need a conservative timing policy that can anticipate whether a moment will still look appropriate from the group's later perspective. In fact, this issue is not peculiar to group chat: proactive coaching agents face the same problem of decomposing whether, when, and how an agent should contribute [224].

5.7 Related Work

This chapter draws on work in proactive and mixed-initiative systems, turn-taking and conversational relevance, and recent timing benchmarks for AI agents.

Proactive and mixed-initiative systems. The trade-off PROSKI measure was named at the outset of the mixed-initiative literature: a system acting on its own behalf must weigh the value of acting against the cost of interrupting, and defer when that calculus is unfavorable [92]. Early proactive assistants modeled what to volunteer and when [162], and a parallel HCI line built models of interruptibility showing that an interruption's cost depends on where it falls in the

structure of the ongoing task [5, 17]. LLM-based agents revived the question at larger scope [254]. They ask clarifying questions before acting [264], hold candidate contributions during a group conversation [142], and intervene in negotiation [144]. Several accounts survey this design space [55, 136, 205], and deployments find that the same behavior can be welcome or unwelcome depending on delivery [141, 200, 137, 260, 16]. Within multi-party systems timing is now an explicit architectural concern: MUCA separates "what to say, when to respond, and who to answer" [150], systems put a dedicated gate before generation [62, 23, 40, 253, 97], and others throttle participation by policy [249, 98], which is where users say they want control [94, 160]. Two things distinguish this work. Most of that literature studies an agent addressing an individual, who can dismiss a badly timed contribution privately, whereas a message posted into a group channel spends everyone's attention at once and joins the shared record, which Yeo et al. mark as a distinct design category [256]. Teams that include an agent build weaker shared mental models and extend less trust than all-human teams performing the same task [208]. An agent's contribution therefore arrives without the standing a member's would carry and has correspondingly less margin for a badly chosen moment. These systems establish that a timing gate can be built and evaluate it by its consequences, rather than by whether the moments it selects are the moments the people in the conversation would have selected.

When a contribution belongs, and what existing resources evaluate. Conversation analysis argues that turn-taking is organized around transition-relevance places, and a contribution ignoring that organization is heard as an interruption independent of its content [202], which extends to machines [191]. Grice's maxims make the complementary point about substance: a contribution is judged against the current purpose of the exchange, including whether it is appropriately informative and relevant *now* [82, 226, 118]; our mechanism finding, that models establish content fit while leaving the moment unaddressed, restates this in measurable terms.

Computational work on multi-party dialogue treats the structural half as its own task, predicting who speaks next and who is addressed [175, 54, 75, 221], where models and humans are broadly similar [244, 74]. That answers a different question: next-speaker prediction asks who is entitled to the floor and is well posed before the contribution is known, whereas ours presupposes a specific contribution and asks whether it is worth the floor now. A model could solve turn-taking completely and still be wrong about every message here, because most structurally permitted turns are turns where this particular message does not belong. The stakes are empirical: interruption timing affects performance and perceived disruption [185, 53], agent interjections reshape collective attention [273], and unprompted contributions require justification to be accepted [57, 16, 6, 194]. The closest precedent asks users when an agent should participate in their group chats and finds appropriateness situational rather than absolute [258], establishing which *kinds* of moment people accept, whereas PROSKI asks the same people to name the turn. Benchmarks for conversational and social capability score what a model says [**?** 134, 250, 270, 96, 184, 224] but presuppose that a turn has already been allocated to the model.

Where the timing labels come from. A recent line of work poses the question directly: DiscussLLM trains models to emit a silent token when no intervention is warranted [180], LlamaPIE builds a semi-synthetic corpus to train a small model deciding whether to respond [40], and When2Speak generates over 215,000 examples from 16,000 synthesized conversations with a SPEAK or SILENT decision at every turn [163]. We take these as convergent evidence for the premise. What PROSKI adds is the source of the labels: in each of the above resources, the correct moment is a property the construction procedure installed, so the measurement is whether a model recovers a planted moment. That inversion has a consequence for how such resources are built, since a per-turn SPEAK/SILENT label presumes the correct moment is a point and Section 5.4 finds that presumption fails on roughly half the messages here: such a label records

one acceptable answer and marks the others wrong. Our measurement commitments accordingly rest on work arguing that annotator disagreement is often signal rather than noise [9, 190]. Aggregating systematic divergence away can produce a target no annotator endorsed, which is what averaging our annotators' best turns would do. They also follow the standard that alignment should be judged against the agreement among the humans themselves, reported alongside it [1], which is what endorsement share measures. These commitments also answer a concern raised about multi-party corpora specifically, that the collection method determines what a corpus can support and convenience sources leave researchers without the contextual information multi-party phenomena depend on [148, 111]. ProSKI takes that to its conclusion: since a timing label is only meaningful from inside the conversation, the people who produced the conversation are also the people who label it.

5.8 Ethics statement

The study was reviewed and approved by our institution's IRB. Participants gave informed consent to the public release of the conversation they held and the annotations they produced, were compensated $35 for the session, and could withdraw at any point, including after the fact. Participants' names are removed from the released dataset.

5.9 Limitations & Future Work

The study reported in this chapter has several limitations that should be considered when interpreting the results. First, annotators and models both worked from recorded conversations, so the feedback loop by which a live contribution would likely reshape what follows lies outside the design [42]. Second, the timing ground truth collected in this chapter assumes the setting

of an informational chat message inserted directly into a group's conversation, which is what makes timing urgent, whereas gentler deliveries and facilitation moves would impose a softer constraint, and text withholds the cues people use to time spoken interventions [133, 266]. Third, the groups recruited for the study are small (2–4 people) and broadly egalitarian, and groups with sharper status differences may hold different expectations. Models also received only the conversation and the candidate message, not the participant information the annotators had, so part of the gap may reflect information withheld rather than judgment not exercised. Supplying a group profile is a straightforward ablation that would separate the two [125]. The annotation is correspondingly thin, at 2–4 judgments per message, and it forces a single contiguous window. The most informative addition would be a human baseline under the model's conditions, since no one in this study judged timing online and the chapter therefore cannot separate the difficulty of the online task from model failure at it. Running a group member and then an outside reader through recorded conversations turn by turn would bound how much of the human reference comes from hindsight and how much from membership.

Chapter 6

Conclusion

6.1 Thesis Argument

This dissertation is motivated by the tension in collaborative work, where groups need shared information to maintain common ground, but attention is limited, and every artifact that enters a collaboration demands people's attention. Generative AI changes this tension by making artifacts cheap to produce. It can summarize meetings, recommend papers, retrieve missing facts, and generate messages on behalf of an agent. Yet abundance alone does not solve the collaborative problem. An artifact helps only when it selects the right information, grounds that information in the group's context, and arrives at a moment when the group can use it.

Chapter 2 framed this problem within the literature on Social Computing and human-AI interaction. Common ground is not only a matter of storing information; it is maintained through selective, situated, and timely communication. AI-generated artifacts, therefore, cannot be evaluated solely by factual correctness or fluency. They must also be evaluated by how well they fit the social setting in which they appear.

The empirical chapters develop this argument in different settings. Chapter 3 examined

information in remote meetings, which is a familiar case where groups already struggle to carry common ground from synchronous discussion into later work. Interviews with 13 remote workers and a survey of 198 participants showed that people are wary of records curated by others, rarely return to raw recordings, and use meeting information for more than just recall. Through a co-design study with 16 workers across five groups, we proposed to reframe meeting artifacts as *bridges*, which are designed artifacts that carry common ground forward into follow-up work.

Chapter 4 then asked what happens when an AI agent contributes new information to an existing group space. Social-RAG treats prior group interaction as a social knowledge base and retrieves social facts from that history to ground generated messages. In PAPERPING, this meant recommending scholarly papers in Slack with explanations tied to the group's past conversations, members, and reactions. A three-month deployment in 18 Slack channels reaching 500+ researchers showed that socially grounded messages can fit existing group practices, while also revealing the limits of treating group traces as a complete account of group preference.

Chapter 5 pushed the argument to timing. A proactive agent may know what information is relevant and still fail by speaking too early or too late. PROSKI therefore asks when an agent should contribute external knowledge to a group conversation. The benchmark contains 384 participant-authored messages from 35 real group conversations, labeled by the people who held those conversations. Across 15 models and 28 configurations, models approached human-level placement only when given the whole conversation in hindsight. Under the deployable replay setting, performance fell to roughly a third of its retrospective value, with the dominant error being premature firing. Together, these chapters show that collaborative AI artifacts require more than generation: they require selection, grounding, and timing.

6.2 Core Contributions

Overall, in this thesis, I make the following contributions:

1. **AI information artifacts as tools for maintaining common ground.** I argue that AI-generated artifacts in collaborative work should be judged not only by whether they are accurate or fluent, but by whether they help groups maintain common ground under limited attention. This contributes to social computing and human-AI interaction by making selection, grounding, and timing central design and evaluation concerns for AI tools in collaborative and knowledge work.

2. **Design principles for carrying common ground forward.** Through Meeting Bridges, I show that meeting artifacts should not be treated only as records for later recall. They can instead be designed to connect meetings to the subsequent asynchronous work. The chapter contributes empirical findings on how workers use meeting information and design principles to design artifacts that bridge synchronous and asynchronous collaboration.

3. **Social grounding through group interaction traces.** Through Social-RAG and PaperPing, I contribute a workflow for grounding AI-generated messages in the social history of a group. This contributes to human-AI interaction by showing how retrieval can be extended beyond factual knowledge to social knowledge, and to social computing by showing both the promise and limits of learning group preferences from interaction traces.

4. **Human-judged evaluation of proactive agent timing.** Through ProSKI, I contribute a benchmark for evaluating when proactive agents should contribute knowledge to a group conversation. The benchmark contributes a human-grounded account of timing as a distribution rather than a point, and it shows that current models can identify relevant content with hindsight but struggle to decide when to speak under deployment conditions.

6.3 Discussion

The common thread across these contributions is a shift from generating information to communicating well. Research on communication has long shown that being understood requires more than being correct: framing, timing, and source all shape whether people take information in [82, 66, 186]. AI systems now act in settings where those same concerns become design requirements. A meeting summary, a paper recommendation, and a proactive contribution are all information artifacts, but their value depends on whether they preserve attention and support the work of maintaining common ground.

This shift matters especially for collaborative and knowledge work because there are social consequences of an unfitting artifact. A private tool can be wrong without disturbing anyone else, but a message in a shared channel demands the group's attention, changes the shared record, and can redirect the conversation. Proactivity and visibility, therefore, raise the criteria for how good the artifact must be. The more an agent decides for itself when to act, and the more unavoidably its output occupies shared attention, the more precisely it must ground and time its contribution.

The thesis also shows that curation is an opinionated act. Social-RAG chooses which traces count as evidence of group interest. PROSKI shows that choosing a moment implicitly chooses what the contribution is for. This is not a defect: selection is what makes an artifact useful. The failure mode is an unexamined selection, one whose basis the group cannot see, contest, or correct. AI artifacts for collaboration, therefore, need mechanisms that make their selection basis inspectable and accountable to the people represented by them.

Finally, the boundary between functional and relational support is thinner than it first appears. The chapters focus mostly on informational contributions: a paper, a price, a constraint, a piece of evidence, or a meeting artifact. Yet functional contributions still have relational effects. In PROSKI, 42% of contributions provide evidence for a claim somebody had already staked out, meaning that

the agent would often be strengthening one side of an ongoing exchange. The same message can fill an information gap and alter a social dynamic. This again highlights that an AI system is never only managing information, but also participating in a social setting. It also highlights why social computing remains central to the design of AI tools for knowledge work.

6.4 Future Work

The future direction I am most excited about is building end-to-end proactive agents for collaborative work. ProSKI isolates the timing judgment, but a real agent must integrate monitoring, retrieval, timing, generation, feedback, and adaptation. Studying those components together would make it possible to ask not only whether an agent can identify a useful contribution, but whether the whole interaction gives attention back to people rather than drawing attention toward the agent itself.

This goal is especially important in collaborative settings. Much of today's human-AI interaction still requires people to stop, formulate instructions, inspect outputs, and manage the agent. In group work, that burden competes with attention to other people. A good proactive agent would reduce that burden by noticing when it can help and staying quiet when it cannot. The ideal of proactivity, in this view, is thus to let collaborators remain more present with one another.

Getting there will require better ways to represent social expectations. Many agents still rely on fixed rules for when to act, but social rules are often ambiguous, tacit, and changing. The right moment to speak depends on the topic, the group's history, the current conversational trajectory, and the cost of being wrong. Future systems need more expressive ways for people to specify expectations, for groups to revise them over time, and for agents to reason about them without reducing them to brittle templates.

Future work should also move beyond treating the group as homogeneous. This thesis often

analyzes the group as the relevant unit, but real groups contain roles, expertise, status differences, relationships, and uneven participation. More concrete group modeling and human behavioral modeling could help AI systems represent social context without flattening it into a single group preference. Such models would also make it easier to study who benefits from socially grounded artifacts, whose traces dominate the system's understanding, and when group-level adaptation hides individual disagreement.

These questions point toward a broader research program for social computing and human-AI interaction. As AI tools become more capable, the central challenge is not only how to make them generate better content, but how to make them participate in collaborative settings without consuming the attention and social context they are meant to support. The ultimate vision is an agent that helps groups preserve common ground, use their attention effectively, and continue collaborating with less friction.

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Appendix A

Meeting Bridges Study Materials

A.1 Participant demographic information

Demographics information for interview participants: Table A.1

Id	Job Title
I1	PhD student
I2	PhD student
I3	PhD student
I4	PhD student
I5	PhD student
I6	Program Manager
I7	PhD student
I8	Product Designer
I9	Senior Research Scientists
I10	Design Lead
I11	Open-source Project Leader
I12	Industry Research Scientist
I13	Professor

Table A.1: demographics information for interview participants

Demographics information for survey participants: Table A.2

Categorical Variables							
Gender		***Age Groups***		***Company Types***		***Education Background***	
Female	99	18-24	35	Publicly listed /traded enterprise	24	Mater's degree or above	45
		25-34	71	Large private enterprise	46	Bachelor's degree	103
Male	99	35-44	61	Small and medium -sized enterprises	60	High School	
		45-54	22	Micro enterprise	Prefer not to say	1	
		Above 55	9	Other	47		
Employment Status		***Top 3 Industry Roles***		***Management Experience***		***Hours spent in remote meetings***	
Full-Time	147	Trained Professional	48	Has experience	103	<4 hours/week	90
Part-Time	41	Junior Management	24	No experience	94	4-12 hours/week	82
Not specified	10	Middle Management	22	Prefer not to say	1	12-20 hours/week	19
		Other Roles	104			20+ hours/week	7

Table A.2: demographics information for survey participants

Demographics information for co-design study participants: Table A.3

Id	Job Title	Organization	Group	Type of preparatory meeting
C1	Program director	Student services at University	A	Mock brainstorm meeting
C2	Associate director	Student services at University	A	Mock brainstorm meeting
C3	Associate director	Student services at University	A	Mock brainstorm meeting
C4	Staff member	Student services at University	A	Mock brainstorm meeting
C5	Associate director	Student services at University	A	Mock brainstorm meeting
C6	PhD student	Research Lab	B	Real progress report meeting
C7	Undergrad student	Research Lab	B	Real progress report meeting
C8	Undergrad student	University	C	Mock brainstorm meeting
C9	PhD student	University	C	Mock brainstorm meeting
C10	Masters student	University	C	Mock brainstorm meeting
C11	Designer	Masters program in design	D	Mock brainstorm meeting
C12	Designer	Masters program in design	D	Mock brainstorm meeting
C13	Designer	Masters program in design	D	Mock brainstorm meeting
C14	Designer	Masters program in design	D	Mock brainstorm meeting
C15	Research scientist	Research Institute	E	Real weekly project meeting
C16	Product designer	Research Institute	E	Real weekly project meeting

Table A.3: demographics information for co-design study participants

A.2 Additional survey results

The proportion of meetings that use notes or recordings: Figure A.1

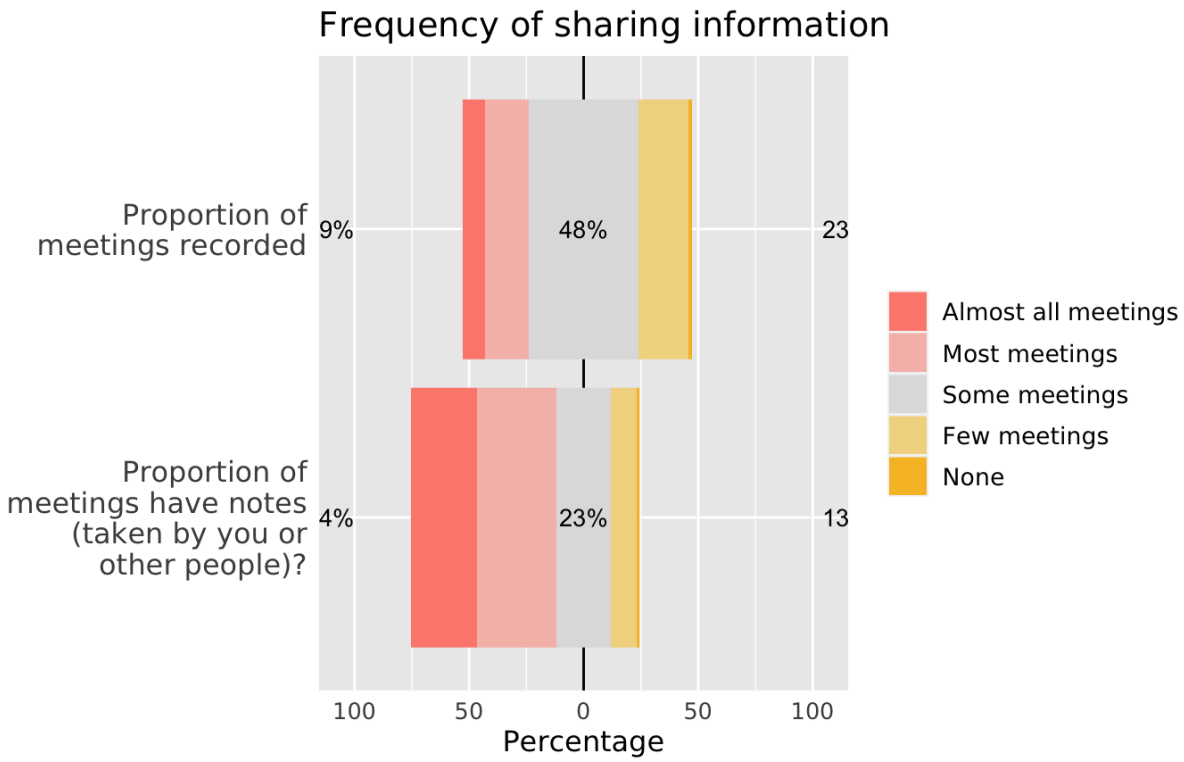
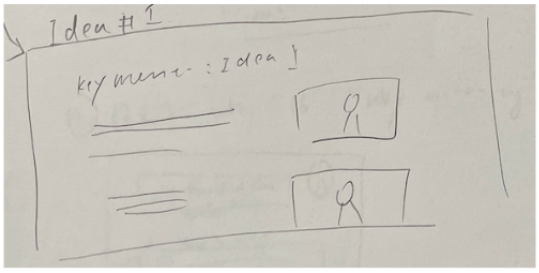


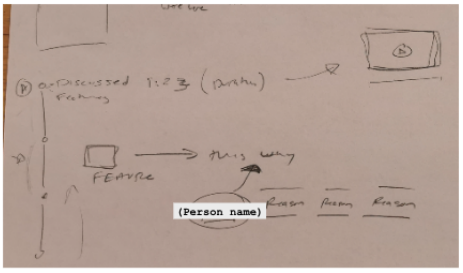
Figure A.1: Proportion of meetings that use notes or recordings

A.3 Additional co-design results

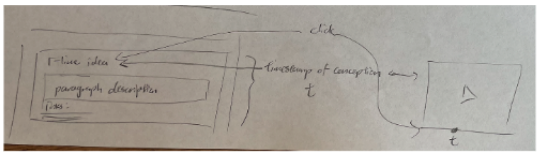
Examples of sketches demonstrating the desired characteristic 1: incorporation of multiple data types and rich media presentations: Figure A.2.



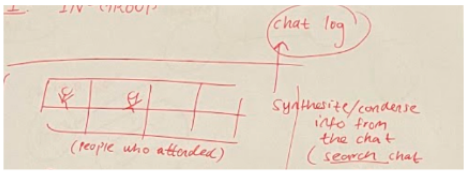
(a) Provide recordings for key moments (P11-S3)



(c) Attribution of notes to speaker and video clips (P16-S3)



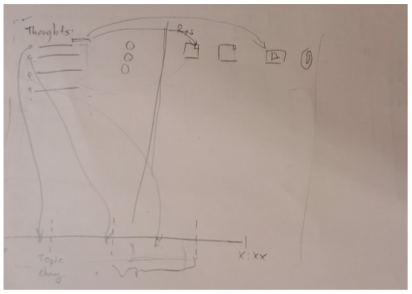
(b) Link between key points in notes and time stamps in recordings (P9-S3)



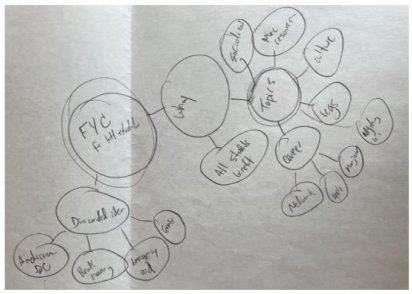
(d) Include attendance and chat log (P7-S3)

Figure A.2: Examples of sketches demonstrating Desired Characteristic 1: Incorporation of multiple data types and rich media presentations.

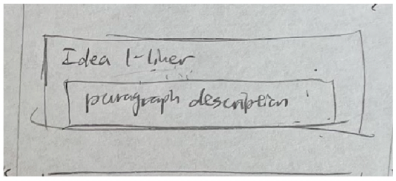
Examples of sketches demonstrating the desired characteristic 2: structure(s) to help users explore information: Figure A.3



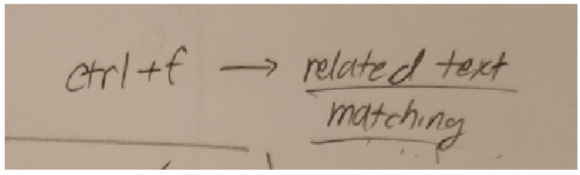
(a) Discussion themes and academic papers as index (P6-S3)



(b) Discussion topics as index (P1-S3)



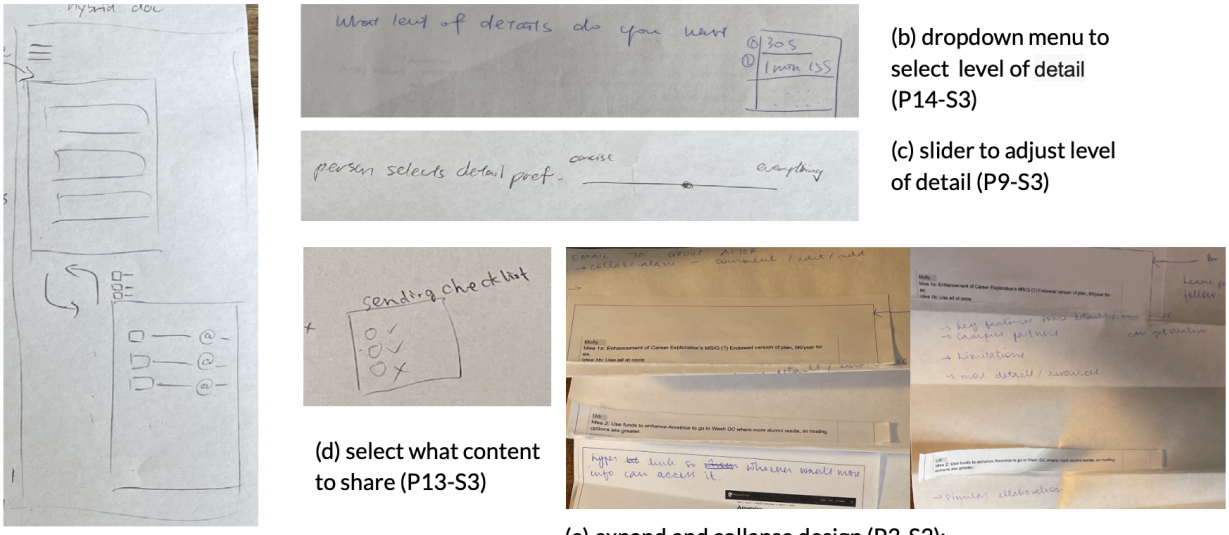
(c) Provide one-linear and paragraph summary (P9-S3)



(d) Semantic search of content (P14-S3)

Figure A.3: Examples of sketches demonstrating Desired Characteristic 2: Structure(s) to help users explore information

Examples of sketches demonstrating the desired characteristic 3: ability to view, share custom artifact presentations for varied contexts, and audiences: Figure A.4



(a) switch between indexes (P9-S3)

(b) dropdown menu to select level of detail (P14-S3)

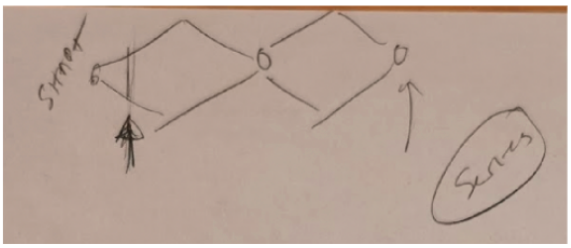
(c) slider to adjust level of detail (P9-S3)

(d) select what content to share (P13-S3)

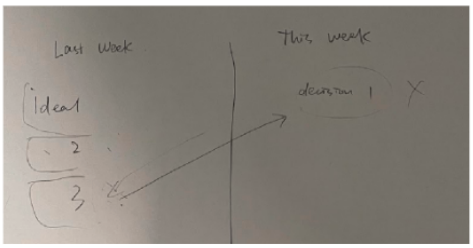
(e) expand and collapse design (P3-S3); left is folded view and right is expanded view

Figure A.4: Examples of sketches demonstrating Desired Characteristic 3: Ability to view, share custom artifact presentations for varied contexts, and audiences.

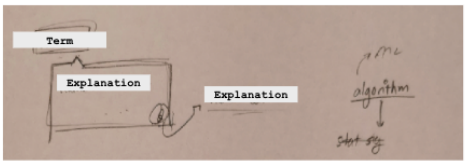
Example of sketches demonstrating the desired characteristic 4: connection to contextual information outside the meeting Figure A.5



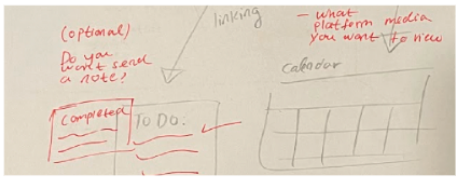
(a) Project timeline in double diamond representation (P15-S3)



(b) Compare and connect last week meeting summary artifacts to this week's (P16-S3)



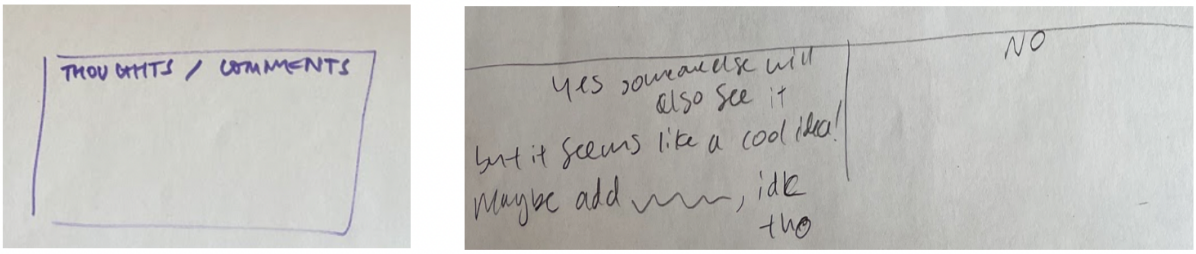
(c) Clickable word with pop up bubble explanation (P14-S3)



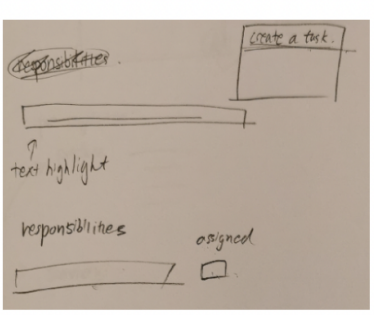
(d) Sync todo into calendar (P7-S3)

Figure A.5: Example of sketches demonstrating Desired Characteristic 4: Connection to contextual information outside the meeting

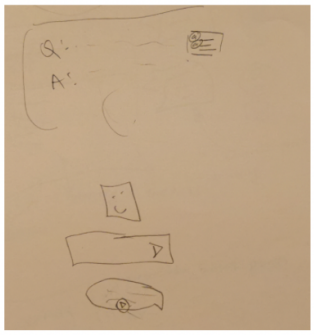
Example of sketches demonstrating the desired characteristic 5: ability to evolve after the meeting Figure A.6

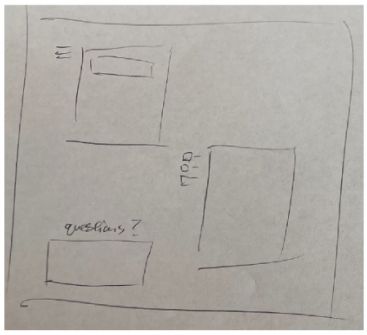


(a) Feedback section to express ideas about the meeting decisions (P3-S3 & P8-S3)



(b) Create and assign tasks (P14-S3)





(c) Q&A section (P9-S3) & (P10-S3)

Figure A.6: Example of sketches demonstrating Desired Characteristic 5: Ability to evolve after the meeting

A.4 Protocol for the interview study

Note: the interviews are semi-structured. The protocol below contains example questions asked in the interviews.

1. **Participant Background**
    - To get us started, would you tell us more about the different types of things that you have generated to capture/keep track of information/discussion in small group/team meetings?
    - What were the purposes of the meetings? Who was in the meetings? How often did the meetings happen?
    - For the meetings that you couldn't be in, how did you catch up?
2. **Continue Discussions After Meetings**
    - Outside of synchronous meetings, how else do people in your team typically communicate? (e.g., Slack, email, discord)

- Can you recall a time when you started a discussion in the meeting and continued after, or vice versa (discussion asynchronous that moved to synchronous)?
- What challenges have you encountered in continuing the discussion outside of the meetings that you were in?
- Were there times when you tried to participate in a discussion that happened in a meeting that you missed? What was your strategy? What challenges did you face/did you succeed or not?
- Have you heard of interesting strategies?

3. **If They Mention Recording Meetings:**

   - For each type of meeting that has been recorded, why did you record the meeting?
   - How often do you go back to the recordings? What information do you look for when going back to the recordings?
     - How much do you want to go back and find something specific vs get a summary over a lot of things?
     - Strategies for retrieval (scrubbing over the timeline, looking at transcripts)?
     - Struggles with getting the info you wanted via the strategies you took.
     - Have you heard of interesting strategies?
   - When do you go back to the recordings? (how long after the meeting happened)
   - When you don't go back to the recordings, why is that?
   - Were there times when you wanted to go back and you hadn't recorded the meetings?

4. **If They Mention Notetaking in Meetings:**

   - For each type of meetings that have been taken notes on, why did you take notes in the meeting?
   - How often do you go back to the notes? What information do you capture in taking notes?
     - How much do you want to go back and find something specific vs get a summary over a lot of things?
     - Strategies for capture (personal vs collaborative note-taking, designated note-taker, meeting outline beforehand, etc.)?
     - Struggles with capturing the info you wanted via the strategies you took.
     - Have you heard of interesting strategies?
   - When do you go back to the notes? (how long after the meeting happened)
   - When you don't go back to the notes, why is that?

- Were there times when you wanted to go back and there isn't anything that you can refer to for the meetings?

5. **Tradeoffs and Preferences Regarding Recording vs. Notes/Other Methods**

   - Can you recall times where you wanted to review audio, video, or a transcript of the audio specifically as opposed to just notes and why? Vice versa?
   - What about the notes and transcripts made them more/less preferable, compared to meeting recordings?
   - If it is easier to retrieve information from recordings, will you go back?

A.5 Example instruction for preparatory mock meetings in the co-design study

Task Description:

Come up with three ideas for the [a fund to sponsor creative ideas]. A group of you are planning to submit proposals to the [a fund to sponsor creative ideas], which grants $5000 for projects that help students feel at home with the [a public university] and engage with the community, for the duration of a year. Your task for today is to come up with three potential ideas for the application.

You will write a short paragraph of description for each idea [examples]. Since your group is planning to meet again later to decide on a final idea, the idea descriptions you come up with in this session do not need to be perfect.

You should take proper notes so that you can continue the discussion next time when you all meet together to decide on the final idea. One member of your group also has to miss the meeting this time, so your notes can also hopefully help them catch up before the next meeting.

Discussion guideline:

- Someone in the group should take on the scribe role, but you should also feel free to take individual notes if needed. All notes will be shared with the research team for research purposes.
- You will share the three project descriptions at the end of the discussion with the research team.
- The session will be recorded for research purposes.
- You can also feel free to build on the ideas you discussed today and submit them to the actual fund next year

A.6 Instruction for the sketching activity in the co-design study

Your task is to imagine the ideal meeting summary artifact, so try not to be constrained by the things that current technologies can achieve.

Task 1: How would you want to design the summary artifact if you want to share it with your group after the meeting via email (in-group)?

Task 2: How would you want to design the summary artifact if you want to keep it for your own uses in the future?

Task 3: How would you want to design the summary artifact if you want to share it with someone who missed the meeting, but only needs to learn about the discussion briefly?

Task 4: How would you want to design the summary artifact if you want to share it with someone who missed the meeting, but will need to join the working group and take on future tasks?

Task 5: How would you want to design the summary artifact if you want to share it with people outside of [your group] after the meeting via email (out-group)?

Tips:

- There is no right or wrong answer. Don't be afraid to draw incomplete or "bad" ideas. Feel free to sketch multiple ideas.
- Feel free to play with and edit the printed materials or sketch/annotate on paper.

A.7 Interview protocol for the co-design study

Note: the interview is a follow-up after the sketching activity. The interviews are semi-structured. The protocol below contains example questions asked in the interviews.

Previous Experience with Meeting Summary Artifacts

- How do you usually keep track of information in small group meetings?
- Do you differ your way of tracking info in different scenarios? For example, by the purposes of the meeting? Expected audience of the meeting? Expected involvement after the meeting? Who was in the meeting? Your role in the meeting?
- Do you face any challenges with the current ways of tracking info?

Discussion of the Mock Meeting

- As the note taker for yesterday's meeting, was there information that you would like to keep track of but weren't able to?
- What did you think was important about the meeting?

Co-Design Activities

- **Task 1:** How would you want to design the summary artifact if you want to share it with your group after the meeting via email? Think aloud during the sketch.
  - Consider the following dimensions: Effort to create, Effort to retrieve, Fidelity of information, Degree of curation, Generalizability/Interpretive, Interactivity.
  - Additional information you wish the system collected during the meeting.
  - Additional information you wish others have collected that you can use.
  - How do you imagine people using the summary artifacts afterward?
- **Task 2:** [Similar structure to Task 1 but focus on personal use]
- **Task 3:** [Similar structure to Task 1 but focus on brief sharing with someone who missed the meeting]
- **Task 4:** [Similar structure to Task 1 but focus on detailed sharing with future participants]
- **Task 5:** [Similar structure to Task 1 but focus on sharing with people outside of your group after the meeting via email (out-group)]

Other Types of Scenarios/Audiences

- Audiences: in-group vs. out-group; manager vs. colleagues; self vs. others; high-involvement vs. low-involvement.

Appendix B

PaperPing Prompts and Evaluation Materials

B.1 Prompts designed for PaperPing

To anonymize, social context information in the following examples is hypothetical. The paper abstracts in the example prompts have been shortened for readability in this appendix. The actual prompts contained the full abstracts.

Prompt 1 in Fig. 4.4 (*before* adaption):

```
You are a helpful assistant for paper summarization.
The paper has the following details:
* Abstract: [ABSTRACT_OF_RECOMMENDED_PAPER]
* Authors: [AUTHORS_OF_RECOMMENDED_PAPER]
In your summarization, you must
* Congratulate the following authors: [AUTHOR_NAME_TO_CONGRATULATE].
* Mention the following authors: [AUTHOR_NAME_TO_HIGHLIGHT].
* Mention [AUTHOR_AFFILIATION_TO_HIGHLIGHT].
* Mention the paper's conference or journal: [VENUE_TO_HIGHLIGHT].
* Keep your output less than [CHARACTER_LIMIT] characters.
* Be informative.
* If applicable, be specific about the numbers in the abstract that may refer to the
    ↪ proposed method's performance.
```

Example Prompt 1 (*after* adaption):

```
You are a helpful assistant for paper summarization.
The paper has the following details:
* Abstract: {By simply composing prompts, developers can prototype novel generative
    ↪ applications...[remainder of the abstract omitted for readability]}
* Authors: Tae Soo Kim, Yoonjoo Lee, Jamin Shin, Young-Ho Kim, Juho Kim
In your summarization, you must
* Congratulate the following authors: Tae Soo Kim.
* Mention the following authors: Juho Kim.
```

```
* Keep your output less than 350 characters.
* Be informative.
* If applicable, be specific about the numbers in the abstract that may refer to the
    ↪ proposed method's performance.
```

Output of example Prompt 1:

> Congratulations to @tsk and team, including Juho Kim, on developing EvalLM! This system aids in refining LLM prompts by evaluating outputs against user-defined criteria, showing a 59% reduction in needed revisions. It enables examining twice as many outputs, enhancing prompt development efficiency.

Prompt 2 in Fig. 4.4 (*before* adaption):

```
You are a helpful assistant.
Title: [TITLE_OF_RECOMMENDED_PAPER]
Abstract: [ABSTRACT_OF_RECOMMENDED_PAPER]
Title: [TITLE_OF_RELEVANT_PAPER]
Abstract: [ABSTRACT_OF_PREVIOUS_PAPER]
The first paragraph of your answer should explain and specify the relationship between [
    ↪ TITLE_OF_RECOMMENDED_PAPER] and [TITLE_OF_RELEVANT_PAPER]. Be informative.
* In this first paragraph of your answer, you must explain and specify how [
    ↪ TITLE_OF_RECOMMENDED_PAPER] is related to [TITLE_OF_RELEVANT_PAPER] in one short
    ↪ sentence.
* In this first paragraph of your answer, you must start with "This paper might be related
    ↪ to [TITLE_OF_RELEVANT_PAPER] because".
* In this first paragraph of your answer, you must specify how [TITLE_OF_RECOMMENDED_PAPER]
    ↪  cites [TITLE_OF_RELEVANT_PAPER] in one short sentence. The content from [
    ↪ TITLE_OF_RECOMMENDED_PAPER] that cites [TITLE_OF_RELEVANT_PAPER]: [CITATION_CONTEXT].
    ↪
* In this first paragraph of your answer, you must mention the following shared authors of
    ↪ the two papers: [SHARED_AUTHOR_NAMES].
* The first paragraph should have no more than [CHARACTER_LIMIT_P1] characters.
The message from [USERNAME] who shared [TITLE_OF_RELEVANT_PAPER]: [MESSAGE]
People's comments about [TITLE_OF_RELEVANT_PAPER]: [COMMENTS]
People's reactions about [TITLE_OF_RELEVANT_PAPER]: [EMOJIS]
The second paragraph of your answer should specify what people think about [
    ↪ TITLE_OF_RELEVANT_PAPER] and who these people are. Be informative.
* In this second paragraph of your answer, you must start with "thoughts about [
    ↪ TITLE_OF_RELEVANT_PAPER]". Note that user A 'cc' user B means that A thought [
    ↪ TITLE_OF_RELEVANT_PAPER] is related to B's research, projects, or interests.
    ↪ Thoughts would not be negative.
* In this second paragraph of your answer, you must appreciate that [USERNAME] shared [
    ↪ TITLE_OF_RELEVANT_PAPER].
* In this second paragraph of your answer, you must NOT add in-line citations and citation
    ↪ numbers.
* The second paragraph should have no more than [CHARACTER_LIMIT_P2] characters.
Your answer should replace [TITLE_OF_RECOMMENDED_PAPER] with "this paper".
```

Example Prompt 2 (*after* adaption):

```
You are a helpful assistant.
Title: EvalLM: Interactive Evaluation of Large Language Model Prompts on User-Defined
    ↪ Criteria
Abstract: {By simply composing prompts, developers can prototype novel generative
    ↪ applications...[remainder of the abstract omitted for readability]}
The first paragraph of your answer should explain and specify the relationship between
    ↪ EvalLM: Interactive Evaluation of Large Language Model Prompts on User-Defined
    ↪ Criteria and Herding AI Cats: Lessons from Designing a Chatbot by Prompting GPT-3.
    ↪ Be informative.
* In this first paragraph of your answer, you must explain and specify how EvalLM:
    ↪ Interactive Evaluation of Large Language Model Prompts on User-Defined Criteria is
    ↪ related to Herding AI Cats: Lessons from Designing a Chatbot by Prompting GPT-3 in
    ↪ one short sentence.
* In this first paragraph of your answer, you must start with "This paper might be related
    ↪ to Herding AI Cats: Lessons from Designing a Chatbot by Prompting GPT-3 because".
* In this first paragraph of your answer, you must specify how EvalLM: Interactive
    ↪ Evaluation of Large Language Model Prompts on User-Defined Criteria cites Herding AI
    ↪  Cats: Lessons from Designing a Chatbot by Prompting GPT-3 in one short sentence.
    ↪ The content from EvalLM: Interactive Evaluation of Large Language Model Prompts on
    ↪ User-Defined Criteria that cites Herding AI Cats: Lessons from Designing a Chatbot
    ↪ by Prompting GPT-3: {However, most designers (P1-7) mentioned how they were unsure
    ↪ about how they should revise their prompts' wellknown challenge with LLM prompts
    ↪ [72]., However, as the space of possible natural language instructions is near
    ↪ infinite, designers need to test as many possibilities as possible to identify high-
    ↪ performing prompts [38, 72].", "As LLMs are non-deterministic and even partial
    ↪ changes in a prompt can significantly influence generated outputs [37, 41],
    ↪ designers need to iterate on their prompts multiple times to achieve satisfactory
    ↪ results [27, 39, 57, 69, 72, 73]., to automatically generate, evaluate, and revise
    ↪ outputs that satisfy the criteria--without the designer needing to "herd" the LLM
    ↪ themselves [72].}.
* The first paragraph should have no more than 425 characters.
Your answer should replace EvalLM: Interactive Evaluation of Large Language Model Prompts
    ↪ on User-Defined Criteria with "this paper".
```

Output of example Prompt 2:

> This paper might be related to Herding AI Cats: Lessons from Designing a Chatbot by Prompting GPT-3 because it cites the challenges in revising prompts, the need for extensive testing, and the iterative nature of achieving satisfactory results, referencing the concept of "herding" LLMs as discussed in the cited work.

Prompt 3 in Fig. 4.4 (*before* adaption):

```
You are a helpful assistant in finding the relationships between two papers.
  Title: [TITLE_OF_RECOMMENDED_PAPER]
  Abstract:  [ABSTRACT_OF_RECOMMENDED_PAPER]
  Title: [TITLE_OF_PAPER_SHARED_BY_RELEVANT_USER]
  Abstract: [ABSTRACT_OF_RELEVANT_USER_SHARED_PAPER]
```

```
  Your answer should explain and specify how [TITLE_OF_RECOMMENDED_PAPER] is related to and
    ↪  different from [TITLE_OF_PAPER_SHARED_BY_RELEVANT_USER] with no more than 300
    ↪ characters. If two papers are irrelevant, you should answer "NONE". Your answer must
    ↪  start with "This paper is related to '[TITLE_OF_PAPER_SHARED_BY_RELEVANT_USER]'
    ↪ because both papers". Be informative.
```

Example Prompt 3 (*after* adaption):

```
You are a helpful assistant in finding the relationships between two papers.
Title: EvalLM: Interactive Evaluation of Large Language Model Prompts on User-Defined
    ↪ Criteria
Abstract: {By simply composing prompts, developers can prototype novel generative
    ↪ applications...[remainder of the abstract omitted for readability]}
Title: Safer-Instruct: Aligning Language Models with Automated Preference Data
Abstract: {Reinforcement Learning from Human Feedback (RLHF) is a vital strategy for
    ↪ enhancing model...[remainder of the abstract omitted for readability]}
Your answer should explain and specify how EvalLM: Interactive Evaluation of Large Language
    ↪  Model Prompts on User-Defined Criteria is related to and different from Safer-
    ↪ Instruct: Aligning Language Models with Automated Preference Data with no more than
    ↪ 300 characters. If two papers are irrelevant, you should answer "NONE". Your answer
    ↪ must start with "This paper is related to Safer-Instruct: Aligning Language Models
    ↪ with Automated Preference Data because both". Be informative.
```

Output of example Prompt 3:

> This paper is related to Safer-Instruct: Aligning Language Models with Automated Preference Data because both focus on improving LLMs' outputs. EvalLM refines prompts via user feedback, while Safer-Instruct automates preference data generation for safer AI.

Prompt 4 in Fig. 4.4 (*before* adaption):

```
You are a helpful assistant for paper summarization.
[PROMPT_123_OUTPUTS]
First, you are required to shorten the above content with no more than [CHARACTER_LIMIT]
    ↪ characters. Note that:
- The shortened content must contain the following strings: [TITLE_OF_RELEVANT_PAPER]
- The shortened content must contain the following strings: [
    ↪ TITLE_OF_PAPER_SHARED_BY_RELEVANT_USER]
- The shortened content must contain the following strings: [RELEVANT_USER_SLACK_ID]
- Two papers are mentioned in [PROMPT_2_OUTPUT]. When you specify people's reactions or
    ↪ comments about [TITLE_OF_RELEVANT_PAPER], you should focus more on who reacted or
    ↪ commented and how these reactions or comments infer the potential reactions or
    ↪ comments about another paper based on the two papers' similarities than people's
    ↪ reactions or comments about [TITLE_OF_RELEVANT_PAPER].
- Do not remove any person's name (with or without '@'), institution's name, number, and
    ↪ conference/journal's name.
- Do not change the content's tone when it is low-confidence.
Second, you are required to bold at most three key phrases of the shortened content by
    ↪ adding ONE '*' to the left of the bolded text and ONE '*' to the right of the bolded
    ↪  text.
```

```
You should only bold the following text:
- The text tells what a paper is about.
- The text explains why a paper might be related to another paper or why a user might be
    ↪ interested in a paper, which is often after (does not include) the keywords such as
    ↪ "because" and "due to".
You should not bold the following text:
- [TITLE_OF_RELEVANT_PAPER]
- [TITLE_OF_PAPER_SHARED_BY_RELEVANT_USER]
- [RELEVANT_USER_SLACK_ID]
```

Example Prompt 4 (*after* adaption):

```
You are a helpful assistant for paper summarization.
{Congratulations to @tsk and team, including Juho Kim, on developing EvalLM! This system
    ↪ aids in refining LLM prompts by evaluating outputs against user-defined criteria,
    ↪ showing a 59% reduction in needed revisions. It enables examining twice as many
    ↪ outputs, enhancing prompt development efficiency.  This paper might be interesting
    ↪ to @xinyiz because @xinyiz've liked Safer-Instruct: Aligning Language Models with
    ↪ Automated Preference Data; improving LLMs' outputs. EvalLM refines prompts via user
    ↪ feedback, while Safer-Instruct automates preference data generation for safer AI.}
First, you are required to shorten the above content with no more than 386 characters. Note
    ↪  that:
- The shortened content must contain the following strings: Safer-Instruct: Aligning
    ↪ Language Models with Automated Preference Data
- The shortened content must contain the following strings: @anonymity
- Do not remove any person's name (with or without '@'), institution's name, number, and
    ↪ conference/journal's name.
- Do not change the content's tone when it is low-confidence.
Second, you are required to bold at most three key phrases of the shortened content by
    ↪ adding ONE '*' to the left of the bolded text and ONE '*' to the right of the bolded
    ↪  text.
You should only bold the following text:
- The text tells what a paper is about.
- The text explains why a paper might be related to another paper or why a user might be
    ↪ interested in a paper, which is often after (does not include) the keywords such as
    ↪ "because" and "due to".
You should not bold the following text:
- Safer-Instruct: Aligning Language Models with Automated Preference Data
- @anonymity
```

Output of example Prompt 4, where the title in blue is clickable and links to the article's Semantic Scholar page in our deployment:

> Congrats to @tsk, Juho Kim on **EvalLM**, reducing prompt revisions by 59% and doubling output examination. It refines LLM prompts with user feedback. This might interest @anonymity, who liked EvalLM: Interactive Evaluation of Large Language Model Prompts..., for its focus on **improving LLM outputs** through automated preference data for safer AI.

Table B.1: Social signals used in PaperPing

Category	Heuristic	Example message
Recommended Paper Information (I) *This paper is about...*		
I	**h1 -** Is the paper's author a channel member? *(i.e., Does the author show up at least once in the past 3 months among papers posted, reacted to, or replied to by at least one human user?)*	*Congrats @{UserSlackID}*
I	**h2 -** Was the paper's author mentioned recently in the channel?	*{AuthorName} also authored {K} papers that you've discussed in the past three months in the channel*
I	**h3 -** Does the author's affiliation overlap with a channel member's affiliation?	*A paper from {AuthorAffiliation}*
I	**h4 -** Was the paper's venue mentioned recently in the channel? *(i.e., Does the venue show up at least once in the past 3 months among papers posted, reacted to, or replied to by at least one human user?)*	*You've reacted positively to other papers from CHI*
Paper Connections (II) *This paper might be related to another paper in the channel, because...*		
II	**h5 -** What is the relationship between two papers? *(i.e., Does one cite the other? Do they have shared authors? Are two papers semantically relevant?)*	*Cites: The Participatory Turn in AI Design… (@anonymity shared in a thread), for its methodology*
II	**h6 -** How did channel members react and reply to another paper? *(If two papers are relevant, such reactions and replies could infer how channel members would react and reply to the recommended paper.)*	*Shared authors: Tae Soo Kim, Juho Kim, and Yoonjoo Lee also contributed to a related paper @anonymity1 shared*
II	**h7 -** Was another paper authored by a channel member?	*Related paper shared by @scholar with 2 replies and 3 reactions*
User Connections (III) *This paper might be interesting to someone in the channel, because...*		
III	**h8 -** Is the paper relevant to a channel member? *(i.e., Is the paper's semantic embedding similar to that of papers already of interest to a channel member?)*	*Possibly of interest to @anonymity*
III	**h9 -** What is the relationship between a user's interests and the recommended paper? *(i.e., Has the user liked a previous paper that's similar? Cites or is cited by this paper? Shares the same authors?)*	*Possibly of interest to @anonymity, because you've liked similar papers / authors / venues*

B.2 Social signals used in PaperPing

See Table B.1.

B.3 Numbers of Interactions in Each Channel

See Table B.2.

Table B.2: Numbers of Interactions in Each Channel

Channel index	# human recs before PaperPing	# human recs after PaperPing	# emojis before PaperPing	# emojis after PaperPing	# comments before PaperPing	# comments after PaperPing	# PaperPing recs
1	31	6	27	21	12	4	51
2	18	4	0	8	0	2	65
3	2	7	4	27	5	4	20
4	3	2	0	1	0	0	12
5	4	2	3	5	0	0	69
6	26	25	22	20	2	0	67
7	20	89	23	127	2	14	64
8	19	10	18	14	6	4	54
9	3	1	0	3	0	0	102
10	2	8	0	21	0	7	106
11	2	0	0	0	0	1	15
12	7	18	5	12	4	2	62
14	6	13	2	13	0	0	71
15	233	41	249	40	183	41	6
16	7	0	4	10	0	0	22
17	19	0	34	3	3	0	21
18	12	0	22	0	11	3	25